



HERNIA REPAIR: GENERAL PRINCIPLES—TENSION-FREE VERSUS TENSION

Thomas D. Kimbrough

STEP 1: SURGICAL ANATOMY

- ◆ Although the widespread adoption of mesh to repair hernias may lead the inexperienced and reckless to presume otherwise, a comprehensive understanding of the anatomy of the inguinal region is of paramount importance in inguinal hernia repairs.
- ◆ See **Figures 71-1 and 71-2** for surface and superficial anatomic structures. The typical location of each of the three inguinal hernia types is shown in Figure 71-2.

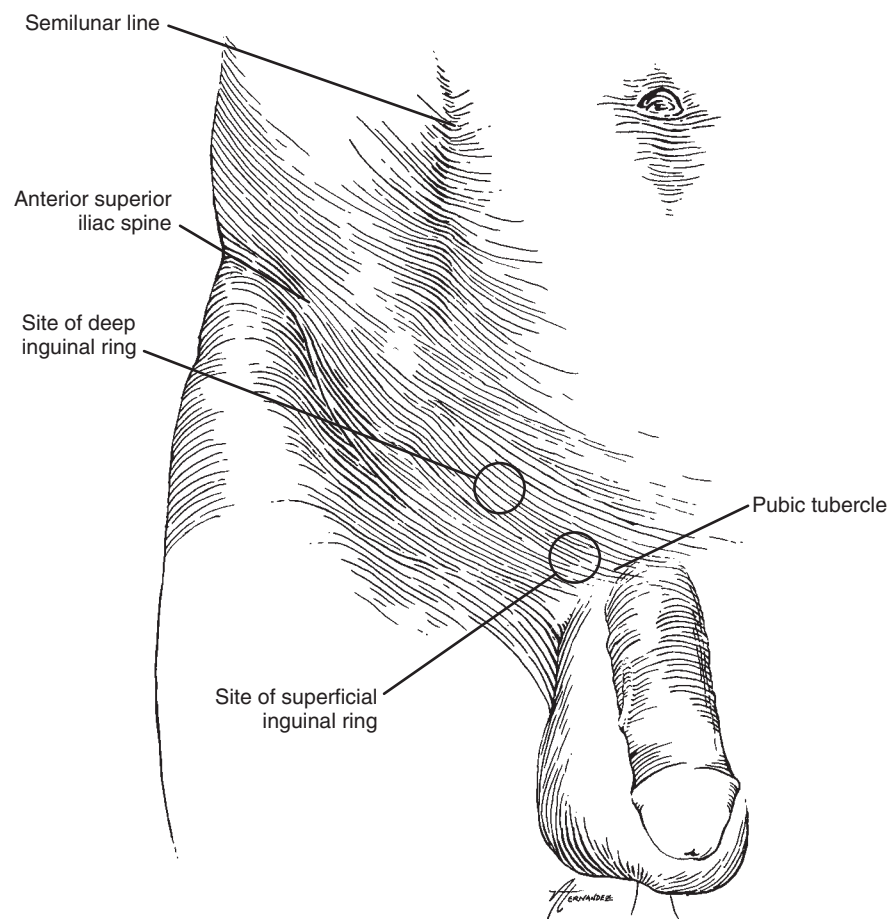


FIGURE 71-1

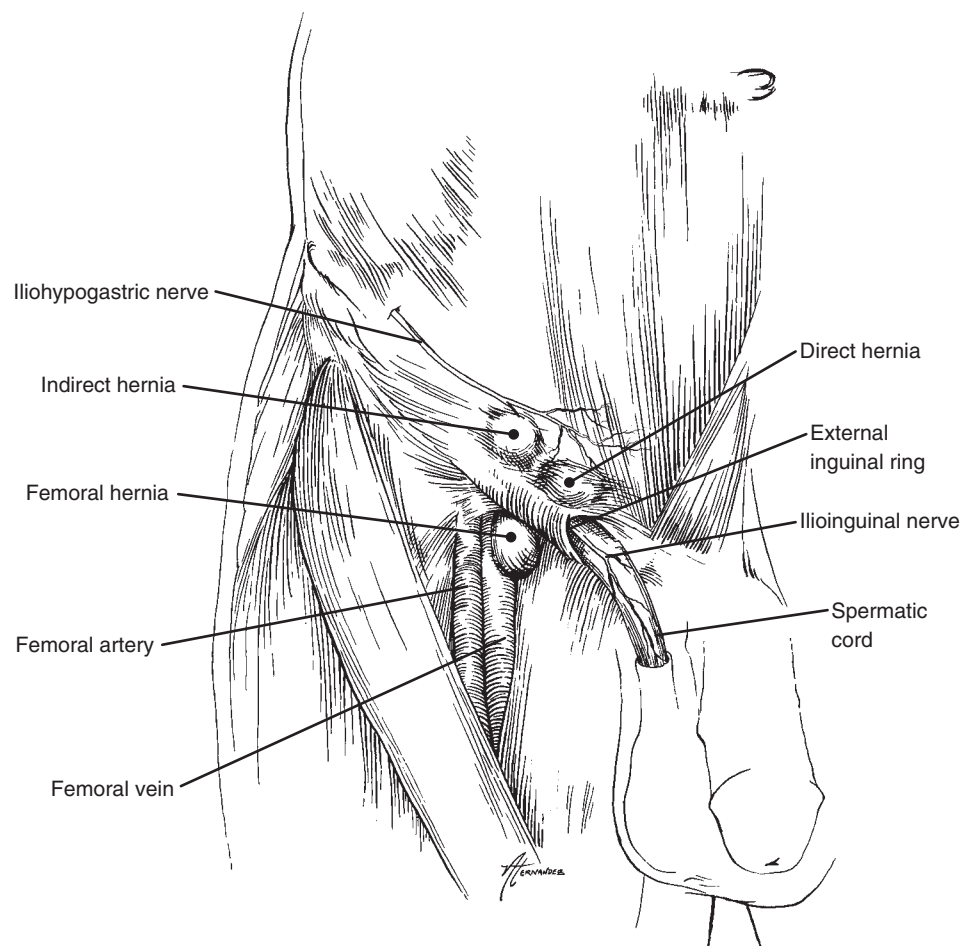
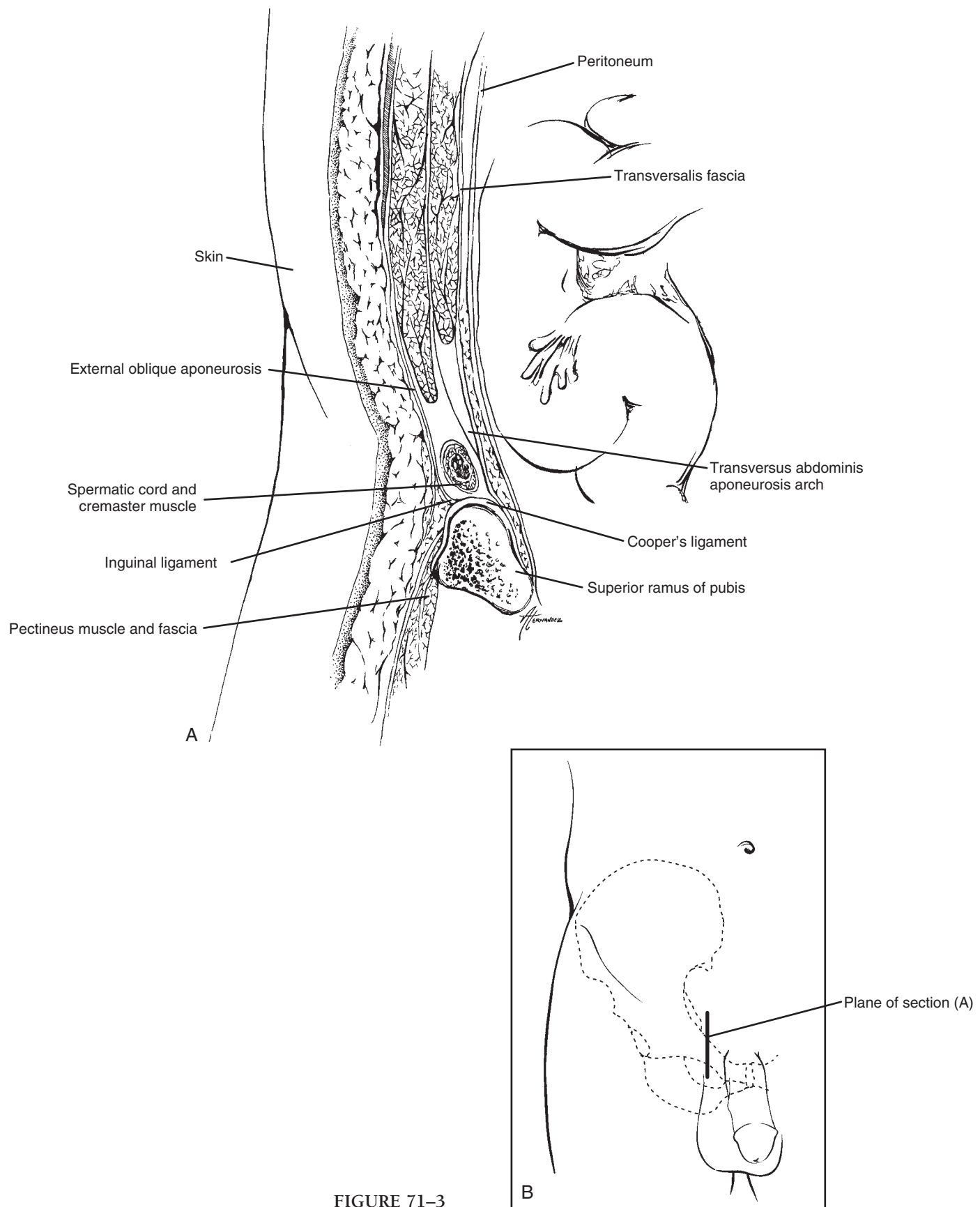


FIGURE 71-2

- ◆ **Figure 71-3** is a parasagittal representation of the various layers at a point midway in the right inguinal canal.



◆ **Figure 71-4** is a view from the inner surface of the abdominal wall—that is, from the preperitoneal space looking out.

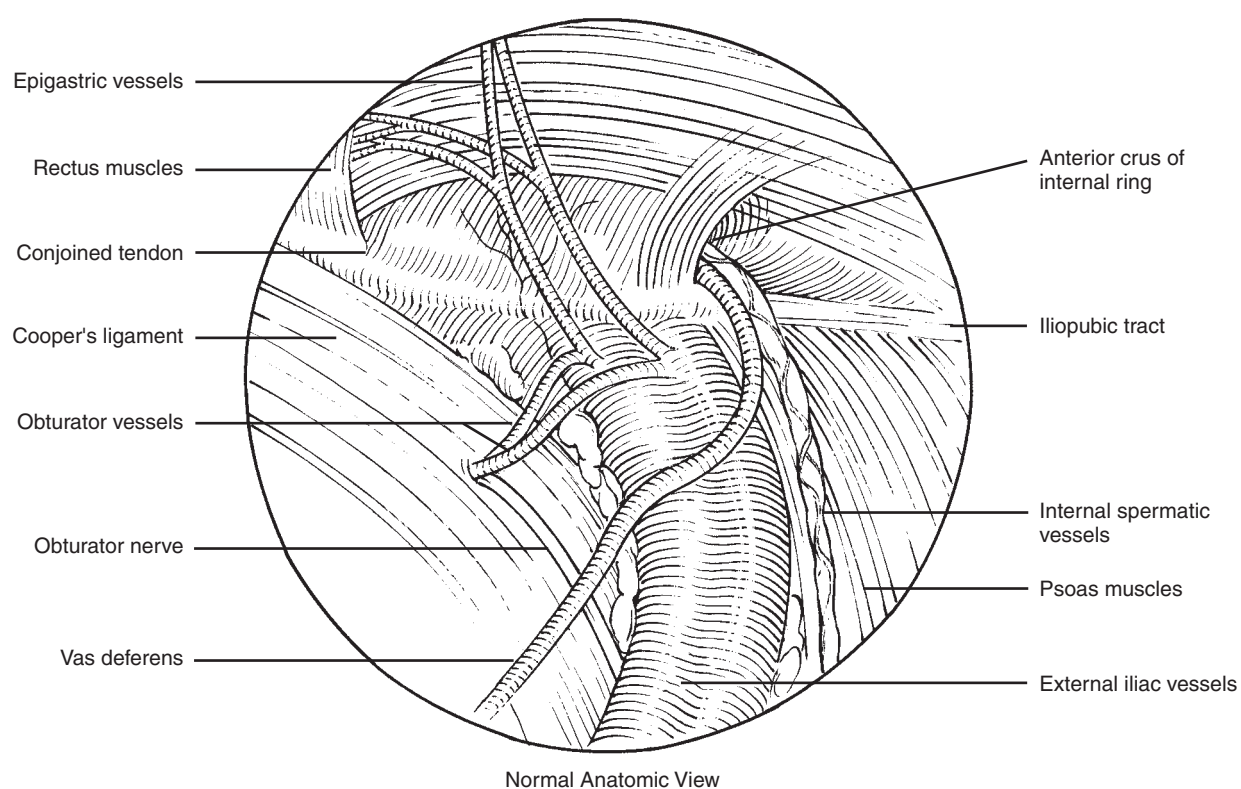


FIGURE 71-4

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ The actual incidence of dangerous complications of untreated hernias is quite low, and the mortal risk from these complications, when they do occur, is similarly low.
- ◆ The few deaths that result from hernia repair are far more likely to be due to complications of comorbidities than operative complications, so any evaluation of a candidate for repair should include careful attention to other medical problems. Appropriate evaluation and referral for evaluation and treatment of other significant medical problems should take precedence over a recommendation for operative repair in elective cases.
- ◆ On the other hand, the probable natural course of an untreated inguinal hernia over time is an increase in size and symptoms, so it is not unreasonable to offer repair to a young individual with no other medical problems.
- ◆ The classic recommendation to evaluate and treat those conditions that might chronically increase intra-abdominal pressure, including chronic cough, constipation, and difficulty with urination, fits under the previous admonition to evaluate comorbidities and does not otherwise bear special consideration.
- ◆ In the final analysis, the recommendation for a hernia repair requires of the surgeon a careful balance and consideration of the natural history of untreated hernias, their symptoms and complications, the patient's age and comorbidities, and the presence of symptoms and their immediate and anticipated effect on quality of life.

TENSION VERSUS TENSION-FREE

- ◆ A basic precept of surgery is that wounds closed under tension are less likely to heal well than those closed with little or no tension.
- ◆ All of the classic tissue repairs require the approximation of tissues that do not exist in that state naturally and thereby to one degree or another create tension on wound closures.
- ◆ Recognition of this has led to the use of mesh to bridge hernia defects and reinforce what has been increasingly recognized as an often attenuated, hernia-prone portion of the abdominal wall, even in those areas away from the actual hernia defect at the time of operation.
- ◆ The purported advantages of the tension-free repairs obtained with mesh include the following:
 - ◆ Mesh repairs have lower recurrence rates than tissue repairs.
 - ◆ Postoperative pain is less, and recovery to full activity is faster.
 - ◆ Long-term morbidity is the same as with tissue repairs.
 - ◆ Although there are case reports of complications arising from the mesh itself, these are sufficiently rare to not preclude its use.

- ◆ The ideal material for tension-free repair has not been identified, and further research holds the promise of developing materials, whether chemical or biologic, that will decrease even further those problems associated with use of mesh.
- ◆ There are arguably some cases in which mesh should or cannot be used. For this reason, surgeons should be familiar with classic tissue repairs, although one has to recognize that the exposure of surgeons in training to these repairs will be quite limited in most cases.

STEP 3: OPERATIVE STEPS

1. INCISION

- ◆ The steps outlined here will be those used in the classic anterior approach to the inguinal canal, which is satisfactory for the performance of most mesh and tissue repairs.
- ◆ **Figure 71-5** illustrates two possible skin incisions for approaching the inguinal region. Incision A lies in the lower inguinal skin fold. Incision B directly exposes the external ring.

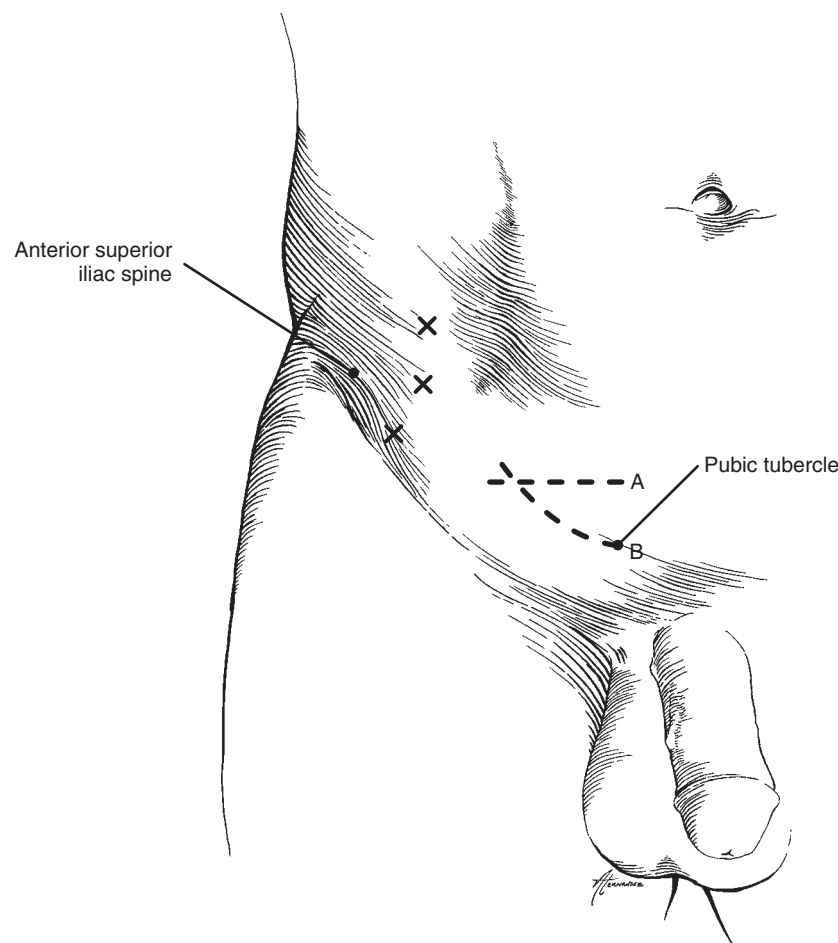


FIGURE 71-5

- ◆ Before the incision is made, 0.25% bupivacaine with epinephrine should be injected medial to the anterior superior iliac spine and deep to the external oblique aponeurosis. Approximate areas of injection are marked with an “X” on Figure 71-5. An additional 10 mL should be injected at the site of the incision.
- ◆ The skin incision should be carried through the superficial subcutaneous fat, Scarpa’s fascia, and the deep subcutaneous fat until the easily recognized oblique fibers of the external oblique aponeurosis are identified.

2. DISSECTION

- ◆ The surgeon should then find the external inguinal ring, obtaining exposure as illustrated in **Figure 71-6**. Such exposure can be achieved with hand-held retractors or, as shown, with self-retaining ones.
- ◆ As shown in Figure 71-6, the aponeurosis of the external oblique should be divided in the line of its fibers through the external inguinal ring. Some advocate the use of a knife for the initial incision through the aponeurosis. I prefer scissors because they are less likely to injure underlying structures.
- ◆ Using a combination of blunt and sharp dissection, the external oblique aponeurosis should be freed superiorly and laterally from the underlying internal oblique muscle and fascia. Inferiorly and laterally, the cord structures should be swept off the undersurface of the external oblique aponeurosis down to the shelving edge of the inguinal ligament.
- ◆ The next step is longitudinal division of the fibers of the cremaster muscle that invests the structures of the spermatic cord as shown in **Figure 71-7**. Care should be taken to divide the cremaster away from the ilioinguinal nerve.

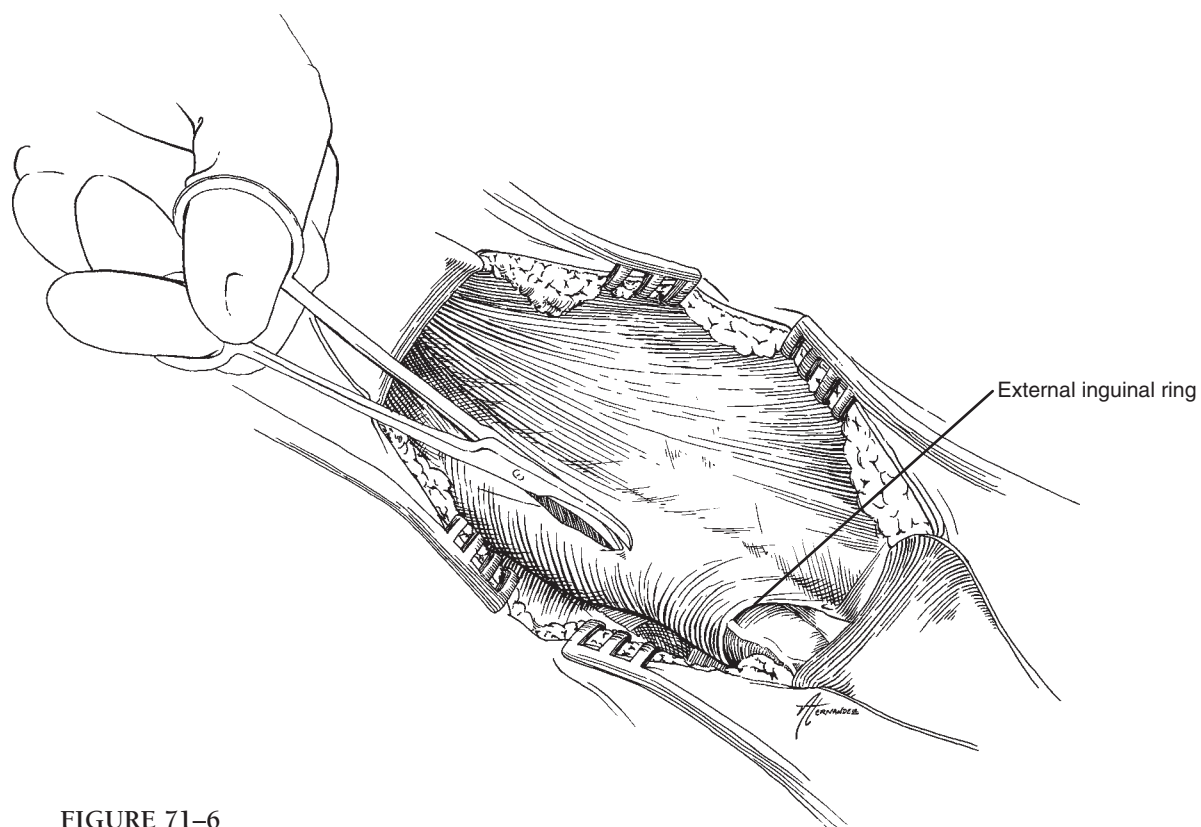


FIGURE 71-6

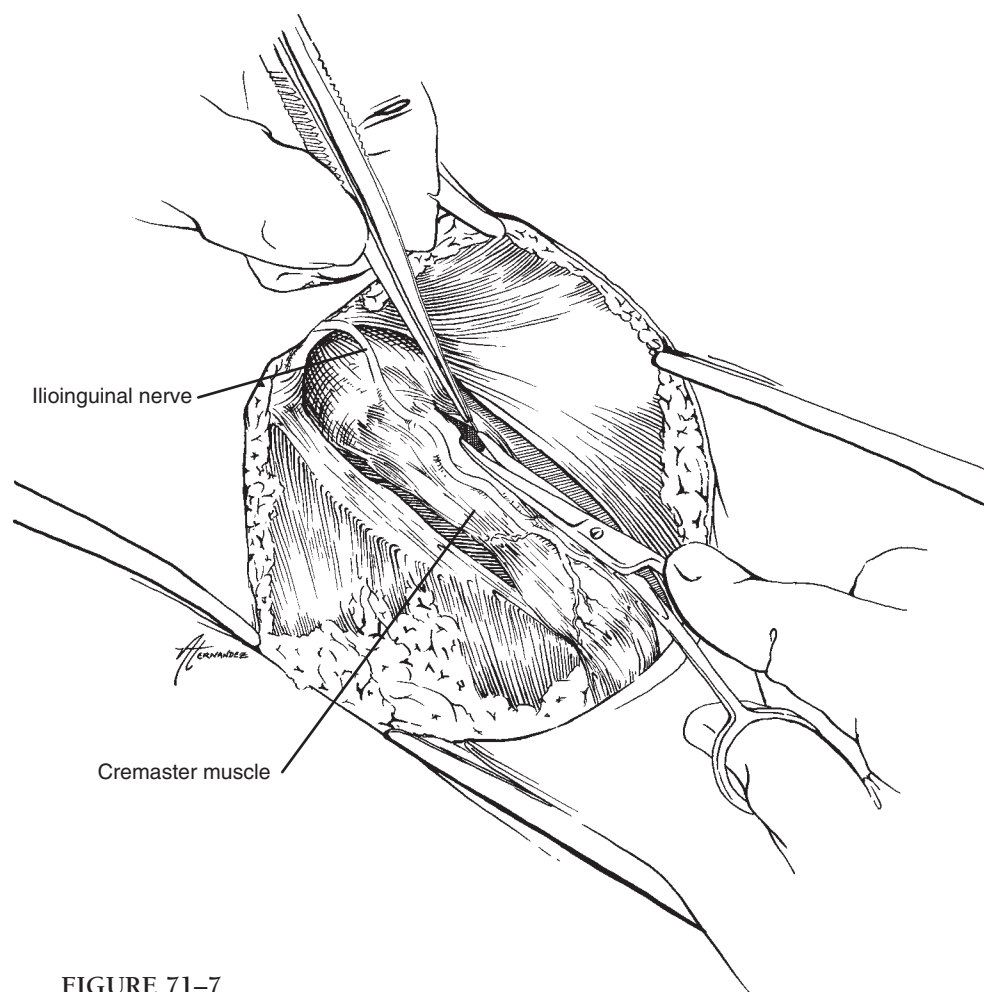


FIGURE 71-7

- ◆ Superiorly, inferiorly, and posteriorly, the cremasteric fibers should be separated circumferentially from the structures of the spermatic cord. A Penrose drain can then be placed around the spermatic cord for traction as shown in **Figure 71-8**.
- ◆ In the case of an indirect hernia, this process will expose the hernia sac lying on the anterior superior part of the cord, as illustrated in Figure 71-8.
- ◆ The indirect hernia sac should then be separated carefully from the spermatic cord well up into the internal ring. Care should be taken to take down any soft tissue connections between the sac and the borders of the internal ring so that the sac can be restored to the preperitoneal space. Figure 71-8 illustrates the view on completion of this process. As shown, the inferior epigastric vessels can be seen medial to the sac and spermatic cord.
- ◆ **Figure 71-9** shows the appearance of a direct hernia. Once the spermatic cord is retracted with a Penrose drain, the direct sac can be easily dissected free of the cord and cremasteric fibers.

3. CLOSING

- ◆ Various options for repair of the hernia defect will be described in the following chapters.
- ◆ Once the repair is completed, the aponeurosis of the external oblique is reapproximated with a running absorbable suture, in the process recreating the external inguinal ring.
- ◆ Scarpa's fascia and superficial subcutaneous tissues are closed with interrupted absorbable suture.
- ◆ The skin is closed with running, subcuticular, absorbable suture reinforced with Steri-Strips.

STEP 4: POSTOPERATIVE CARE

- ◆ Most of these repairs are outpatient procedures, and patients can be discharged with a prescription for a mild analgesic agent.
- ◆ Activity instructions will vary with the type of repair and the opinion of the individual surgeon. In most cases, I advise patients to advance their level of activity as tolerated.

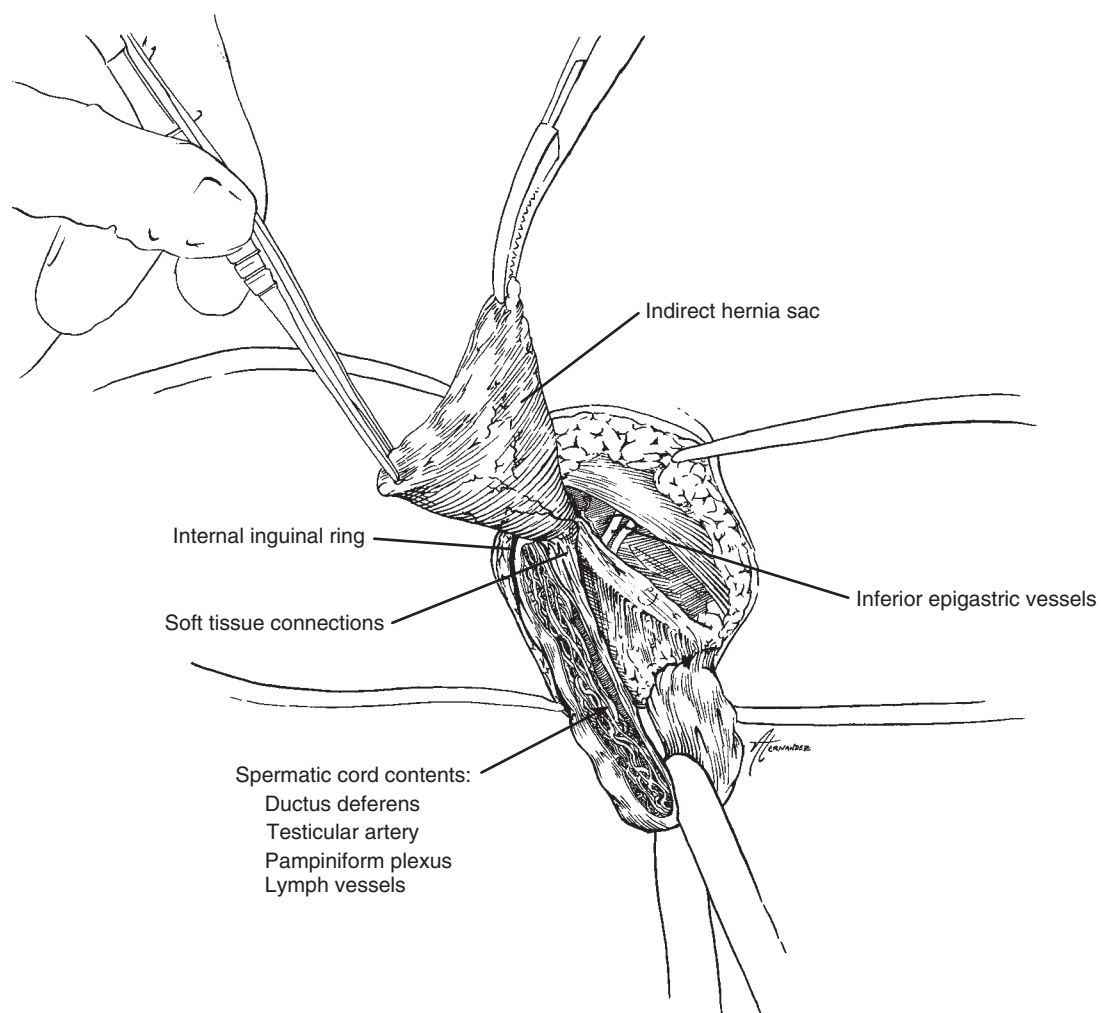


FIGURE 71-8

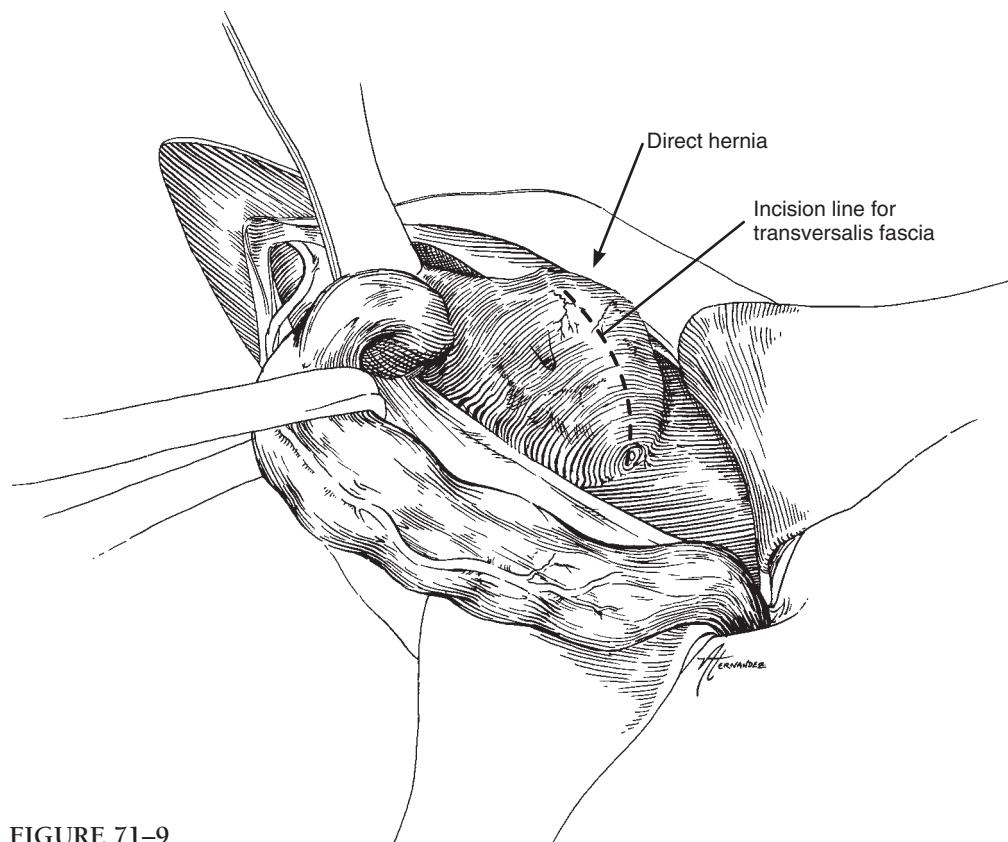


FIGURE 71-9

STEP 5: PEARLS AND PITFALLS

- ◆ Although long-known as “intern cases,” a careful review of the outcome literature on hernia repair shows conclusively that experience and knowledge produce better results.
- ◆ Superficial postoperative hematomas generally arise from a failure to secure the superficial epigastric veins that are encountered in exposing the external oblique aponeurosis. Although not always necessary, the safest course is to divide them between an appropriate suture ligature, such as 3-0 silk or Vicryl.
- ◆ In separating the structures of the spermatic cord from the investing cremasteric fibers, it is helpful to remember that the ductus deferens is usually the most posterior element that needs be mobilized and is easily identified by palpation. Although the genital branch of the genitofemoral nerve is posterior to the ductus deferens, I have not found it necessary to mobilize the nerve to obtain necessary exposure.
- ◆ Sharp dissection of an internal hernia sac off the spermatic cord with good exposure by traction and countertraction is less likely to lead to damage to the pampiniform venous plexus than blunt dissection.

SELECTED REFERENCES

1. Zinner MJ, Schwartz SI, Ellis H: Hernias. In Maingot R, Zinner M (eds): *Maingot's Abdominal Operations*, vol 1, 10th ed. Stamford, Conn, Appleton & Lang, 1997, pp 479-580.
2. Condon RE: The anatomy of the inguinal region and its relation to groin hernias. In Nyhus LM, Condon RE (eds): *Hernia*, 4th ed. Philadelphia, JB Lippincott, 1995, pp 16-72.



MESH REPAIR

Thomas D. Kimbrough

STEP 1: SURGICAL ANATOMY

- ◆ See Chapter 71.

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ There are several types of mesh repairs, and to date no studies have shown clear superiority of one over another.
- ◆ The Prolene Hernia System (PHS) repair, developed under the guidance of Dr. Arthur Gilbert, is illustrated here.

STEP 3: OPERATIVE STEPS

1. INCISION

- ◆ See Chapter 71.

2. DISSECTION

- ◆ The mesh used in the repair is illustrated in **Figure 72-1**. It comes in three sizes: extended, large, and medium.
- ◆ The technique of repair differs for indirect and direct hernias. The indirect repair will be described first. The initial operative steps for both indirect and direct hernias are outlined in Chapter 71.
- ◆ Mesh repair of an indirect hernia
 - ◆ After the hernia sac has been dissected free from all attachments to the spermatic cord and internal ring (as illustrated in Chapter 71), the next step is to return the sac to the preperitoneal space. It is not necessary to open the sac unless it contains irreducible contents, which require visual inspection. There is no advantage to resection and high ligation and that is not done. Associated cord lipomas, which are actually preperitoneal fat that has protruded through the internal ring, can be amputated or restored to their original location.
 - ◆ The preperitoneal space should be developed by bluntly pushing the peritoneal sac away from the transversalis or endoabdominal fascia. This can be facilitated by stuffing a gauze sponge through the internal ring and then withdrawing it.

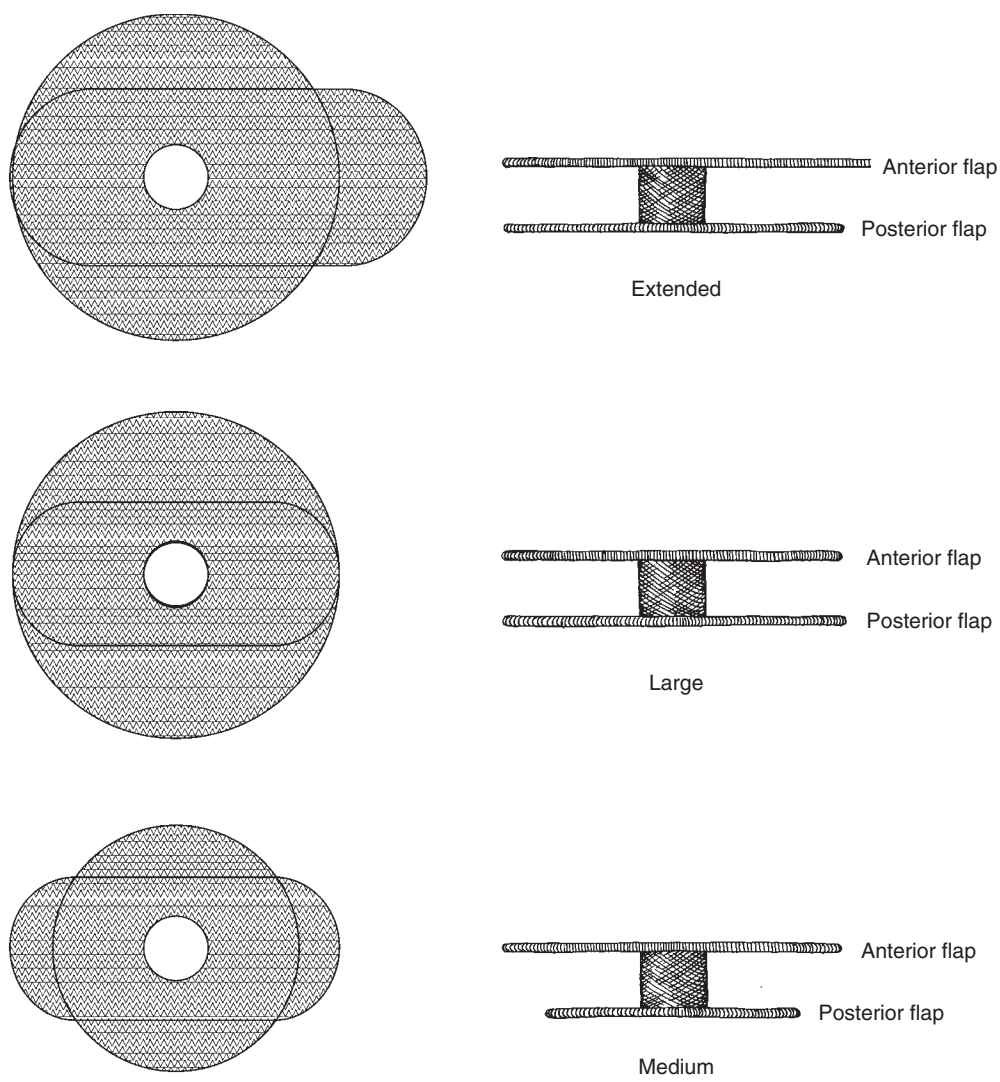


FIGURE 72-1

- ◆ The PHS mesh is loaded onto an empty sponge forceps as illustrated in **Figure 72-2** and inserted through the internal ring. When the extended size is being used, the shorter of the two long anterior flaps should be positioned medially.
- ◆ The direction of insertion should be toward the umbilicus. The index finger of the opposing hand can be inserted through the ring into the preperitoneal space as a guide.
- ◆ The posterior flap of the mesh should be spread out in the preperitoneal space. This can be done with a finger or a pair of forceps, depending on the size of the internal ring. Medially and superiorly, the mesh should lie flat underneath the pubic tubercle and transversalis fascia. Inferiorly and laterally, the mesh will conform to the pelvic architecture and external iliac vessels.
- ◆ The anterior flap is then deployed in the floor of the inguinal canal. It is tacked into place in at least three locations with 2-0 Vicryl. Site *A* on **Figure 72-3** should be over the pubic tubercle with mesh overlap of at least 3 cm. Site *B* on Figure 72-3 should be a superficial site fixing the mesh to the fascia of the internal oblique only. Finally, as shown in Figure 72-3, a notch should be cut in the inferior part of the anterior mesh at the internal ring. The two flaps created should be wrapped around the spermatic cord at the internal ring and secured to each other and the shelving edge of the inguinal ligament with 2-0 Vicryl. The resulting opening through which the spermatic cord passes should be loose enough to allow easy introduction of a pair of forceps.
- ◆ The lateral flap is tucked under the uncut portion of the external oblique aponeurosis. Excess mesh on the anterior flap can be trimmed as desired.

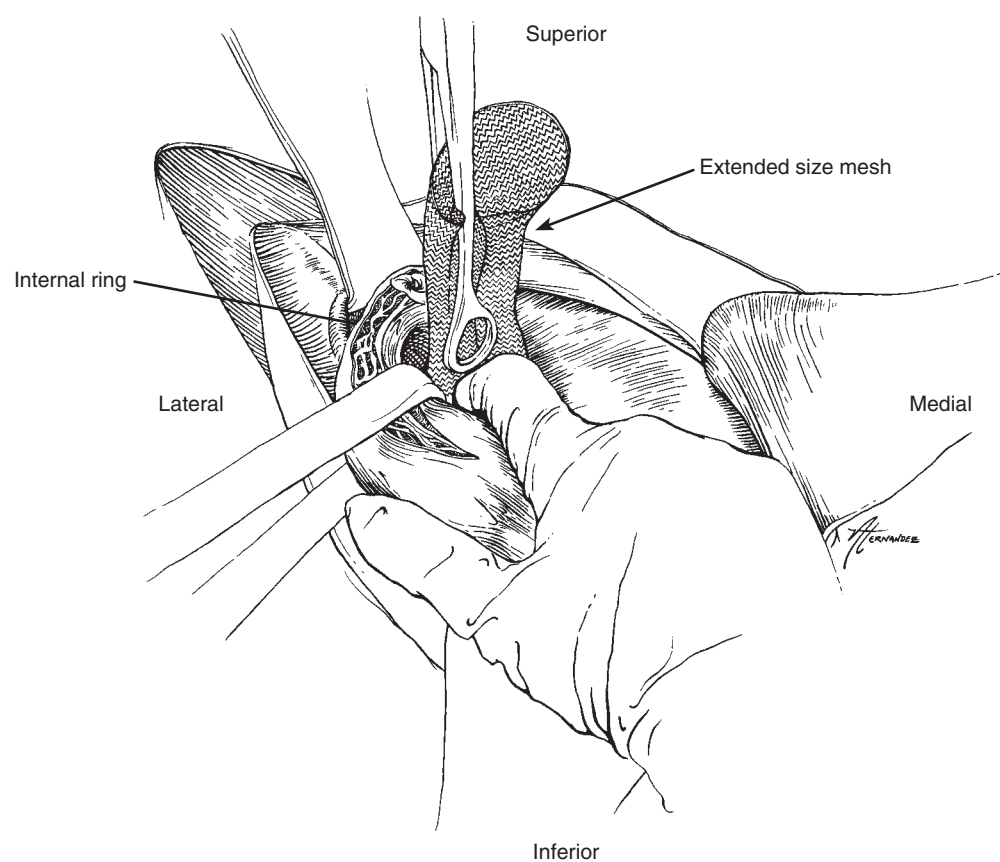


FIGURE 72-2

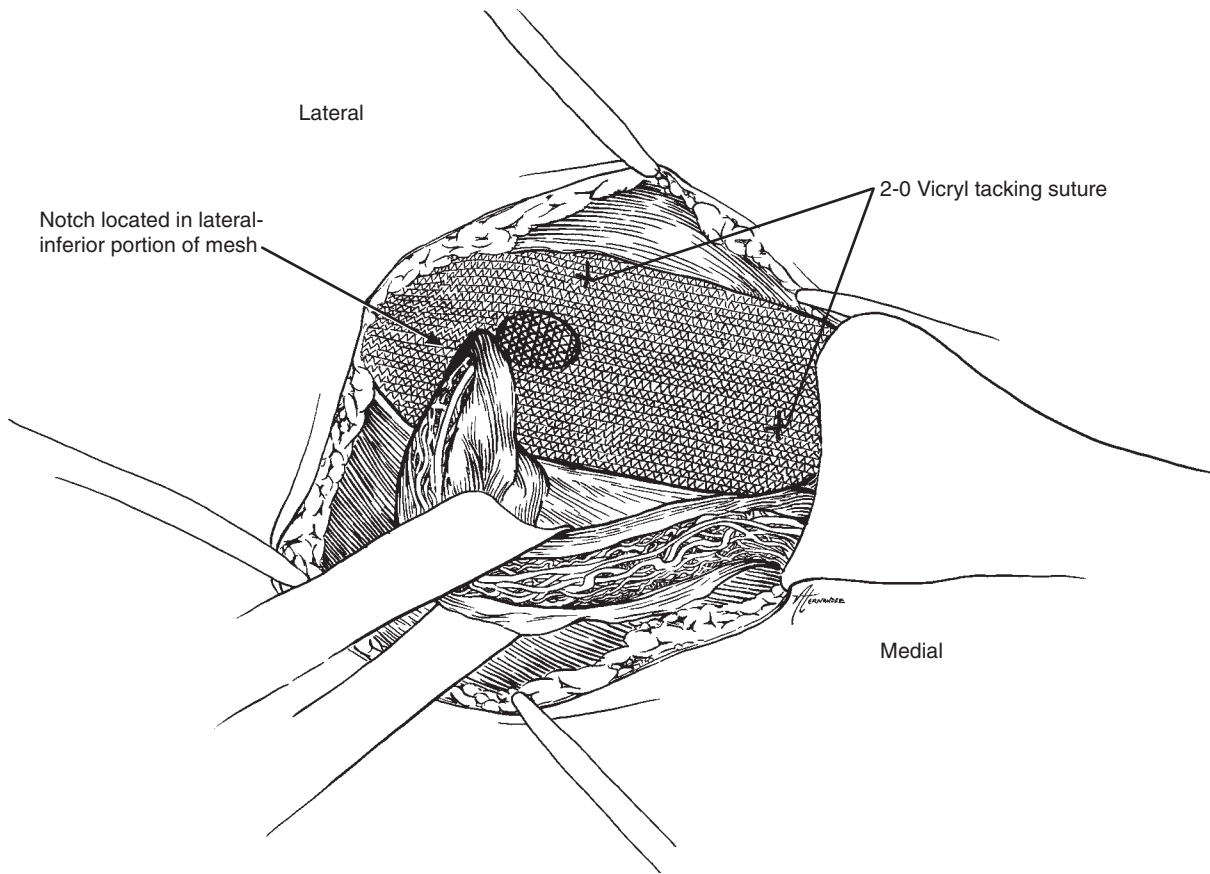


FIGURE 72-3

- ◆ Mesh repair of a direct hernia
 - ◆ In the PHS mesh repair of a direct hernia, the posterior flap of the mesh is deployed into the preperitoneal space through an incision in the attenuated transversalis fascia. If the hernia is a large direct hernia, you may desire to resect some of the attenuated transversalis fascia as shown in **Figure 72-4**.
 - ◆ As has always been the case, the spermatic cord should be inspected and explored to ensure there is not an indirect hernia also. The posterior flap of the mesh is deployed in the preperitoneal space as shown in **Figure 72-5**. Care must be taken to ensure that the mesh extends laterally to cover and protect for herniation through the internal ring. This is sometimes facilitated by dividing and ligating the inferior epigastric vessels.
 - ◆ In larger direct hernias, it is possible for the mesh to be extruded through the transversalis fascia before fibroblastic ingrowth and collagen deposition in the mesh interstices occurs. As a result, it is my practice to do the following: The connecting ring between the anterior and posterior flaps is left seated in the incision through the transversalis fascia after the posterior flap is deployed. The two sides of the cut portions of the transversalis fascia are affixed to each other and to the medial and lateral edges of the connecting ring, as shown in **Figure 72-6**. Should the incision through the transversalis be larger than the diameter of the connecting ring, additional sutures of 2-0 Vicryl are used to approximate the cut edges of the transversalis, again catching the underlying posterior mesh with each suture.

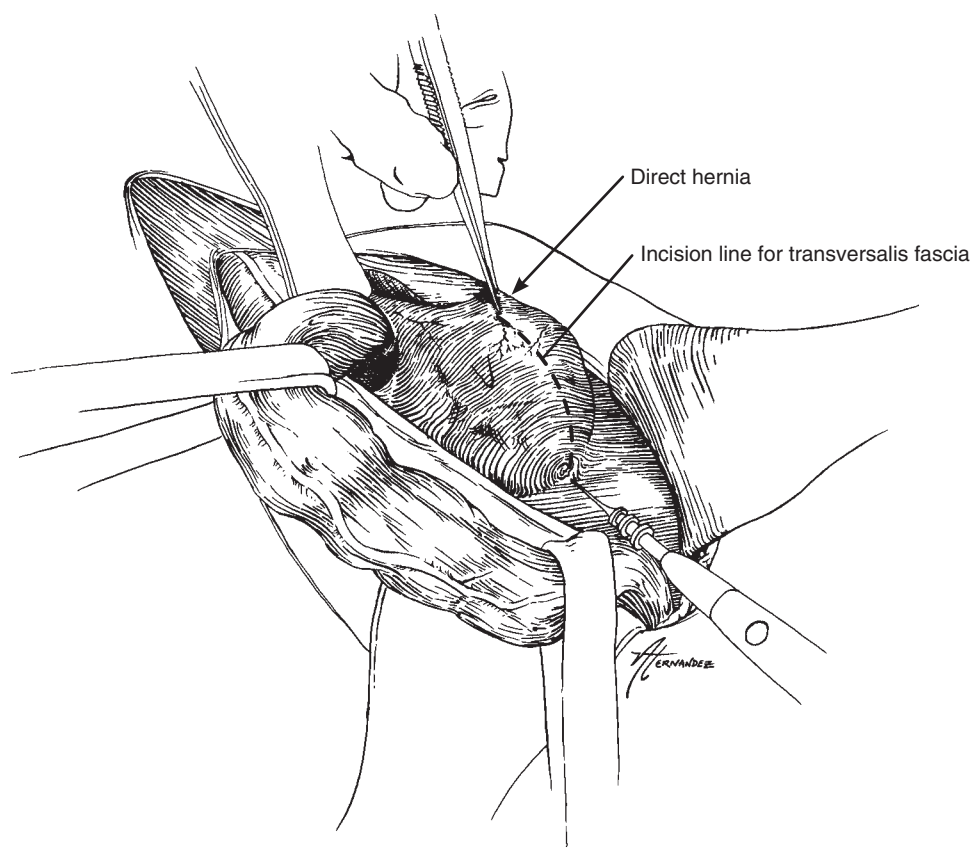


FIGURE 72-4

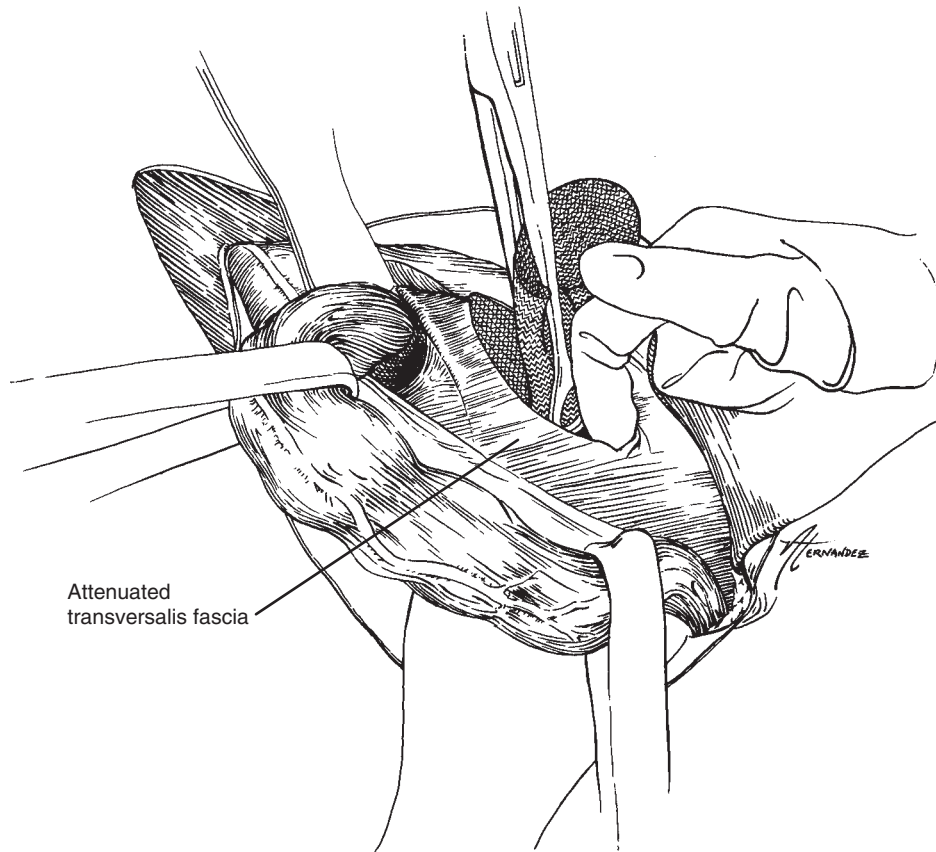


FIGURE 72-5

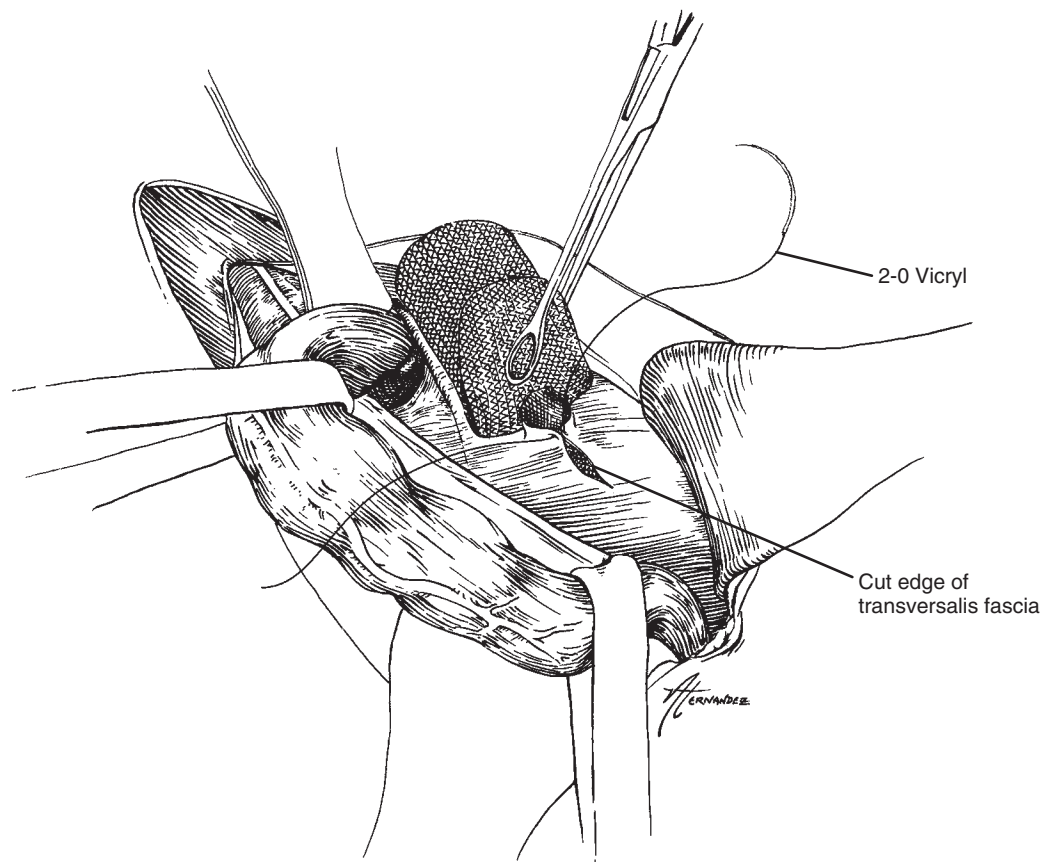


FIGURE 72-6

- ◆ Deployment and fixation of the anterior flap is accomplished much the same as in repair of an indirect hernia with two exceptions (see Figure 72-6).
- ◆ The notch at the internal ring is cut in the lateral inferior part of the mesh because of the more medial placement of the connecting ring (**Figure 72-7**).
- ◆ Most surgeons are more generous with tacking sutures because of the greater propensity for mesh extrusion early in the postoperative course.

3. CLOSING

- ◆ Closure is the same for either direct or indirect hernias with approximation of the external oblique aponeurosis over the completed repair with running 2-0 or 3-0 Vicryl. Care is taken to reconstruct the external ring.
- ◆ Scarpa's fascia and the superficial subcutaneous fat are closed with interrupted 3-0 Vicryl sutures to eliminate dead space and remove tension from the skin closure.
- ◆ The skin is closed with running subcuticular 4-0 Monocryl reinforced with Steri-Strips.

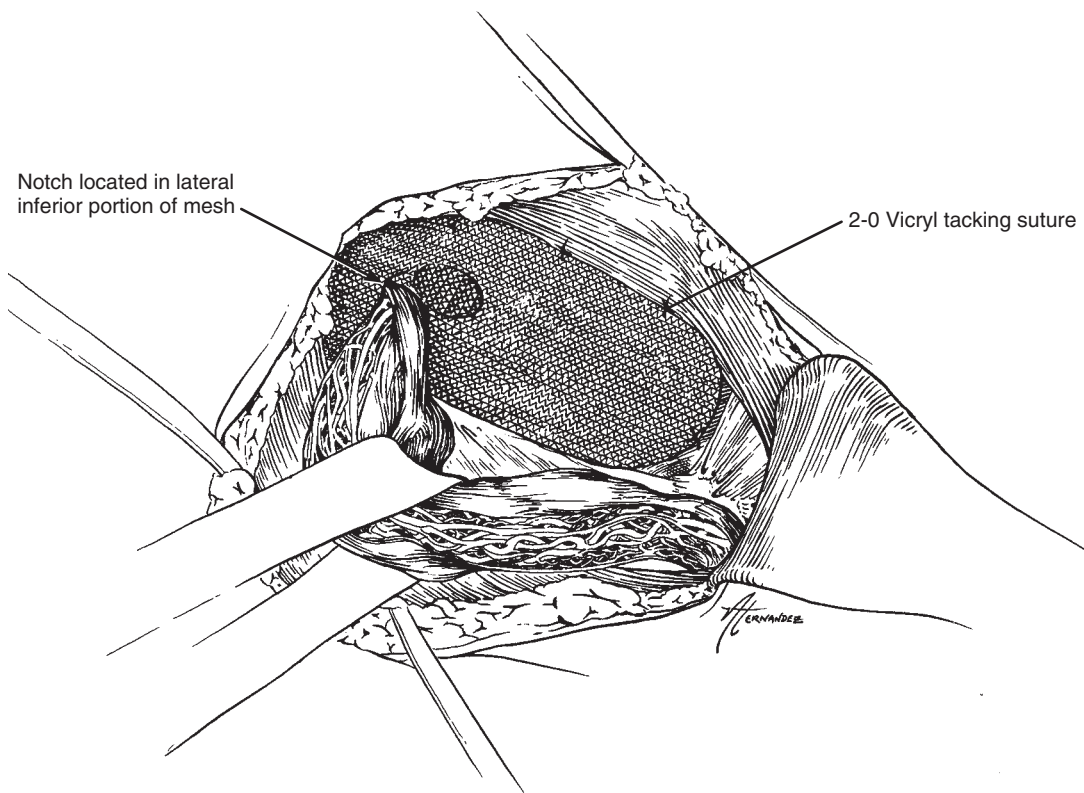


FIGURE 72-7

STEP 4: POSTOPERATIVE CARE

- ◆ Wound care instructions are similar to those of any other abdominal operation. Patients are told that they can begin showering in 24 to 36 hours. Steri-Strips are left on until they fall off or need to be removed for other reasons.
- ◆ Patients are discharged on the day of surgery with prescriptions for oral pain medications.
- ◆ In general, patients are told that they may resume any and all activity in a stepwise progression as their own comfort dictates.
- ◆ The only exceptions are patients with large, blowout, direct hernias. In these cases, most patients limit activity for 3 to 4 weeks to help prevent mesh extrusion.

STEP 5: PEARLS AND PITFALLS

- ◆ I always use the extended size in adult males, preferring to trim any unnecessary mesh from the anterior flap rather than using too small a piece overall. In women, an extended or large size generally suffices. I have never used the medium size in an inguinal hernia repair.
- ◆ The anterior flap of the mesh is designed so that its long axis should parallel the long axis of the inguinal canal. Because the longitudinal orientation of the mesh is difficult to change once the posterior flap is deployed, correct orientation is best achieved by loading the mesh onto the sponge forceps such that the handles lie parallel to the long axis of the mesh and the inguinal canal at insertion.
- ◆ If, as is often the case, it is necessary to pull the anterior flaps out of the preperitoneal space after insertion on the sponge forceps, it is useful to place the index finger of the opposing hand in the connecting ring between the anterior and posterior flaps of the mesh. This will prevent accidental withdrawal of the posterior flap at the same time.

SELECTED REFERENCES

1. Zinner MJ, Schwartz SI, Ellis H: Hernias. In Maingot R, Zinner M (eds): *Maingot's Abdominal Operations*, vol 1, 10th ed. Stamford, Conn, Appleton & Lang, 1997, pp 479-580.
2. Condon RE: The anatomy of the inguinal region and its relation to groin hernias. In Nyhus LM, Condon RE (eds): *Hernia*, 4th ed. Philadelphia, JB Lippincott, 1995, pp 16-72.



CHAPTER

73

INGUINAL HERNIORRHAPHY—BASSINI

William J. Mileski

STEP 1: SURGICAL ANATOMY

- ◆ The anterior superior iliac spine and the ipsilateral pubic tubercle are identified and marked (**Figure 73-1**).

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ To avoid any confusion about the site of surgery, the site should be marked by the operating surgeon while the patient is awake in the preoperative area.
- ◆ Patient is positioned supine.
- ◆ Anesthesia may be general, regional, or local.

STEP 3: OPERATIVE STEPS

1. INCISION

- ◆ A near transverse incision is then made in the skin, paralleling Langer's lines, 1 to 2 inches inferior and medial to the iliac spine to a point 1 to 2 inches lateral and superior to the pubic tubercle (see **Figure 73-1**).

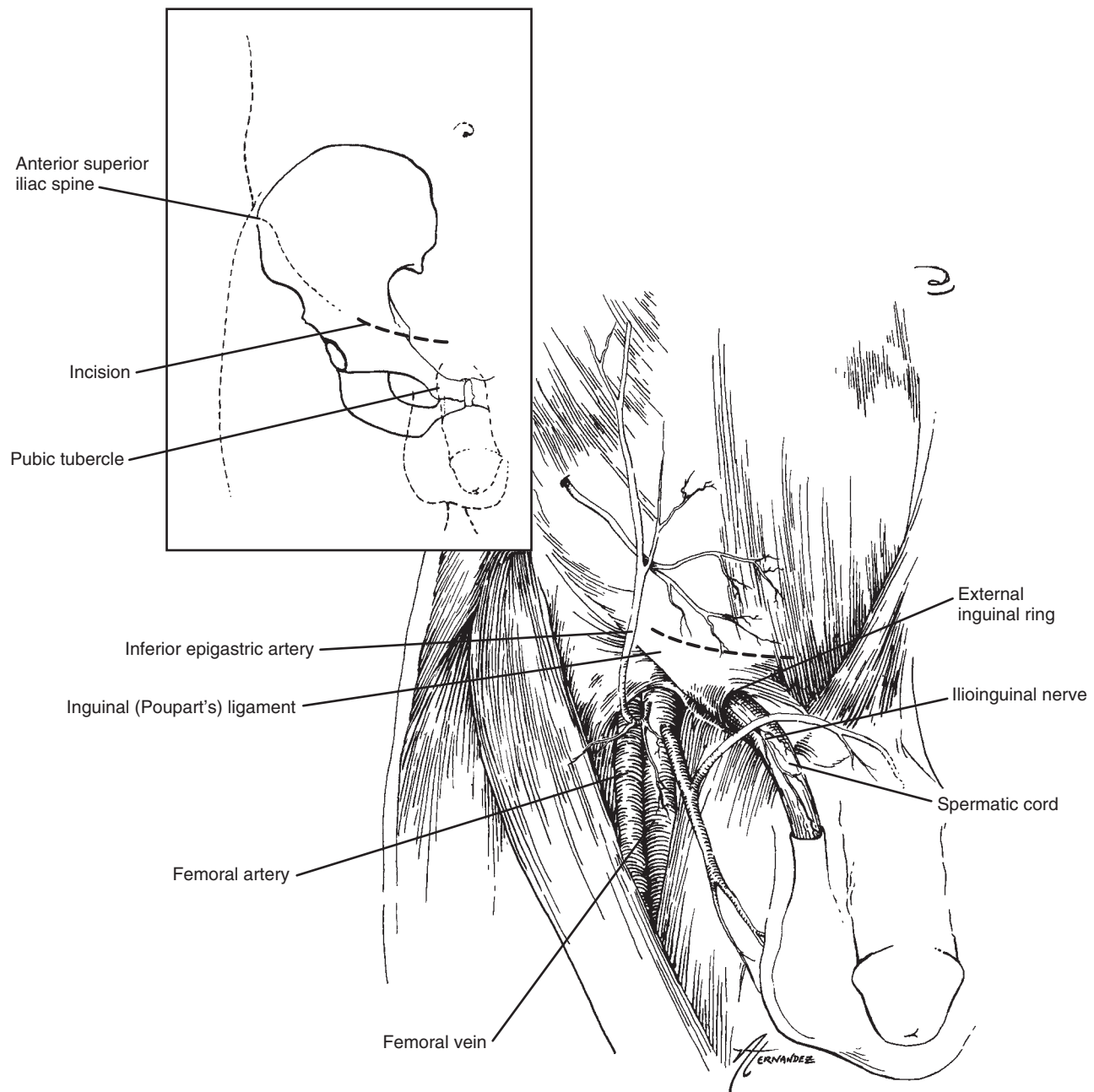


FIGURE 73-1

2. DISSECTION

- ◆ Subcutaneous fat is dissected, and the external oblique aponeurosis, inguinal ligament, and external inguinal ring are exposed (**Figure 73-2**).
- ◆ The external oblique is incised from the external ring to a point lateral to the internal ring. Blunt dissection is used to separate the external oblique fascia from the underlying tissues, and the external oblique fascia is held with a self-retaining retractor (**Figure 73-3**).
- ◆ The ilioinguinal nerve is dissected from the cord structures and cremaster and retracted to the inferior lateral aspect of the wound.

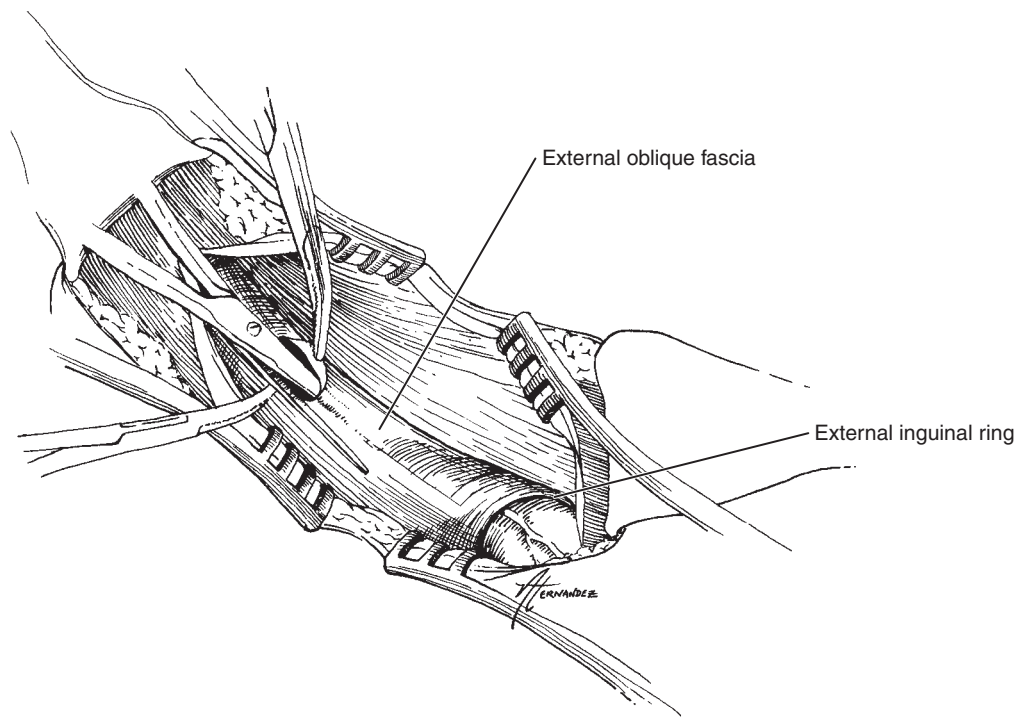


FIGURE 73-2

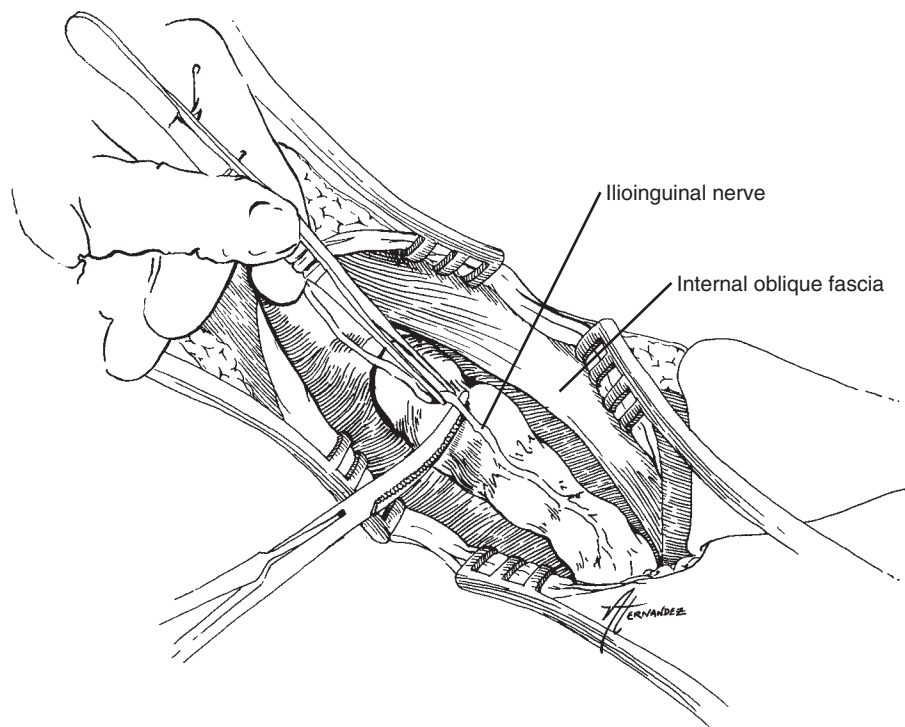


FIGURE 73-3

- ◆ The spermatic cord is then fully encircled at the pubis and retracted with a Penrose drain (Figures 73-4 and 73-5). Cremasteric fibers are dissected from the cord structures, and if present the indirect hernia sac is identified; its contents are reduced; and the sac is ligated with a nonabsorbable suture, such as 2-0 silk (Figure 73-6).

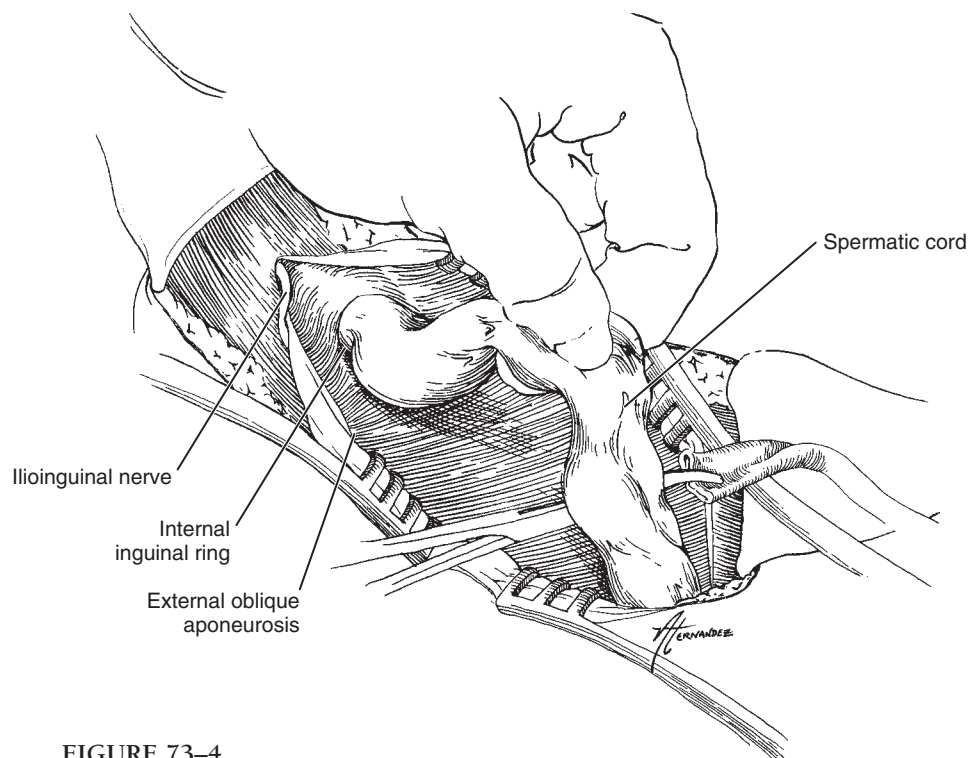


FIGURE 73-4

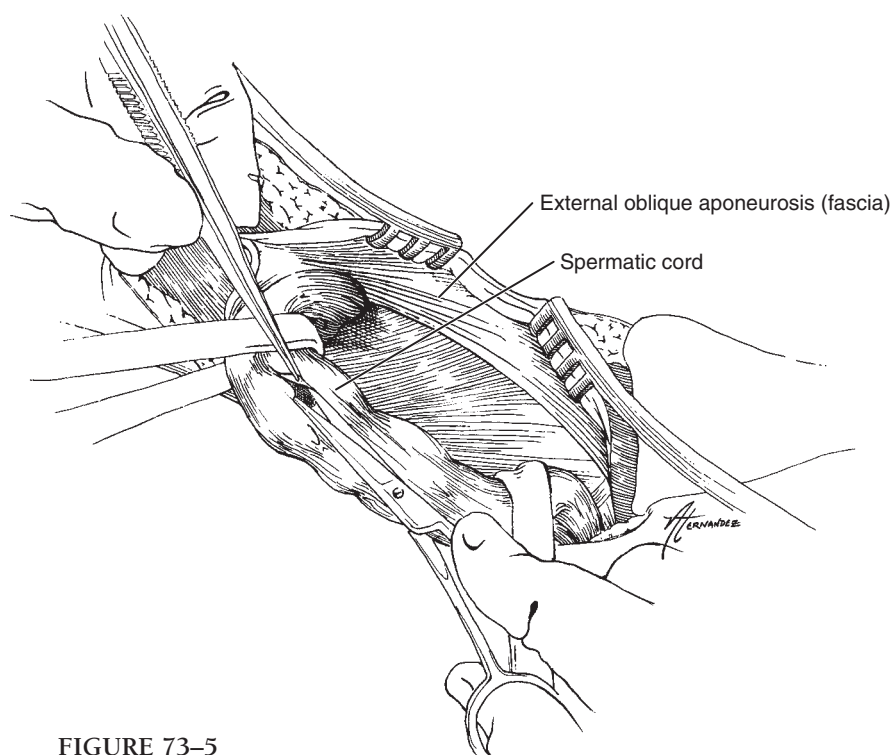


FIGURE 73-5

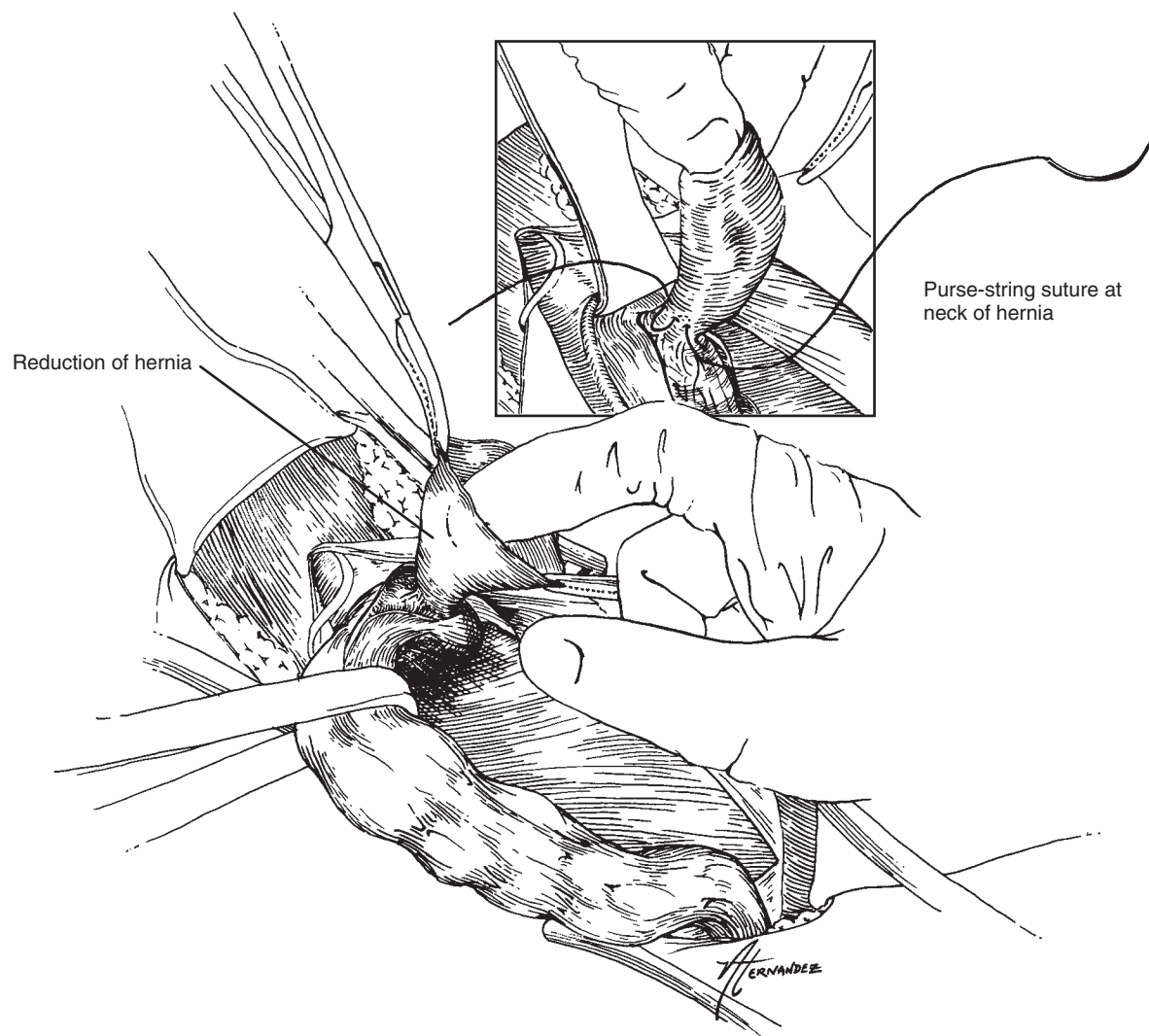


FIGURE 73-6

- ◆ The repair of the inguinal floor can then be accomplished by approximating the reflex portion of the inguinal ligament to the inferior aspects of the transversalis fascia and the transversus abdominis muscles. In some patients this is clearly fused as a conjoint tendon; however most often the tissues of the transversalis and transversus abdominis are difficult to distinguish. This can be aided by retracting the internal oblique muscle with a rake or Senn retractor and grasping the transversalis fascia and transversus abdominis with several Allis clamps. Interrupted nonabsorbable sutures (0 silk or 0 Prolene) are then placed approximately 1 cm apart from the pubic tubercle to the internal ring. It is important to avoid injury to the femoral vessels as the sutures are placed near the internal inguinal ring. Two additional ligatures are placed lateral to the internal ring, leaving enough room for the spermatic cord without causing venous obstruction (**Figure 73-7**).
- ◆ The spermatic cord and ilioinguinal nerve are replaced, and the external oblique muscle is reapproximated using a running absorbable suture, such as 2-0 Vicryl (**Figure 73-8**).

3. CLOSING

- ◆ Skin is closed with a running subcuticular suture of 3-0 Monocryl (**Figure 73-9**).

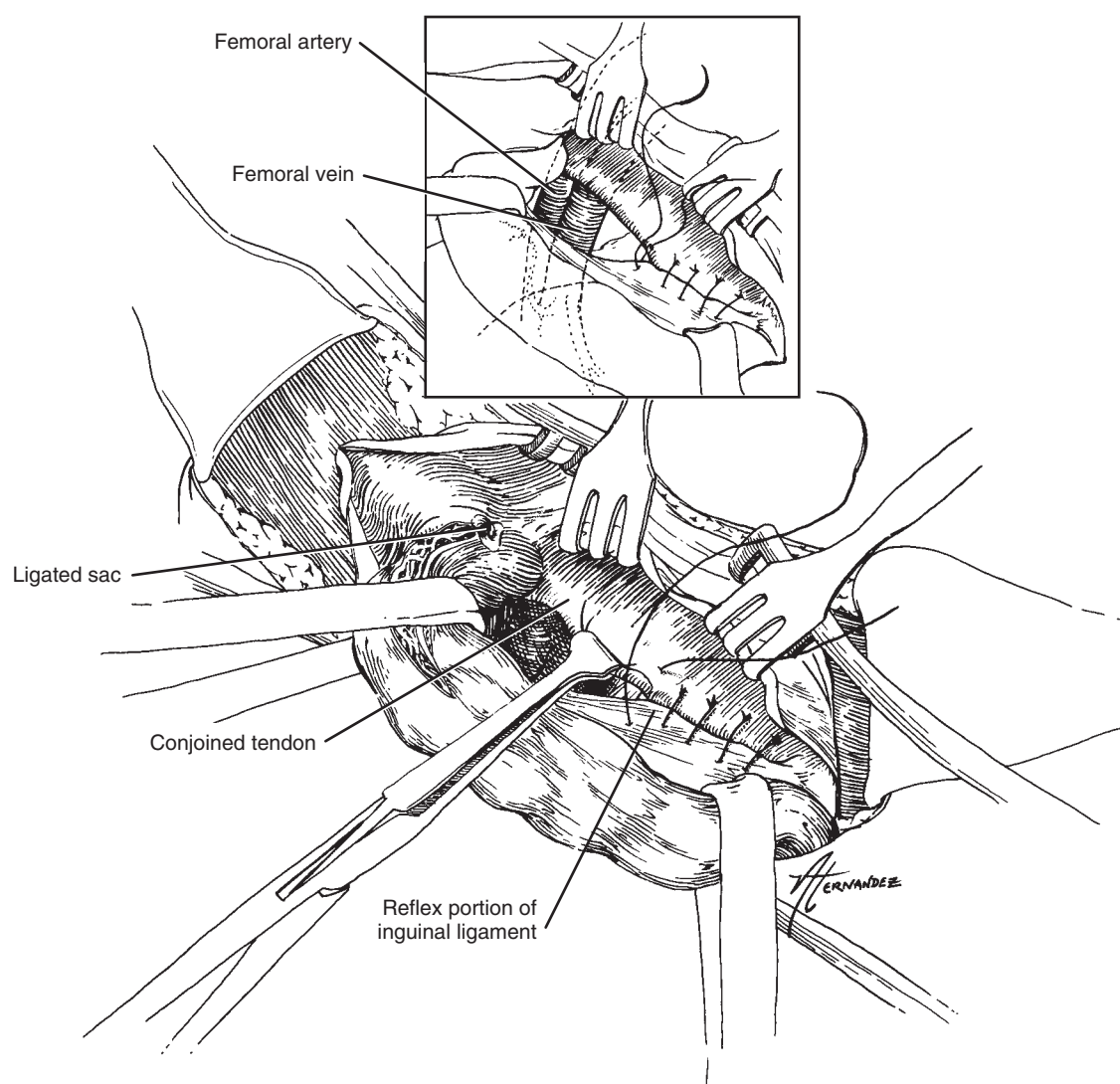


FIGURE 73-7

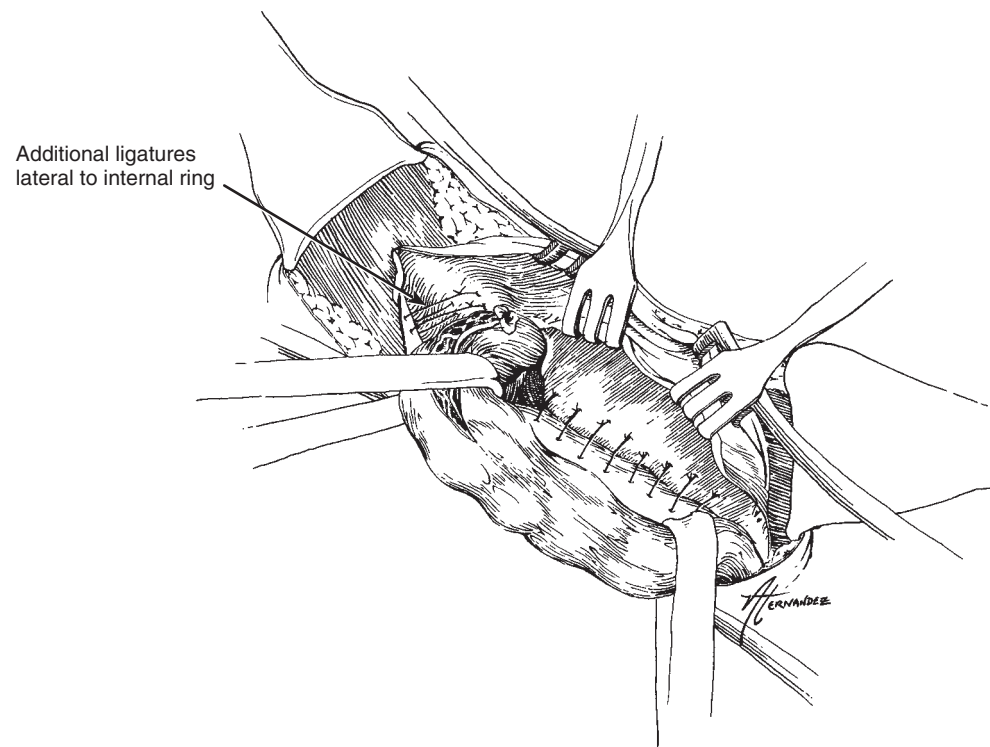


FIGURE 73-8

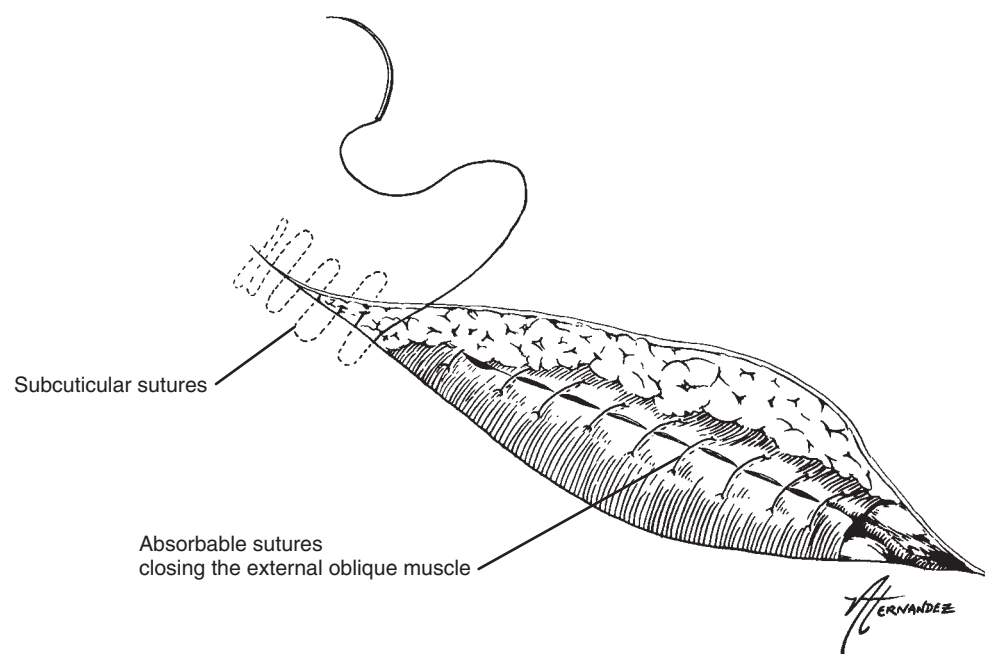


FIGURE 73-9

STEP 4: POSTOPERATIVE CARE

- ◆ The patient should be encouraged to participate in early ambulation and to void before discharge. No specific wound care is required; patients can shower on the first postoperative day. The patient should remain on stool softeners as long as postoperative analgesia requires the use of narcotics. The patient should be instructed to avoid heavy lifting (more than 8 pounds, or a gallon of liquid) for 4 to 6 weeks.

STEP 5: PEARLS AND PITFALLS

- ◆ Careful identification of the conjoint tendon and transversalis fascia are critical before approximation to the reflex portion of the inguinal ligament. Meticulous suture placement at 0.5 to 1.0 cm apart to distribute tension evenly helps ensure a sound repair.

SELECTED REFERENCES

1. Nyhus LM, Condon RE: *Hernia*, 3rd ed. Philadelphia, Lippincott, 1989.
2. Wantz GE, Henselmann C: *Atlas of Hernia Surgery*. New York, Raven Press, 1991.
3. Ponka JL: *Hernias of the Abdominal Wall*. Philadelphia, Saunders, 1980.
4. Wantz GE: The operation of Bassini as described by Attilio Catterina. *Surg Gynecol Obstet* 1989;168: 67-80.

INGUINAL HERNIORRHAPHY (McVAY; COOPER'S LIGAMENT REPAIR)

Dennis C. Gore

STEP 1: SURGICAL ANATOMY

Important landmarks in the inguinal region include the ilioinguinal ligament, which runs from the anterior superior iliac crest to the symphysis pubis. The spermatic cord traverses anteriorly and medially relative to the ilioinguinal ligament, exiting the abdomen at the external ring. The femoral vessels traverse the ilioinguinal ligament posteriorly (**Figure 74-1**).

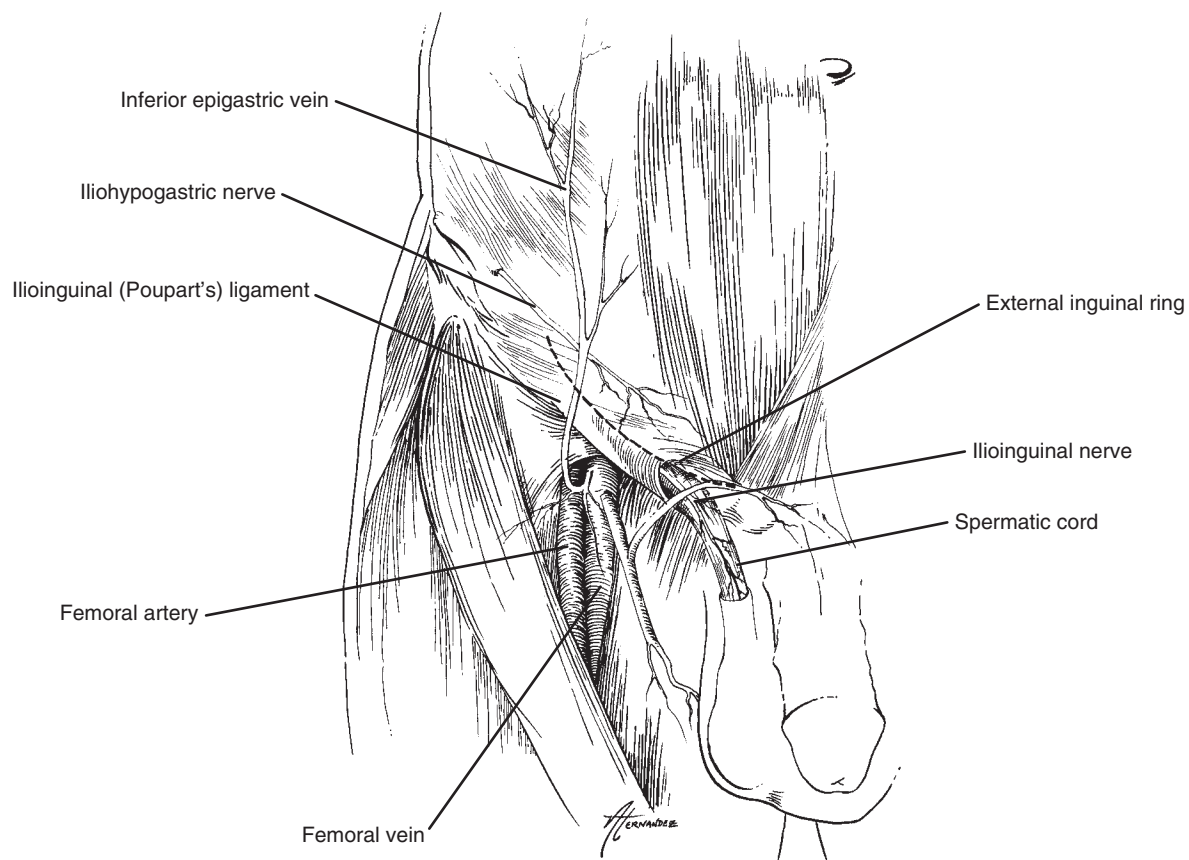


FIGURE 74-1

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ Indications: inguinal hernia
- ◆ Anesthesia: general or spinal
- ◆ Position: supine
- ◆ Before the procedure, exclude and correct as feasible any conditions that may increase intra-abdominal pressure and thereby weaken the herniorrhaphy. For example, correction of prostatic hypertrophy, chronic cough, or constipation may aid in reducing hernia recurrence.

STEP 3: OPERATIVE STEPS**1. INCISION**

- ◆ A skin incision is made 3 cm above and parallel to the ilioinguinal (Poupart's) ligament.

2. DISSECTION

- ◆ Use 3-0 Vicryl ligatures to secure hemostasis on the predominant veins that traverse the subcutaneous tissue along the incision.
- ◆ Use small Richardson retractors to bluntly dissect subcutaneous fat, exposing the external oblique aponeurosis.
- ◆ Manually identify the external ring.
- ◆ Make a small incision in the direction of the external oblique aponeurosis bands, and extend this incision to the external ring using scissors (**Figure 74-2**).
- ◆ Dissect free the ilioinguinal nerve that commonly lies anterior to the spermatic cord. Retract the nerve away from the hernia/spermatic cord (**Figure 74-3**).

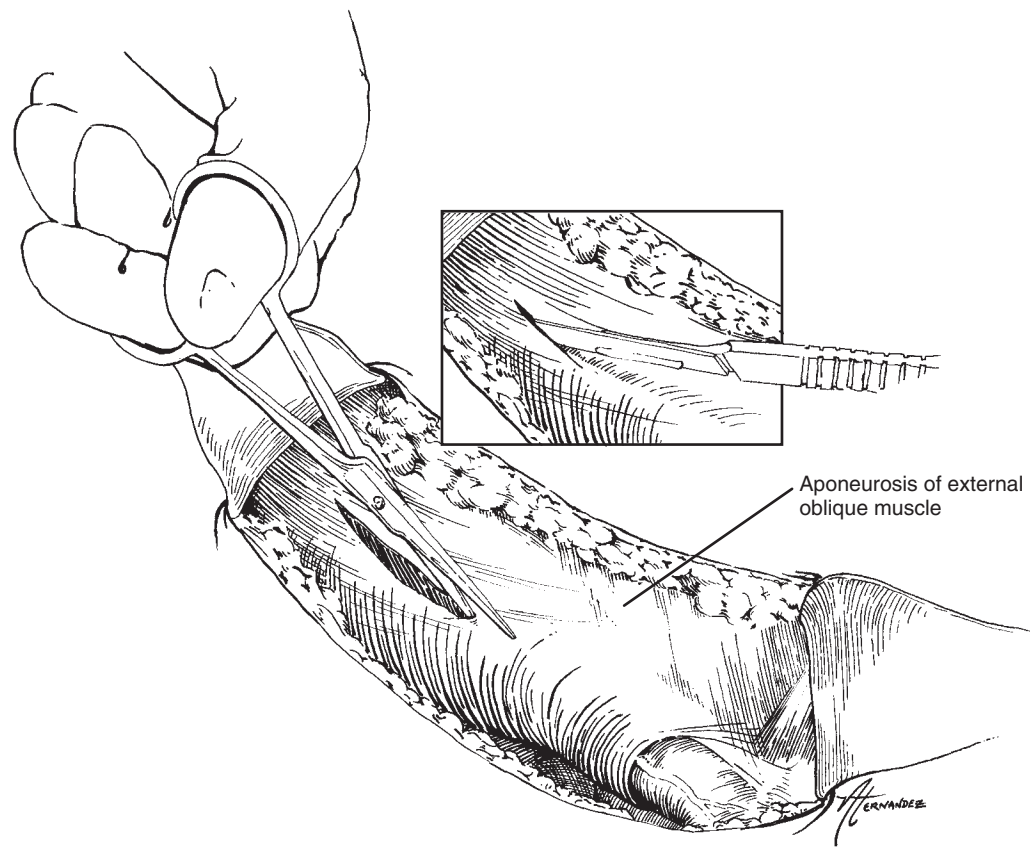


FIGURE 74-2

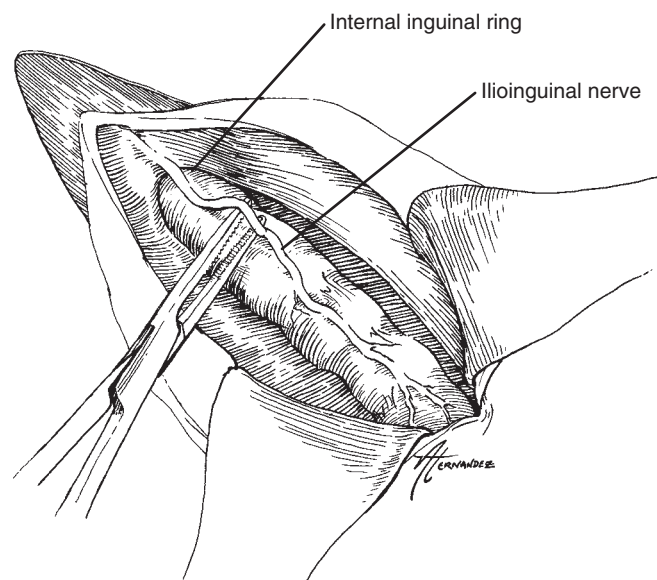


FIGURE 74-3

- ◆ Use manual dissection to free the spermatic cord from the underlying abdominal wall at the level of the prior external ring. Place a Penrose drain around the spermatic cord and retract the cord laterally (**Figure 74-4**).
- ◆ Use sharp dissection with forceps and retraction with hemostats to separate cremaster muscle, vas deferens, and spermatic vasculature away from any indirect hernia sac (**Figure 74-5**).

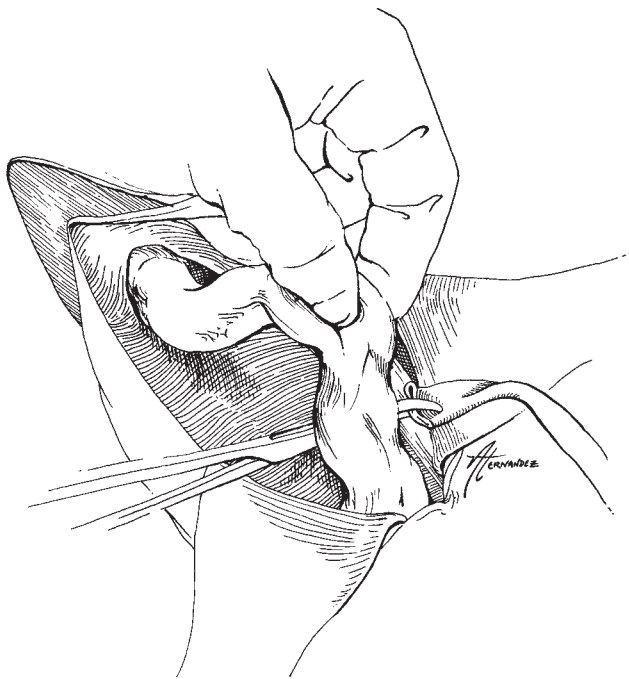


FIGURE 74-4

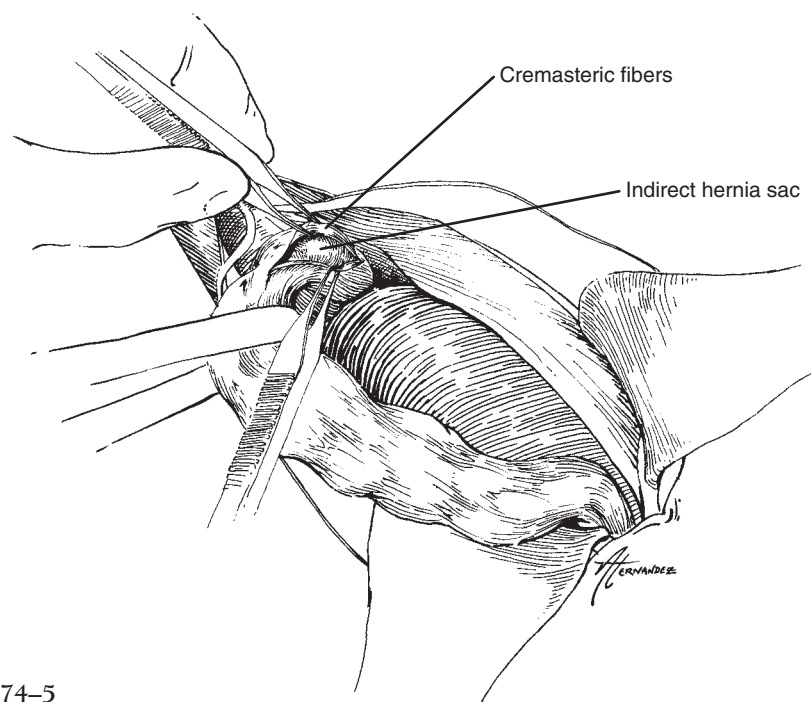


FIGURE 74-5

- ◆ Open the indirect hernia sac and reduce any contents back into the abdomen.
- ◆ Ligate the indirect hernia sac at the internal ring using 2-0 permanent braided suture (Figure 74-6).
- ◆ Incise the anterior surface of the indirect hernia sac distally.

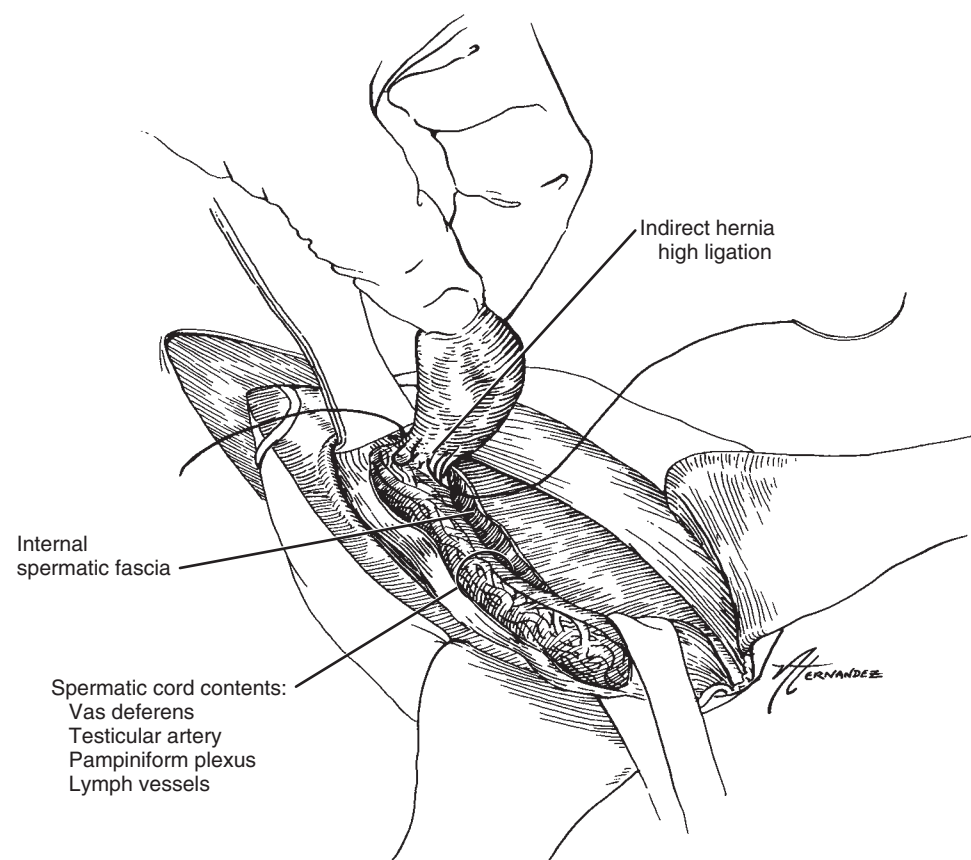


FIGURE 74-6

- ◆ Invert any direct hernia, place several 2-0 permanent braided sutures in a Lembert fashion to hold inversion of the direct hernia sac (**Figure 74-7**).
- ◆ Place two Richardson retractors cupping the external oblique aponeurosis and exposing the junction of the ilioinguinal ligament and symphysis pubis.
- ◆ Manually or with forceps start at this medial portion of the ilioinguinal ligament and dissect along the ligament to expose the underlying Cooper's ligament. Extend exposure laterally to the lacunar ligament.

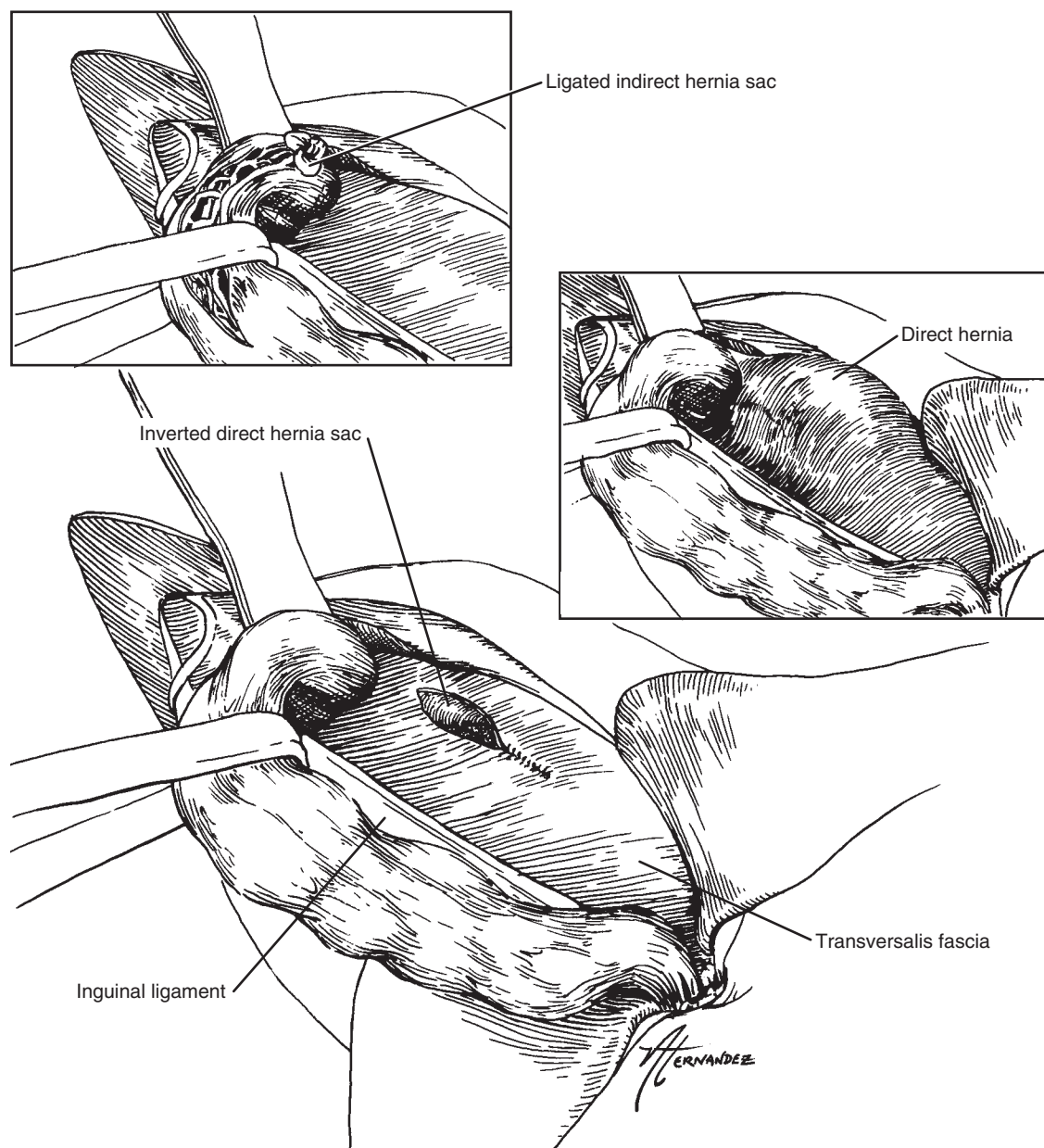


FIGURE 74-7

- ◆ Starting medially, use multiple 2-0 permanent braided sutures to approximate the transversalis fascia to Cooper's ligament (**Figure 74-8**).
- ◆ After 3 to 5 interrupted sutures have secured the transversalis fascia to Cooper's ligament, place additional 2-0 sutures securing the transversalis fascia to the ilioinguinal ligament as this ligament passes anterior to the iliac vessels (**Figure 74-9**).

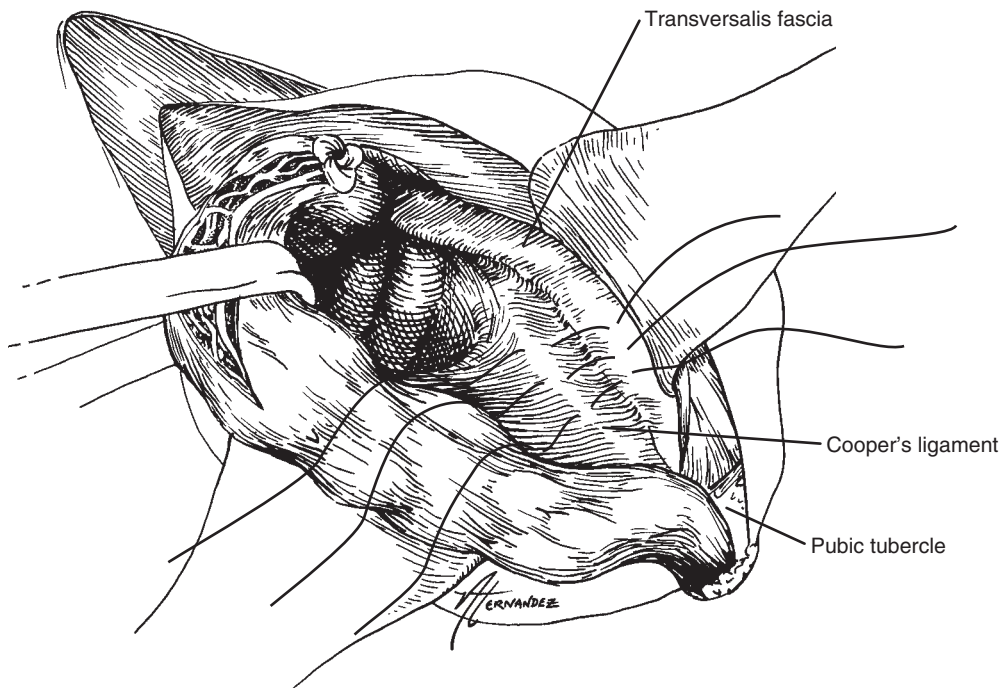


FIGURE 74-8

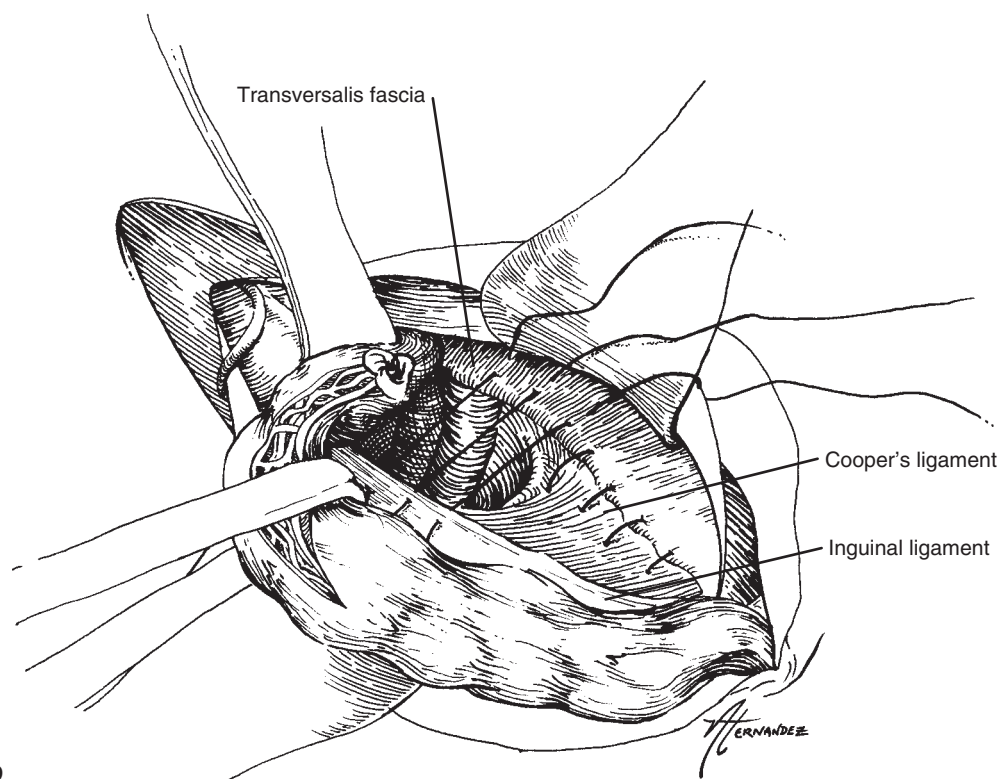


FIGURE 74-9

- ◆ Place multiple small incisions through the fascia overlying the rectus abdominis muscle, thereby relaxing tension on the repair (**Figure 74-10**).
- ◆ Remove the Penrose drain and replace the spermatic cord and ilioinguinal nerve over the pelvic floor.
- ◆ Reapproximate the aponeurosis of the external oblique muscle using a running 3-0 Vicryl suture (**Figure 74-11**).

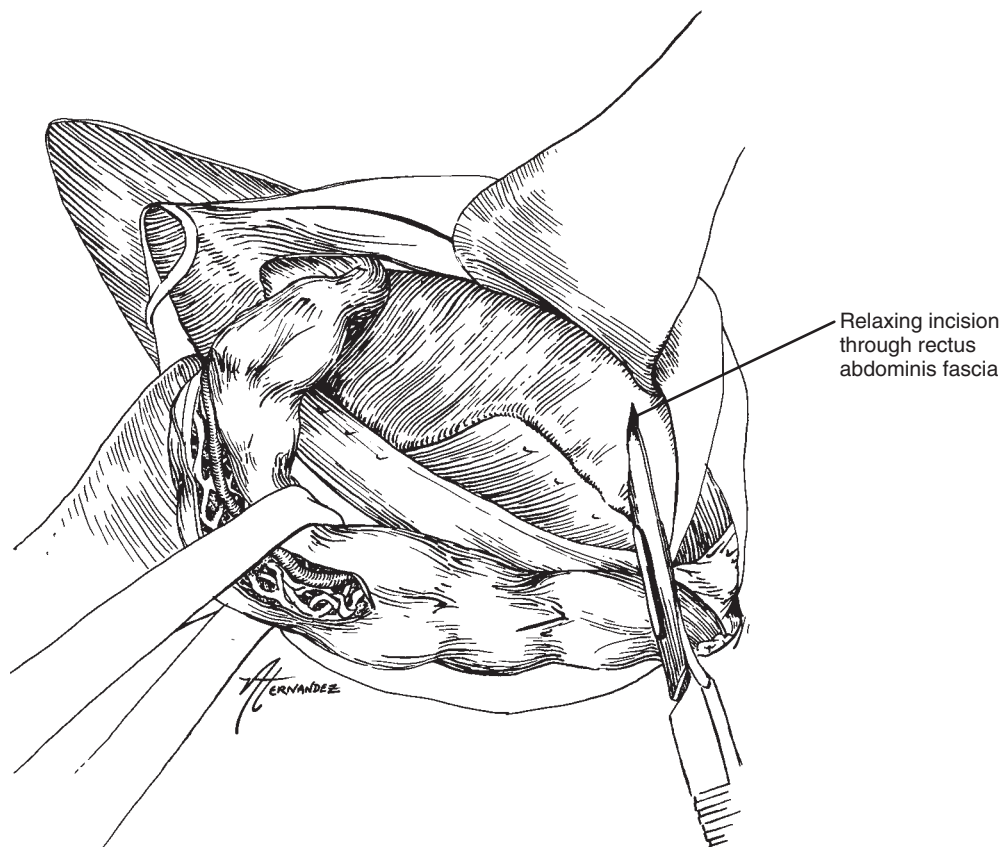


FIGURE 74-10

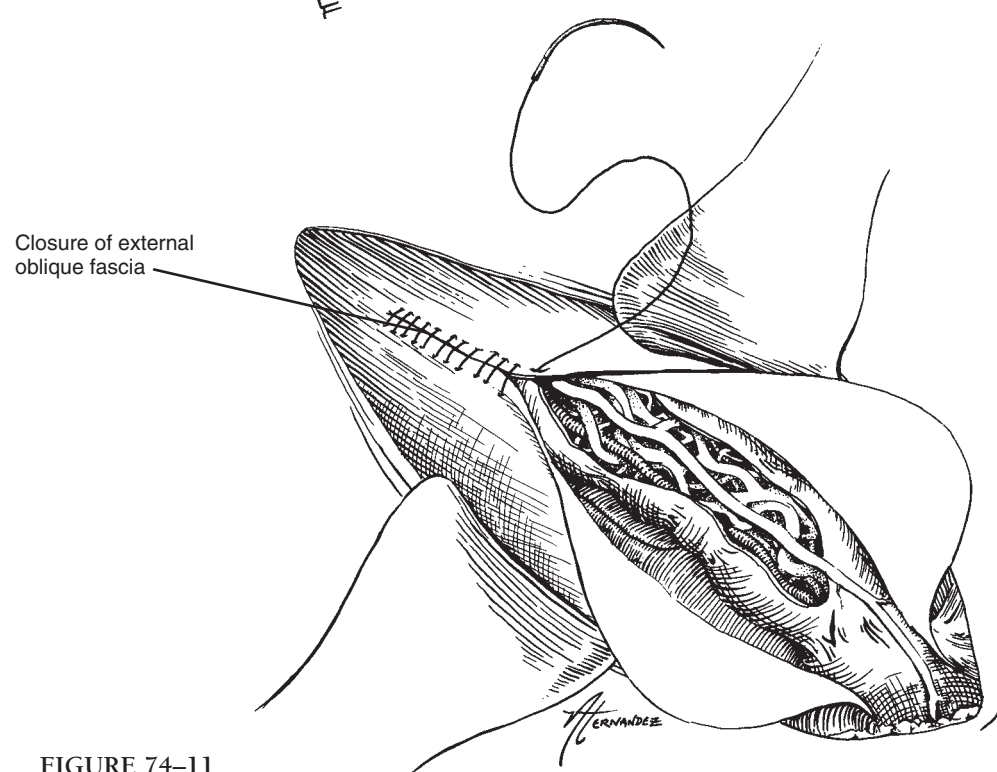


FIGURE 74-11

3. CLOSING

- ◆ Close skin with staples or 3-0 or 4-0 permanent monofilament suture (Prolene).

STEP 4: POSTOPERATIVE CARE

- ◆ Instruct the patient to refrain from heavy lifting or aggressive activity for 3 to 4 weeks.

STEP 5: PEARLS AND PITFALLS

- ◆ For large indirect hernia sacs, simply incising along the anterior surface precludes the need for removal of the indirect hernia sac, thereby greatly reducing the incidence of subsequent hydrocele formation yet minimizing any bleeding associated with complete removal of the sac.
- ◆ Place sutures at unequal depth and distance when approximating the transversalis fascia or external oblique aponeurosis to reduce any sheer effect and thereby strengthen the closure.

SELECTED REFERENCE

1. Zollinger RM Jr, Zollinger RM: Atlas of Surgical Operations, 5th ed. New York, Macmillan, 1983, p 402.



INGUINAL HERNIORRHAPHY— SHOULDICE

Michael D. Trahan

STEP 1: SURGICAL ANATOMY

- ◆ A thorough knowledge and understanding of the anatomy of the inguinal canal and pre-peritoneal space is critical to success in the tissue repair of the inguinal hernia.
- ◆ The site of incision and surface landmarks important to consider are demonstrated in **Figure 75-1**.

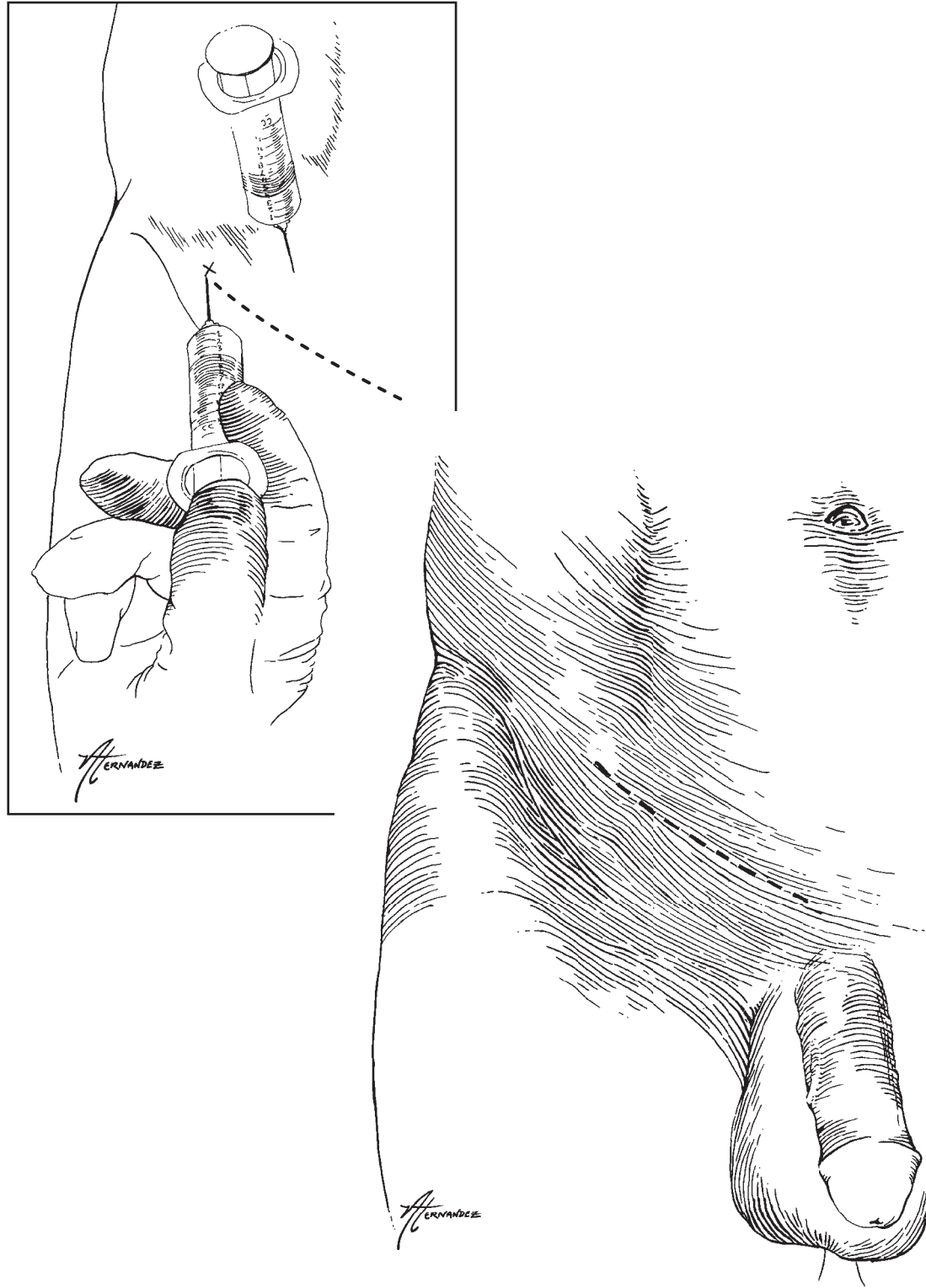


FIGURE 75-1

- ◆ The important structures of the male inguinal canal are illustrated in **Figure 75-2**.
- ◆ It is important to note that these structures may be significantly distorted by large or long-standing hernias, or both.

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ Repair of an inguinal hernia should be considered for the patient with symptoms from the hernia that interfere with daily activities and for those hernias at risk for incarceration or strangulation.
- ◆ An effort should be made to diagnose and treat conditions that result in Valsalva, such as constipation, urinary straining, and chronic cough, before hernia repair.
- ◆ The Shouldice hernia repair has the lowest recurrence risk of all of the tissue repairs.

ANESTHESIA

- ◆ The choice of anesthesia should be tailored to the individual patient after evaluation by the anesthesiologist.
- ◆ This repair can be performed with general, regional (spinal), or local anesthesia.
- ◆ Five to 10 mL of a long-acting local anesthetic, such as bupivacaine, should be injected just medial to the anterior superior iliac spine and deep to the external oblique muscle.
- ◆ Ten milliliters of the local anesthetic is injected subcutaneously at the site of the incision.

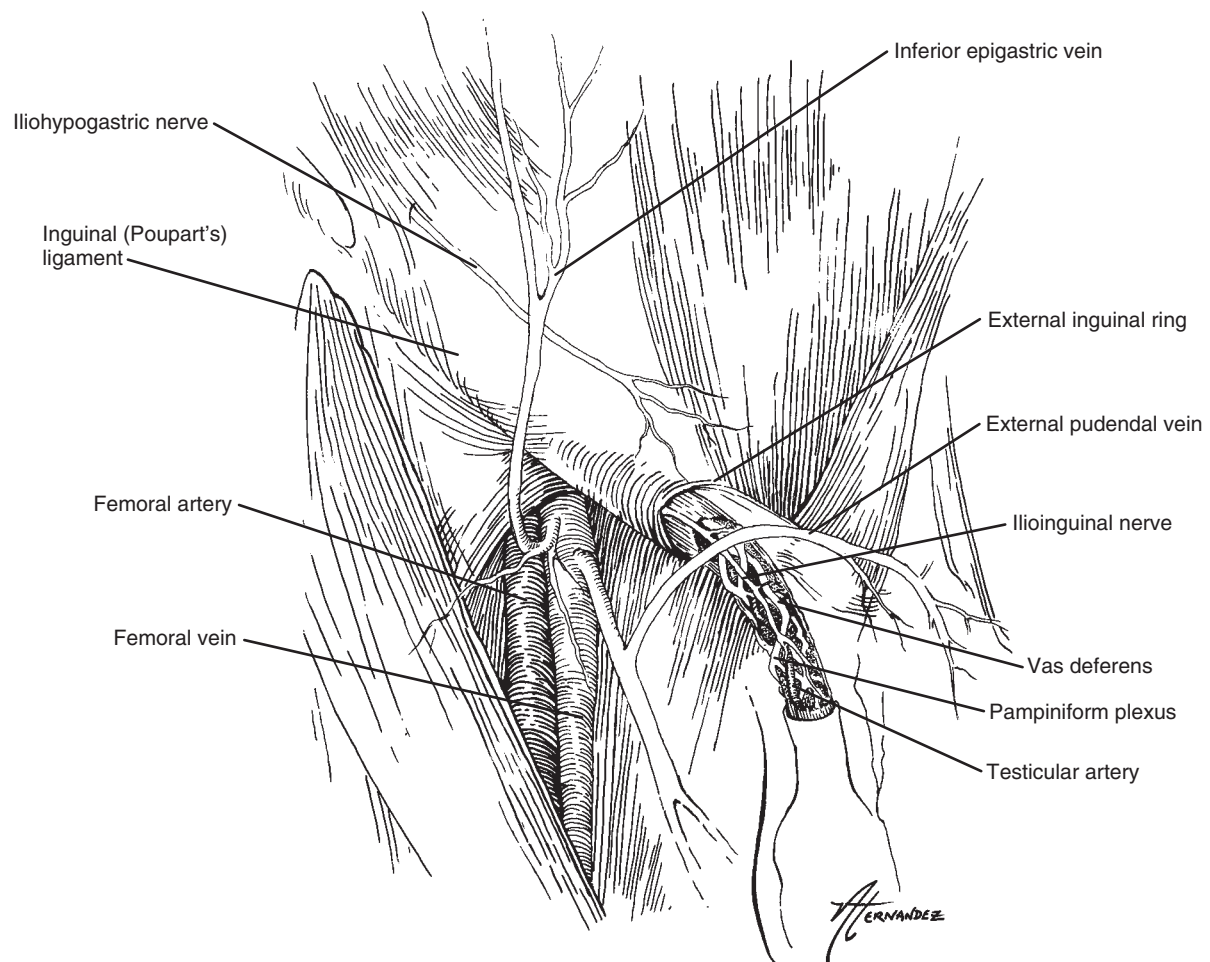


FIGURE 75-2

STEP 3: OPERATIVE STEPS

1. INCISION

- ◆ Shaving should be avoided. If hair removal is necessary, it should be removed with an electric clipper.
- ◆ A linear incision is made over the external inguinal ring parallel to the course of the inguinal ligament.
- ◆ The subcutaneous fat and Scarpa's fascia are divided sharply to expose the external abdominal oblique aponeurosis and spermatic cord (**Figure 75-3**).

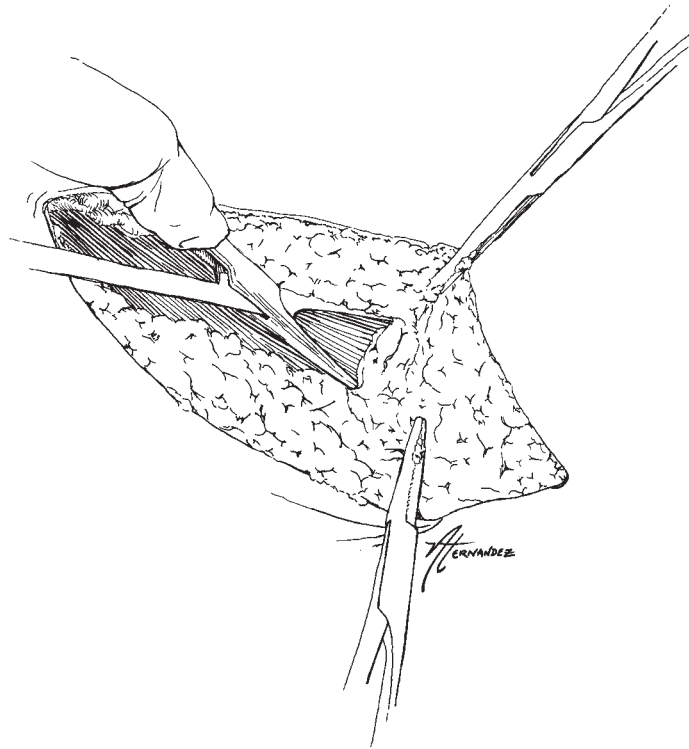


FIGURE 75-3

2. DISSECTION

- ◆ The external oblique aponeurosis is incised obliquely along the lines of its fibers, down through the external inguinal ring (Figure 75-4).
- ◆ The plane between the external and internal oblique muscle layers should be sharply dissected to the rectus sheath medially and the shelving edge of the inguinal ligament inferior laterally (Figure 75-5).

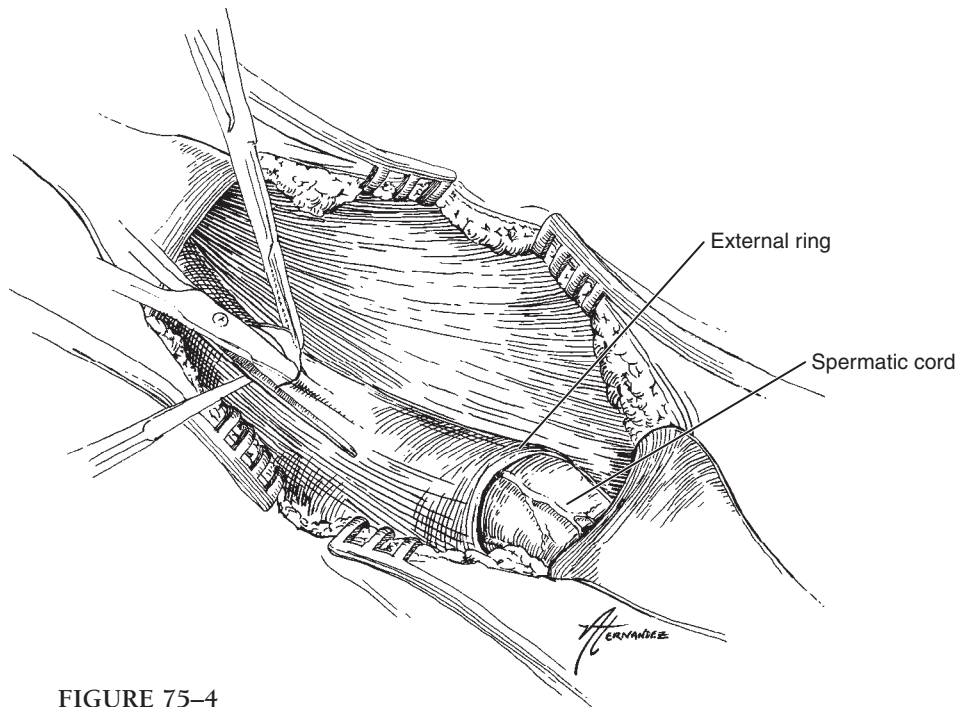


FIGURE 75-4

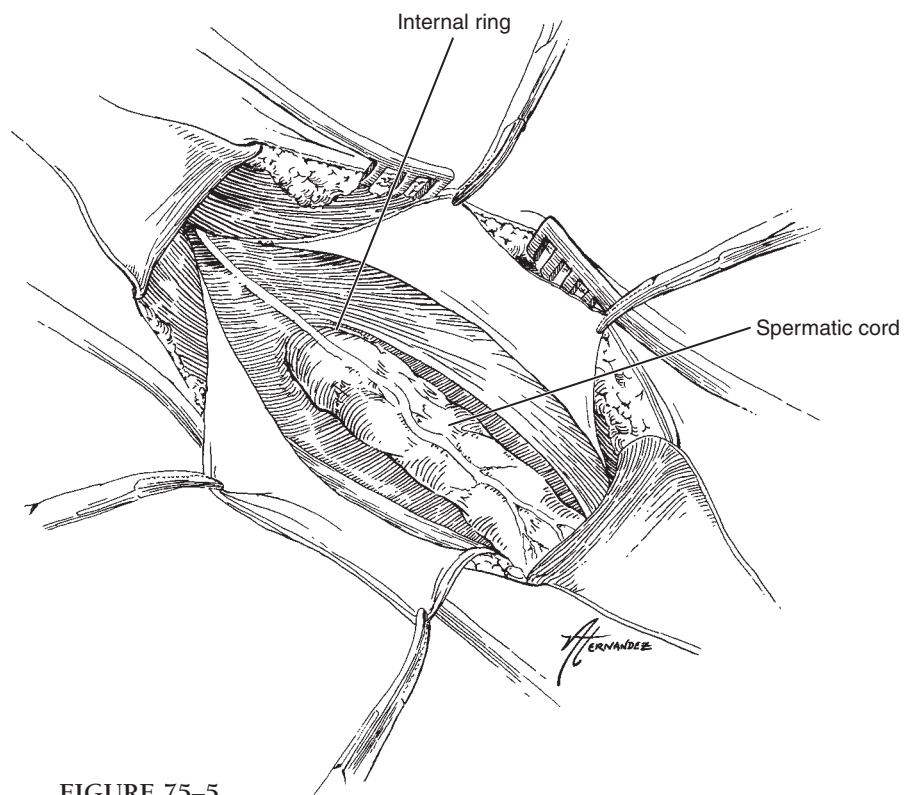


FIGURE 75-5

- ◆ The spermatic cord is mobilized from the floor of the inguinal canal and encircled with a Penrose drain (Figure 75-6).
- ◆ The cremaster muscle is incised longitudinally to reveal the hernia sac and cord structures (Figure 75-7).

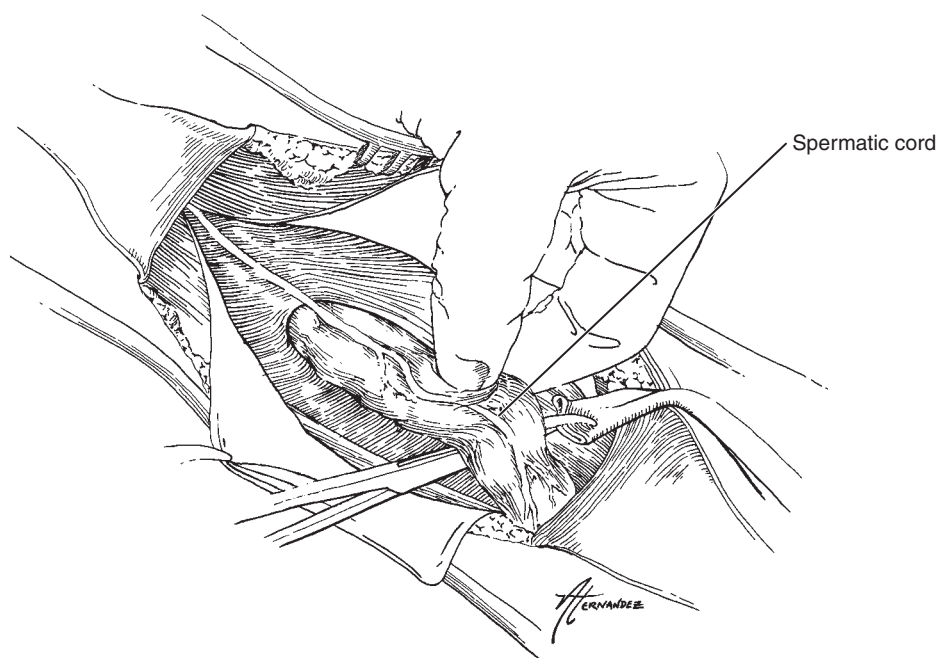


FIGURE 75-6

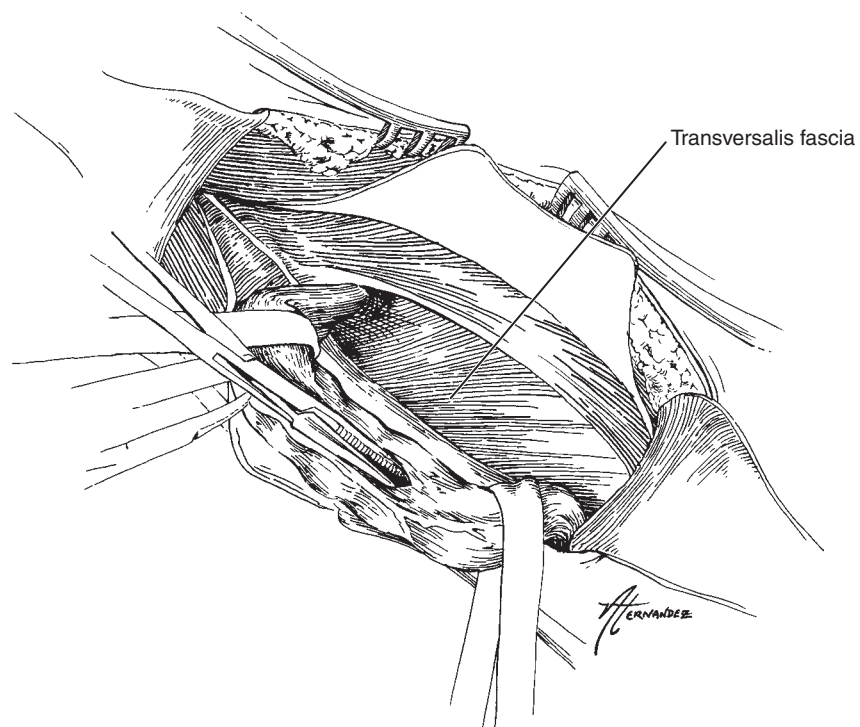


FIGURE 75-7

- ◆ Dissection of the indirect sac and any cord lipomas from the cord structures should be complete up through the internal inguinal ring. Large sacs may be excised, whereas smaller sacs may be reduced through the ring (**Figure 75-8**).
- ◆ The floor of the inguinal canal (transversalis fascia) is incised beginning near the pubic tubercle and proceeding laterally, with care taken to avoid the epigastric vessels as the internal ring is approached. The preperitoneal fat should protrude through the incision to ensure adequate opening of this layer (**Figure 75-9**).

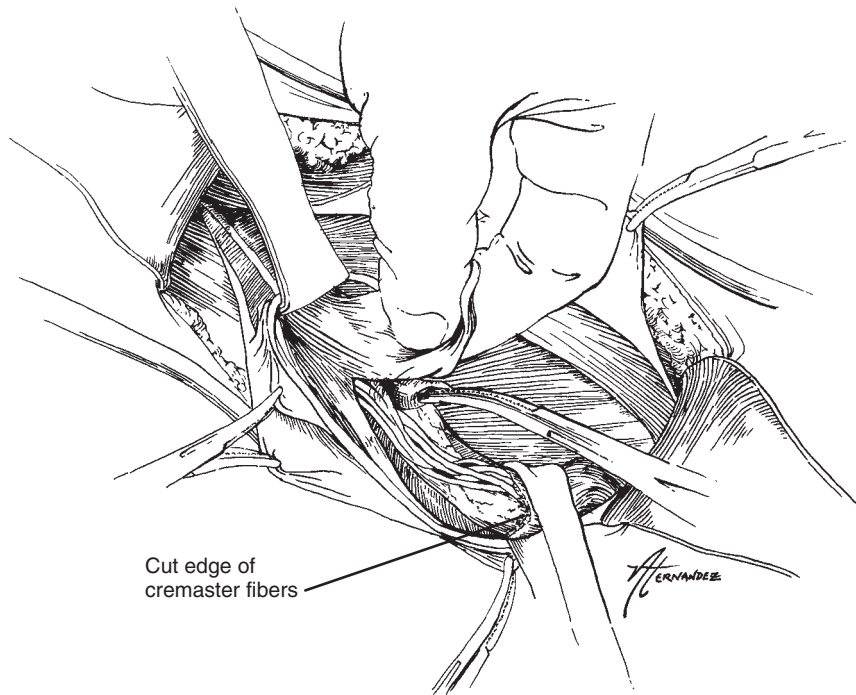


FIGURE 75-8

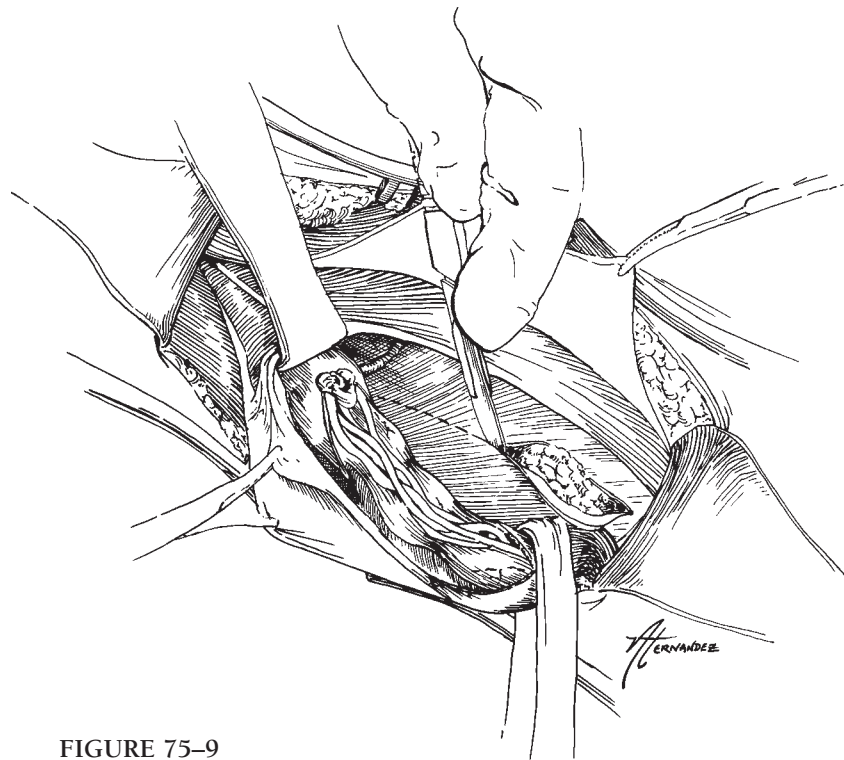


FIGURE 75-9

- ◆ The excess transversalis fascia of a large direct hernia may need to be trimmed.
- ◆ The preperitoneal fat should be swept off of the transversalis fascia to allow adequate mobilization of these flaps to complete the operation.
- ◆ Two nonabsorbable sutures such as 2-0 polypropylene are then used to perform the running four-layer closure.
- ◆ The first suture is started with a healthy bite of the pubic tubercle securing the lower layer of the transversalis fascia to the undersurface of the upper flap incorporating a bite of the rectus sheath. Small, closely spaced bites are used to close this layer progressing to the internal ring (**Figure 75-10**).
- ◆ Without tying this suture, the second layer is started by approximating the upper flap of the transversalis to the shelving edge of the inguinal ligament, with care taken to not strangulate the cord. The second layer of the first suture ends at the pubic bone, where the suture is tied (**Figure 75-11**).

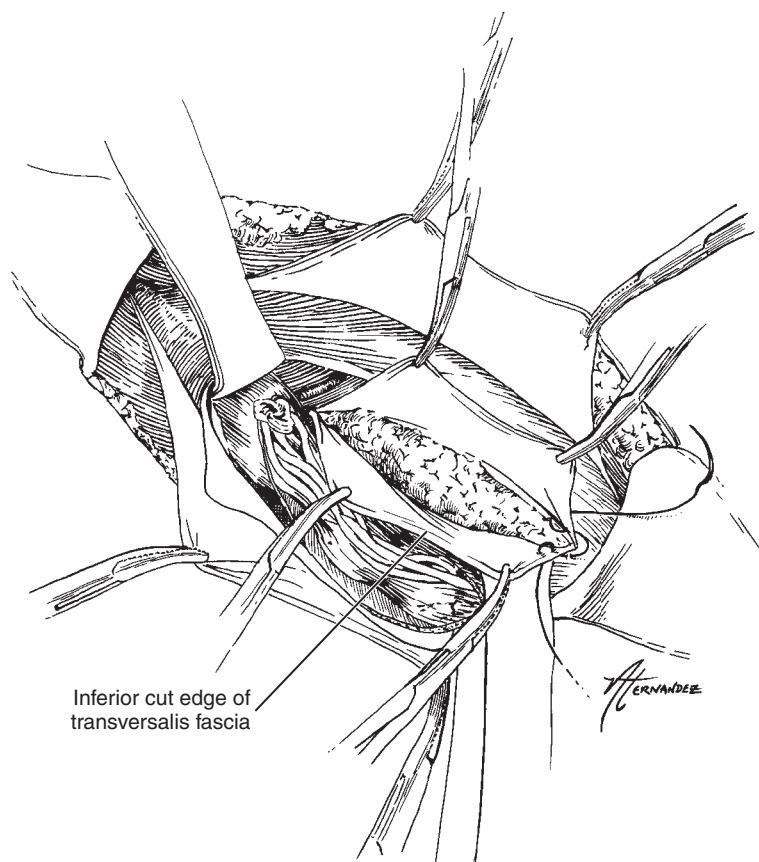


FIGURE 75-10

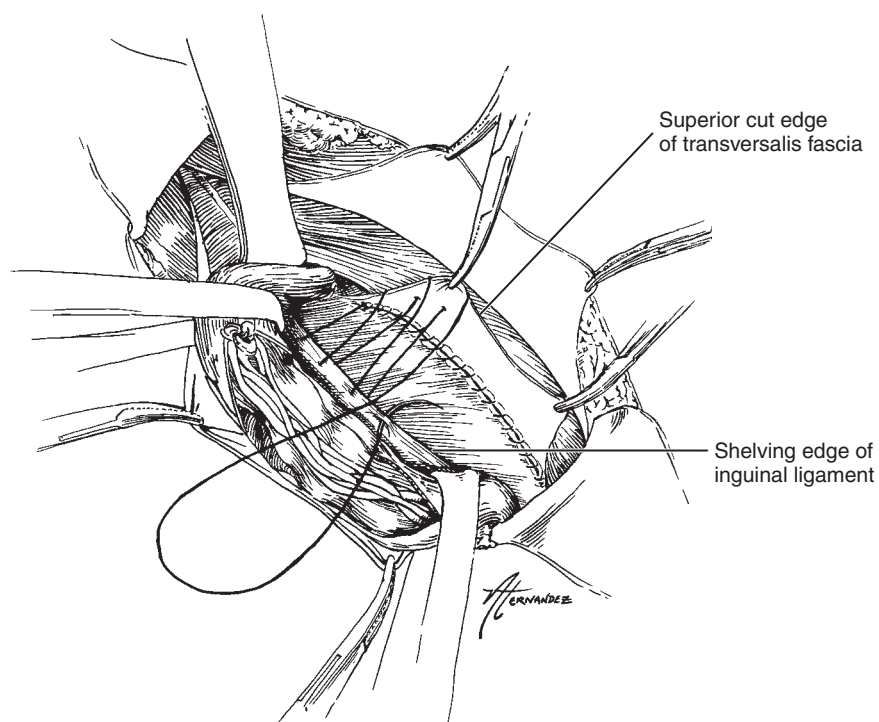


FIGURE 75-11

- ◆ The second polypropylene suture is started laterally at the internal ring, and this third layer is used to oversee the second layer reinforcing the transversalis/internal oblique to inguinal ligament approximation (**Figure 75-12**).
- ◆ After reaching the pubic tubercle again, in the final layer, the internal oblique muscle is sewn to the inguinal ligament. The suture is then tied at the inguinal ring (**Figure 75-13**).

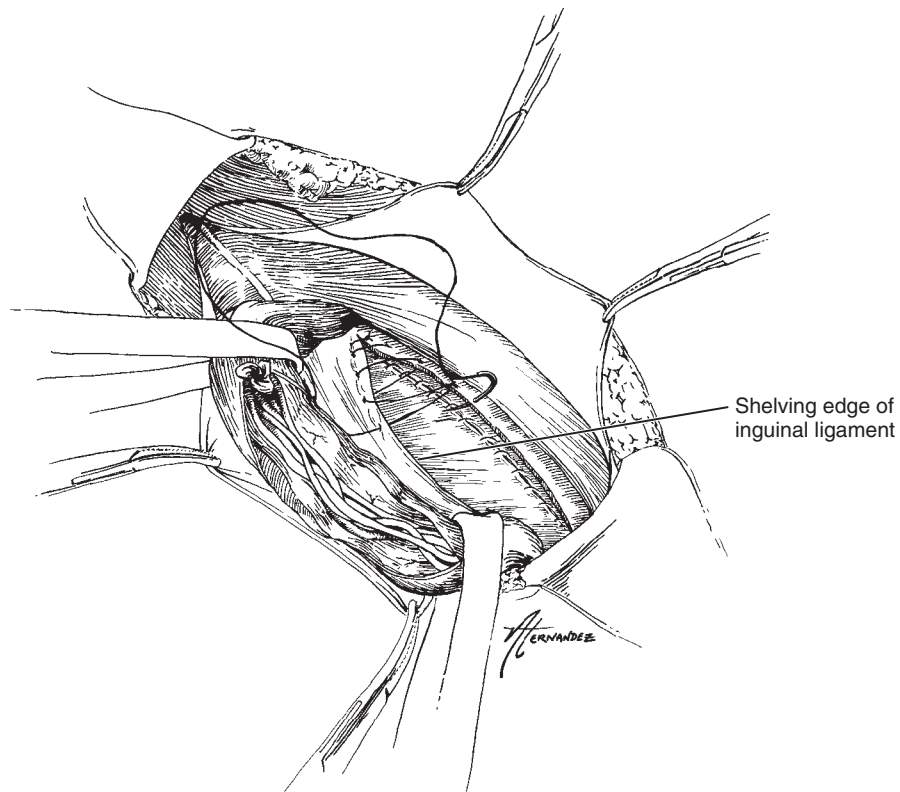


FIGURE 75-12

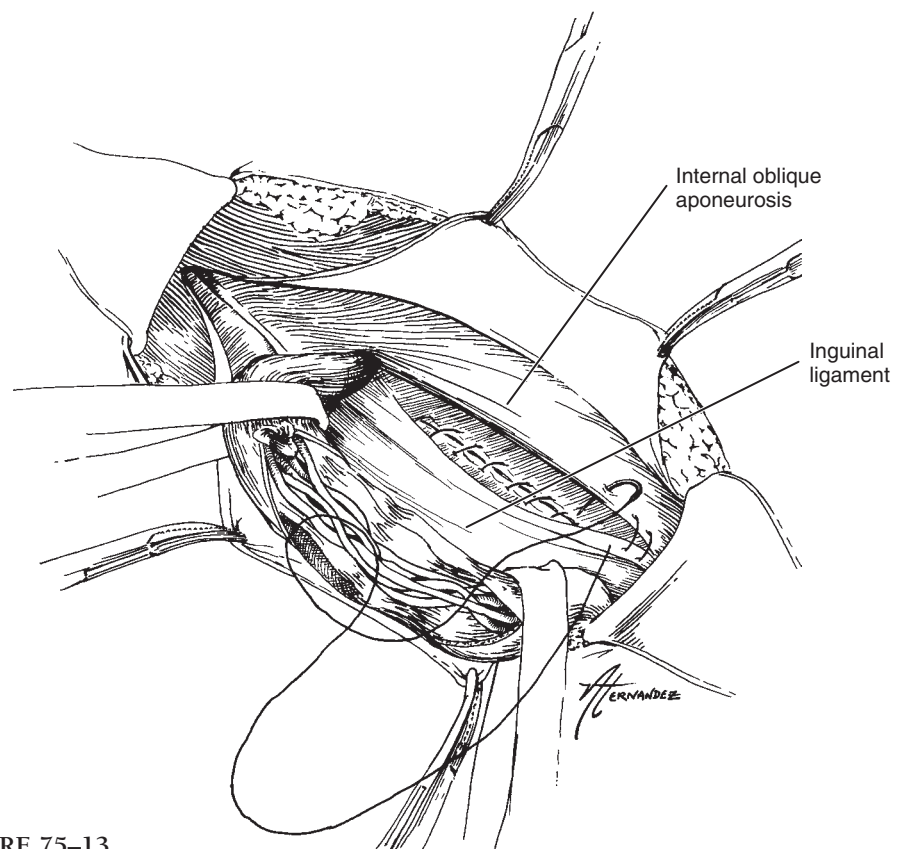


FIGURE 75-13

3. CLOSING

- ◆ The spermatic cord is returned to its anatomic position, and the external oblique is closed with running 3-0 absorbable suture, such as Vicryl (**Figure 75-14**).
- ◆ Scarpa's fascia is reapproximated with 3-0 Vicryl.
- ◆ Standard skin closure is the surgeon's choice of running subcuticular absorbable suture, such as 4-0 Monocryl or interrupted permanent suture, such as staples or 2-0 polypropylene (**Figure 75-15**).

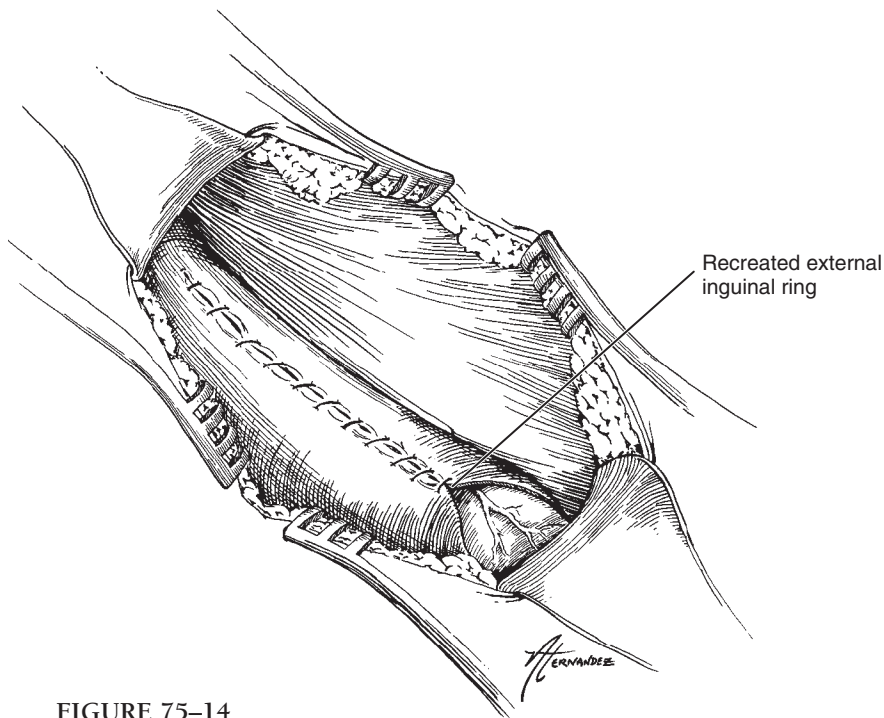


FIGURE 75-14

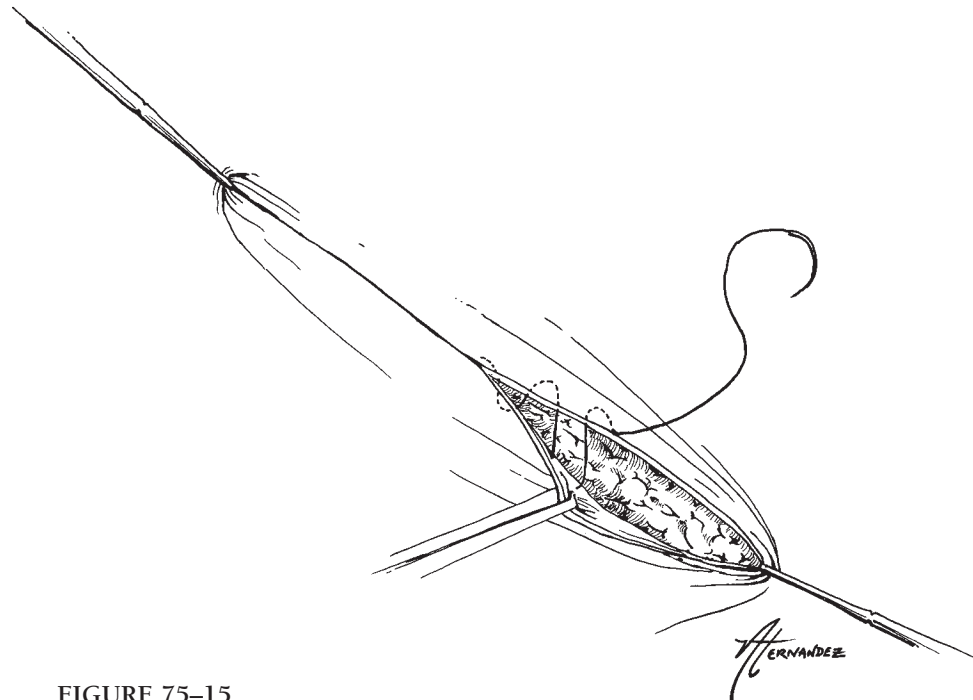


FIGURE 75-15

STEP 4: POSTOPERATIVE CARE

- ◆ Ice pack, oral hydrocodone or propoxyphene, and stool softeners are the standard post-operative orders.
- ◆ Activity and driving should be restricted until the patient is comfortable and no longer in need of pain medication.
- ◆ Severe pain and swelling of the testicle should be immediately evaluated for the possibility of ischemia.

STEP 5: PEARLS AND PITFALLS

- ◆ Hemostasis during sac dissection should be meticulous to avoid injury to the pampiniform plexus and the development of wound hematomas.
- ◆ Care must be taken to avoid crushing, burning, or entrapment of the cutaneous nerves in the inguinal canal to decrease the possibility of postoperative chronic pain syndromes.
- ◆ Patients should be thoroughly educated about the postoperative disability after a tension-producing repair of inguinal hernia.

SELECTED REFERENCES

1. Hay JM, Boudet MJ, Fingerhut A, et al: Shouldice inguinal hernia repair in the male adult: The gold standard? A multicenter controlled trial in 1578 patients. *Ann Surg* 1995;222:719-727.
2. Shouldice EE: The treatment of hernia. *Ont Med Rev* 1953;20:670-684.
3. Welsh DR, Alexander MA: The Shouldice repair. *Surg Clin North Am* 1993;73:451-469.



SLIDING INGUINAL HERNIA

Thomas D. Kimbrough

STEP 1: SURGICAL ANATOMY

- ◆ A sliding inguinal hernia is defined as one in which a viscus or its attendant mesentery constitute a part of the wall of the hernia sac.
- ◆ Although sliding hernias have been reported in all three types of groin hernia, they are most common in the indirect location.
- ◆ Again, although a wide variety of organs have been reported, including female adnexa, appendix, ileum, ureter, and bladder, the sliding component most commonly is the cecum on the right side of the body and the sigmoid colon on the left.
- ◆ The part of the wall of the hernia sac involved is usually the posterior lateral region.

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ Sliding hernias are rarely identified preoperatively, and in fact there is little reason to worry about doing so.
- ◆ Preoperative preparation should follow guidelines outlined in earlier chapters.

STEP 3: OPERATIVE STEPS

1. INCISION

- ◆ The standard incisions described in Chapter 71 are sufficient.

2. DISSECTION

- ◆ The most important step in dealing with a sliding hernia is its recognition. Failure to do so can result in unnecessary dissection leading to injury to the involved organ or its blood supply.
- ◆ It is important to differentiate between adhesions from abdominal organs to the inner surface of the hernia sac and true sliding hernias. Adhesions can be carefully lysed and the freed organ reduced back into the abdominal cavity.
- ◆ A right-sided sliding hernia is illustrated in **Figure 76-1**. Note the cecum forming part of the posterior lateral wall of the opened hernia sac.

3. CLOSURE

- ◆ Once any adhesions are taken down, the peritoneum is sewn shut, closing the hernia sac as shown in **Figure 76-2**. Care should be taken to avoid catching any part of the colon and its mesentery in the closure.
- ◆ Because a sliding hernia is a modification of an indirect hernia, once the sac is closed, it and the attached colon can be reduced back into the preperitoneal space in the fashion described in Chapter 71.
- ◆ Similarly, repair can then proceed by one of the techniques appropriate for an indirect hernia.

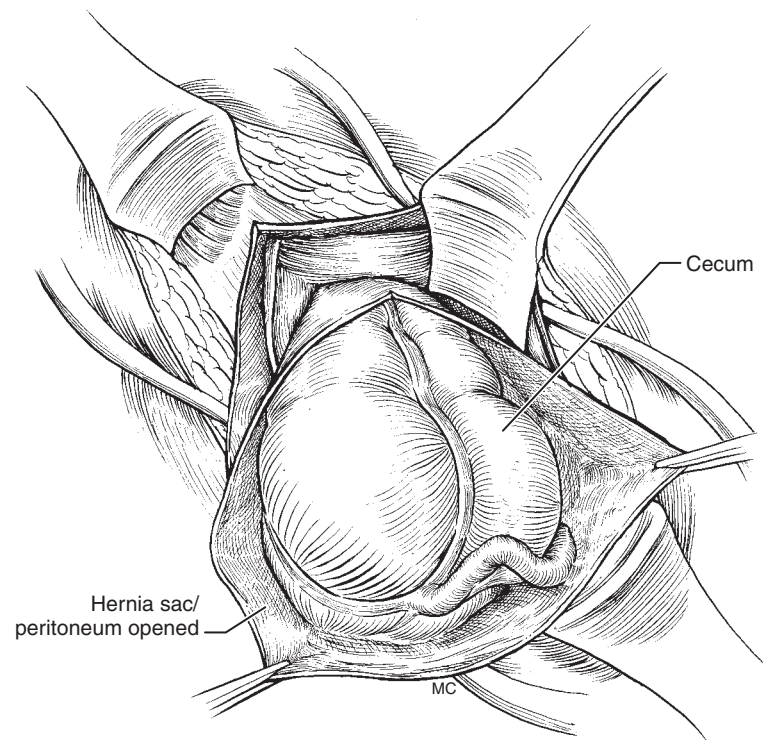


FIGURE 76-1

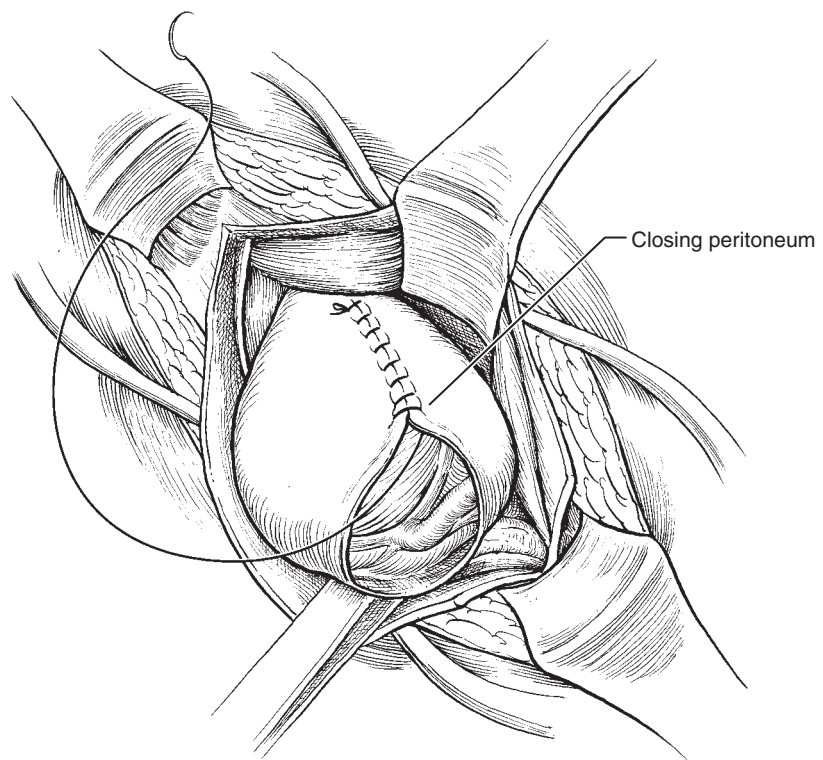


FIGURE 76-2

STEP 4: POSTOPERATIVE CARE

- ◆ There are no special considerations or steps necessary in postoperative care different from those mentioned in Chapter 71.

STEP 5: PEARLS AND PITFALLS

- ◆ Mentioned earlier but worth repeating, prompt recognition of the presence of a sliding hernia and avoidance of unnecessary and potentially damaging dissection is desirable.
- ◆ Because most of these are variants on the standard indirect hernia, any repair appropriate for a large version of that hernia will suffice here.

SELECTED REFERENCE

1. Nyhus LM, Condon RE (eds): Hernia, 4th ed. Philadelphia, JB Lippincott, 1995.

INGUINAL HERNIAS IN INFANTS AND SMALL CHILDREN

Carlos A. Angel

INTRODUCTION

- ◆ The incidence of indirect inguinal hernias (which comprise 99% of the hernias found in small children) ranges from 1% to 5% of the population, with a male-to-female ratio of 8:1 to 10:1. Premature infants are at greater risk for developing inguinal hernias, with reported incidences ranging from 7% to 30% for boys and 2% for girls. The risk of incarceration is inversely proportional to the age of the patient and may exceed 60% in the first 6 months of life. Most neonatologists and pediatric surgeons recommend repair of inguinal hernias in premature babies before discharge from the hospital. The incidence of bilateral inguinal hernias in children and routine contralateral groin exploration at the time of repair are controversial topics. The possibility that bilateral inguinal hernias will be present at operation is greater in younger patients, but the risk of bilaterality subsequently decreases to 41% for children 2 to 16 years of age. Incidence of bilateral inguinal hernias seems to be greater in female patients in all age groups, with reported values ranging from 20% to 50%. Patients with ventriculoperitoneal (VP) shunts, peritoneal dialysis catheters, connective tissue disorders such as Ehlers-Danlos syndrome, and cystic fibrosis have a high enough incidence of bilaterality to justify routine contralateral exploration. Laparoscopic exploration of the contralateral inguinal ring by inserting a small 70-degree scope (or 120-degree, if available) through the hernia sac is a recent approach that is helpful in avoiding unnecessary and potentially morbid contralateral groin explorations. I continue to perform routine contralateral explorations in all premature infants with an inguinal hernia.

STEP 1: SURGICAL ANATOMY

- ◆ The processus vaginalis, which is a peritoneal diverticulum that extends in utero through the internal inguinal ring, is dragged along with testicular descent into the scrotum, where the portion surrounding the testicle will become the tunica vaginalis while the rest of the processus obliterates before the child's birth. Persistence of a patent processus vaginalis may lead to indirect inguinal hernias, hydroceles of the cord, or communicating hydroceles. Most inguinal hernias in children (99%) are indirect; that is, the sac originates lateral to the inferior epigastric vessels (although it may extend past them) and is close (on the anteromedial side) to the spermatic vessels and the vas deferens. All cord structures are enveloped by the deep spermatic fascia, which is very thin and translucent, and more superficially by the cremaster muscle, which originates from the internal oblique muscle.

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ In most children, laboratory examinations or antibiotics are not indicated before hernia repair. The time at which ingesting clear fluids is stopped depends on the age of the patient and ranges from 3 to 6 hours before the procedure. Most anesthesiologists will administer an anxiolytic agent such as midazolam in the preoperative area to reduce separation anxiety and the fear that arises from an unfamiliar environment. Anesthetic technique varies depending on the level of comfort and experience of the anesthesiologist. In most patients, general endotracheal anesthesia will be preferred. In selected cases, such as very small infants with chronic pulmonary disease, in which endotracheal intubation can result in prolonged mechanical ventilatory support in the postoperative period, the procedure can be safely performed with the infant under a regional anesthetic such as a caudal block. Preoperative planning should include 23-hour postoperative observation and monitoring in all patients who were born prematurely within the previous 4 to 6 months.

STEP 3: OPERATIVE STEPS**1. INCISION**

- ◆ The patient is placed on the operating table in the supine position. After anesthesia is induced, the lower abdomen, both groins, penis, and scrotum are prepped with a topical antiseptic solution and draped in a sterile fashion. A small (3- to 4- cm) incision is made along the lower abdominal crease (**Figure 77-1**). The subcutaneous fat is grasped between mosquito hemostats and divided with electrocautery at very low settings. Two crossing veins can be either pushed off the midline, cauterized, or tied. Division of the subcutaneous fat stops when Scarpa's fascia becomes evident. Scarpa's fascia has a pearly, shiny appearance. Scarpa's fascia is opened with scissors, and blunt dissection is used, starting on the lateral aspect of the incision and working medially to expose the external oblique aponeurosis and the external inguinal ring. The external oblique aponeurosis is opened sharply, and this incision is extended until opening the external inguinal ring (**Figure 77-2**). Care must be taken to avoid injury of the ilioinguinal nerve. The spermatic cord is grasped gently, avoiding any manipulation of the vas deferens. At this stage of the operation, bringing the testicle into the operative field should be avoided.

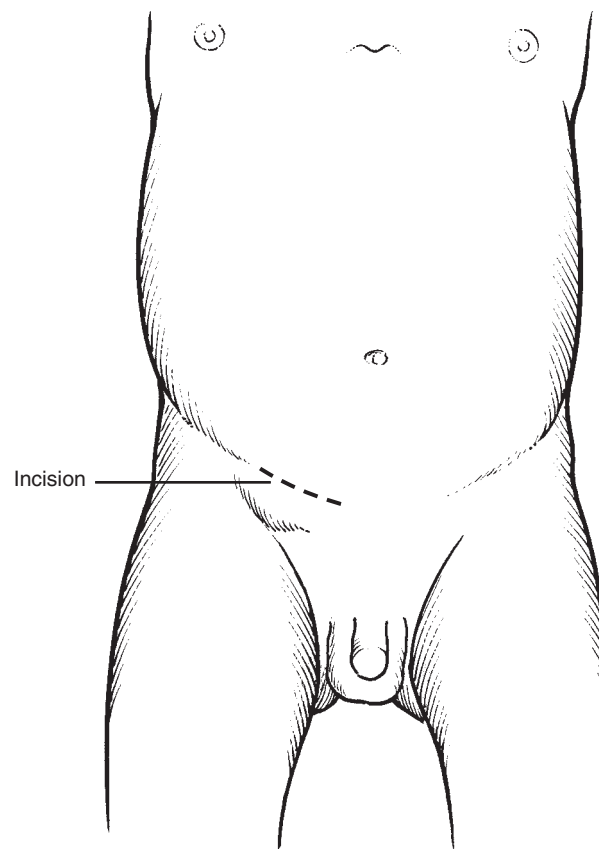


FIGURE 77-1

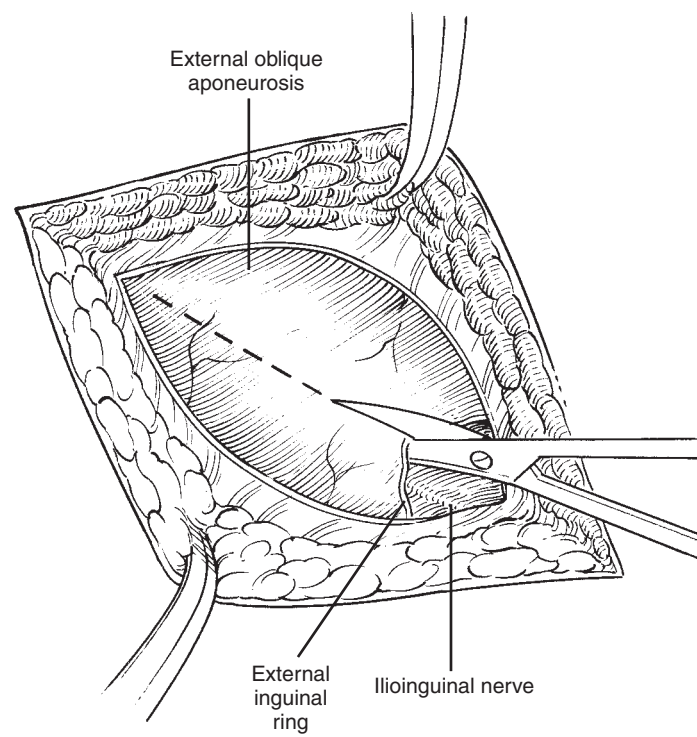


FIGURE 77-2

2. DISSECTION

- ◆ The cremasteric fibers that surround the spermatic cord are bluntly separated. Be aware that use of electrocautery in the vicinity of the spermatic vessels or the vas deferens is very hazardous, because transmitted heat or electrical current may damage these structures and may even result in testicular loss. The hernia sac will be found on the anteromedial aspect of the spermatic cord (**Figure 77-3**). Gentle blunt dissection is used to separate the hernia sac from the spermatic vessels and the vas deferens, avoiding direct manipulation of the latter (**Figure 77-4**). These structures must be positively identified before proceeding with the rest of the operation. Once the hernia sac has been separated from the vas deferens and the spermatic vessels, the hernia sac is divided between hemostats in its midcourse after it is ensured that there are no other tissues inside the sac and that there are no sliding components making part of the wall of the sac. I find it helpful to place the cord structures within a vessel loop for gentle traction to avoid injuries. The operation proceeds with dissection of the proximal portion of the hernia sac up to the level of the internal inguinal ring, where it is suture ligated with nonabsorbable suture and excised (**Figure 77-5**).

If you wish to perform a diagnostic laparoscopy, a short 5-mm trocar is introduced through the sac and secured with a 3-0 Vicryl tie. Pneumoperitoneum is created with a maximum pressure of 4-8 mm Hg. The patient is placed in the Trendelenburg position, and the table is tilted toward the surgeon. A 120° telescope is introduced to inspect the contralateral inguinal ring. After this is done, the trocar is removed, the pneumoperitoneum evacuated, and the ligation of the sac completed.

In most cases, high ligation of the hernia sac is sufficient treatment for an inguinal hernia in a child. The distal portion of the sac is opened widely; no attempts are made to remove the sac because this may result in devascularization of the testicle. In patients in whom the floor of the inguinal canal is weak, repair may be performed using the Bassini technique by approximating the internal oblique muscle to the shelving edge of the inguinal ligament with two to three interrupted stitches. The most medial stitch approximates the internal oblique muscle (or the conjoint tendon when present) to the pubic spine. If a hydrocele is present, the tunica vaginalis is opened and the fluid is evacuated. The testicle can be brought back down into the scrotum by gentle caudad traction of the scrotal skin, which will pull the testicle down along with the gubernaculum testis.

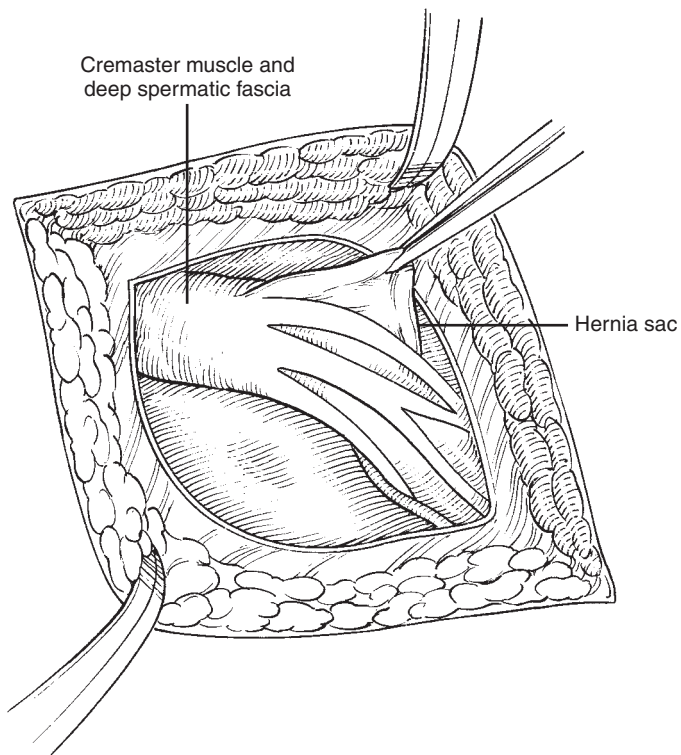


FIGURE 77-3

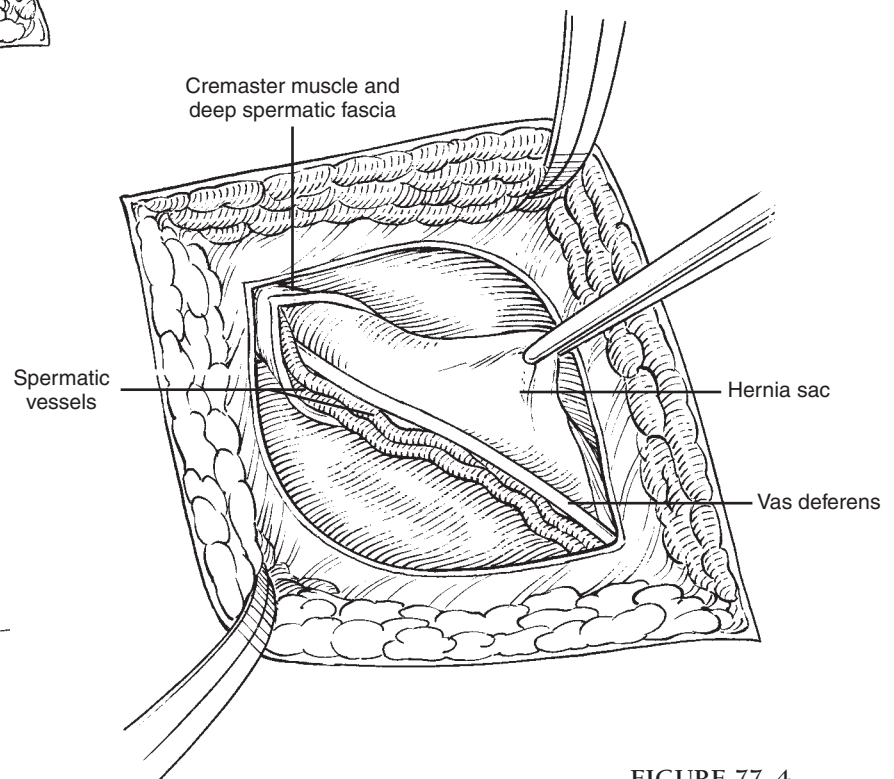


FIGURE 77-4

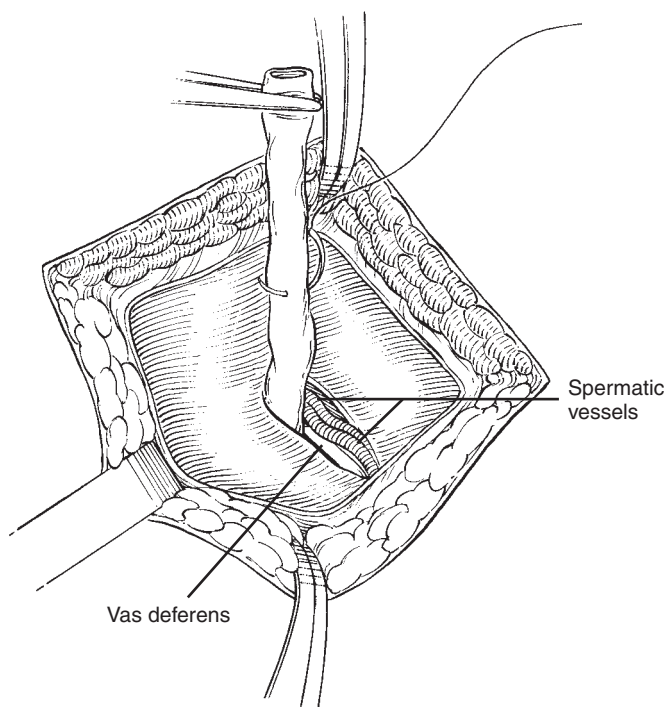


FIGURE 77-5

3. CLOSING

- ◆ The external oblique aponeurosis is closed with interrupted fine absorbable suture, making sure that the external inguinal ring does not constrict the cord structures (**Figure 77-6**). Scarpa's fascia is closed with interrupted fine absorbable suture, and the skin is closed with a running fine absorbable monofilament subcuticular suture. The skin is dressed with adhesive strips.

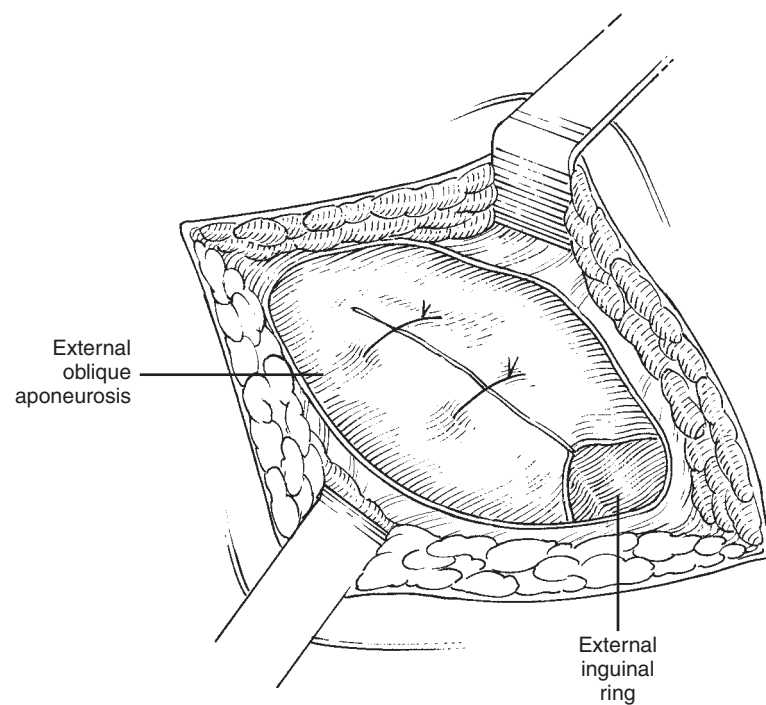


FIGURE 77-6

STEP 4: POSTOPERATIVE CARE

- ◆ Most patients will have either a caudal block, an ilioinguinal block, or subcutaneous infiltration of the incision with local anesthetic in the operating room for postoperative pain control. An oral analgesic such as acetaminophen is prescribed to be given every 4 to 6 hours on the first postoperative day and then administered only as needed. Children are allowed to bathe normally 24 hours after the operation and can resume full activity after 2 weeks.
- ◆ Although rare, the most common complications are wound infections and hematomas. Injury to the vas deferens, epididymis, or spermatic vessels and hernia recurrence are reported in up to 1% of cases.

STEP 5: PEARLS AND PITFALLS

- ◆ Operating immediately after manual reduction of an incarcerated inguinal hernia in a child is technically difficult and fraught with complications, because the hernia sac is edematous and friable, and the structures of the cord are not easily identifiable. A period of 24 hours to allow some of the edema to subside is advisable.
- ◆ As a general rule, no structures should be divided until both the spermatic vessels and the vas deferens have been positively identified and placed within a vessel loop.
- ◆ Use of electrocautery in the vicinity of the spermatic cord is discouraged, because arcs of electrical current may result in thrombosis of the spermatic vessels and loss of the testicle.

SELECTED REFERENCES

1. Weber TR, Tracy TF, Keller MS: Groin hernias and hydroceles. In Ashcraft KW, Holcomb GW, Murphy JP (eds): *Pediatric Surgery*, 4th ed. Philadelphia, Elsevier Saunders, 2005, pp 697-705.
2. Engum SA, Grosfeld JL: Hernias in children. In Spitz L, Coran AG (eds): *Operative Pediatric Surgery*, 6th ed. London, Edward Arnold, 2006, pp 237-244.



LAPAROSCOPIC INGUINAL HERNIA REPAIR

Michael D. Trahan

STEP 1: SURGICAL ANATOMY

- ◆ A thorough understanding of the preperitoneal space and important structures of the retroperitoneal space and inguinal canal is prerequisite to attempting laparoscopic inguinal hernia repair (**Figure 78-1**).

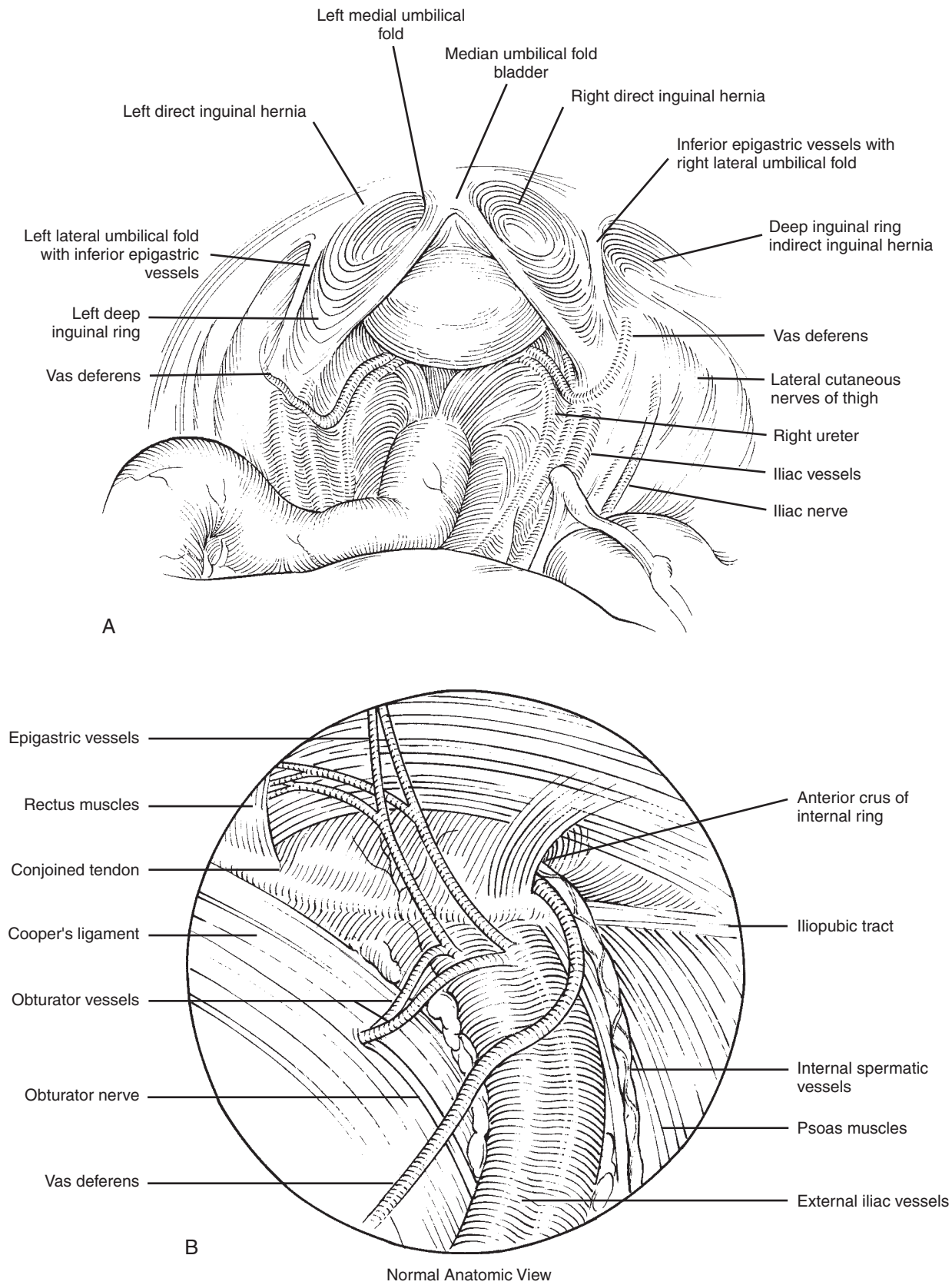
STEP 2: PREOPERATIVE CONSIDERATIONS

INDICATIONS

- ◆ A laparoscopic approach to inguinal hernias is indicated for any indirect, direct, or femoral hernia but is particularly suited to bilateral hernias and recurrences from anterior repairs.
- ◆ Larger hernias, especially with scrotal extension, can make the laparoscopic approach much more difficult.

PREPARATION

- ◆ A urinary catheter is inserted for bladder decompression.
- ◆ The patient is placed in the supine Trendelenburg position with the arms padded and tucked.



STEP 3: OPERATIVE STEPS**1. INCISION**

- ◆ Three ports are used: one 10-mm port and two low-profile 5-mm ports (**Figure 78-2**).

2. DISSECTION

- ◆ Transabdominal preperitoneal (TAPP) repair
 - ◆ An optically guided, bladeless 10-mm port is placed into the peritoneal cavity near the umbilicus. The two 5-mm ports are placed, guided by internal visualization to avoid the epigastric vessels.
 - ◆ The peritoneum is incised starting at the medial umbilical fold and proceeding laterally to or past the anterior superior iliac spine. The incision is made well away from the internal ring to provide ample tissue to cover the peritoneal defect at the end of the procedure (**Figure 78-3**).
 - ◆ The peritoneum is peeled down and bluntly dissected from the underlying structures. An indirect hernia sac is gently pulled out of the internal ring and dissected free from the cord structures.

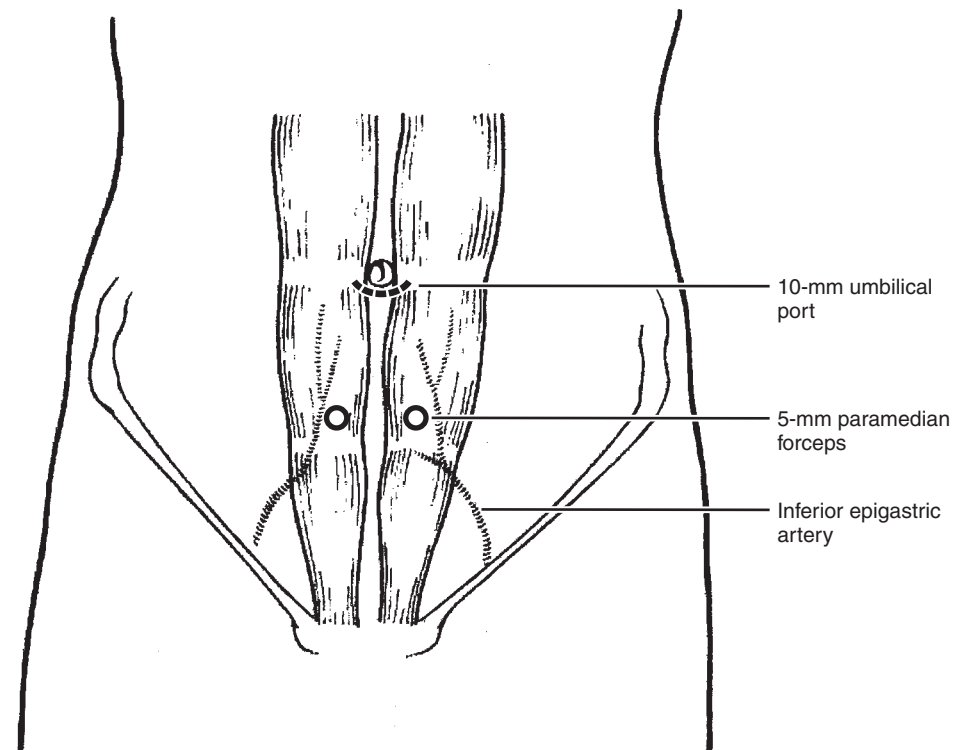


FIGURE 78-2

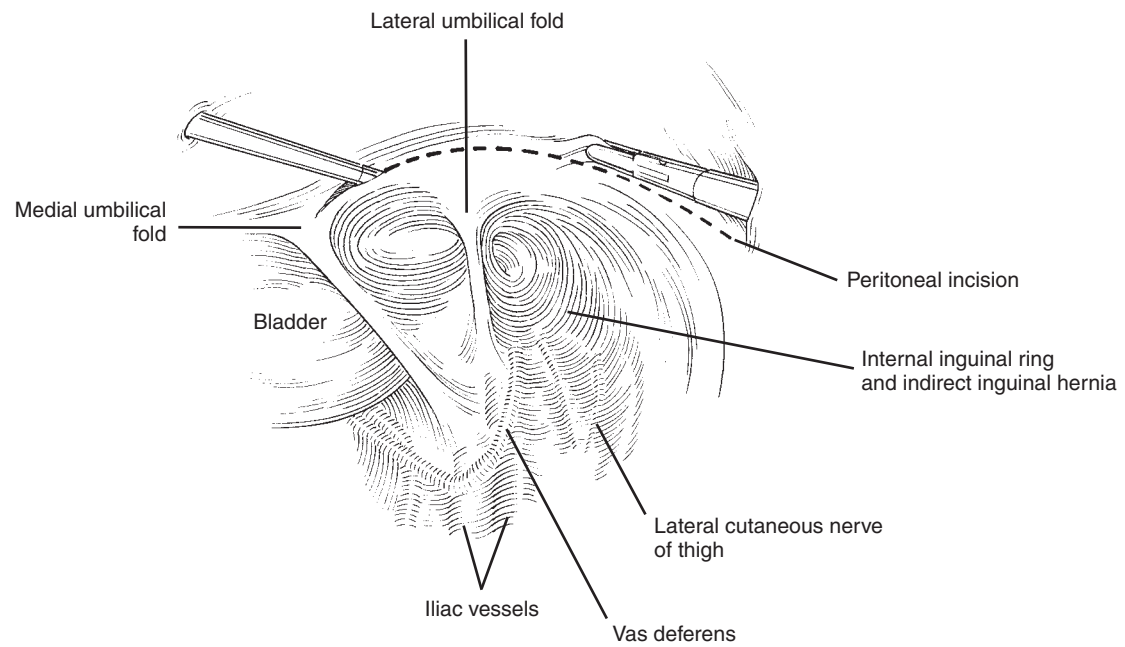


FIGURE 78-3

- ◆ The margins of a complete dissection are the midline medially, the iliac bone laterally, and Cooper's ligament inferiorly (**Figure 78-4, A-B**).
- ◆ The fatty contents of a direct hernia defect are reduced and trimmed away as necessary (**Figure 78-4, C**).

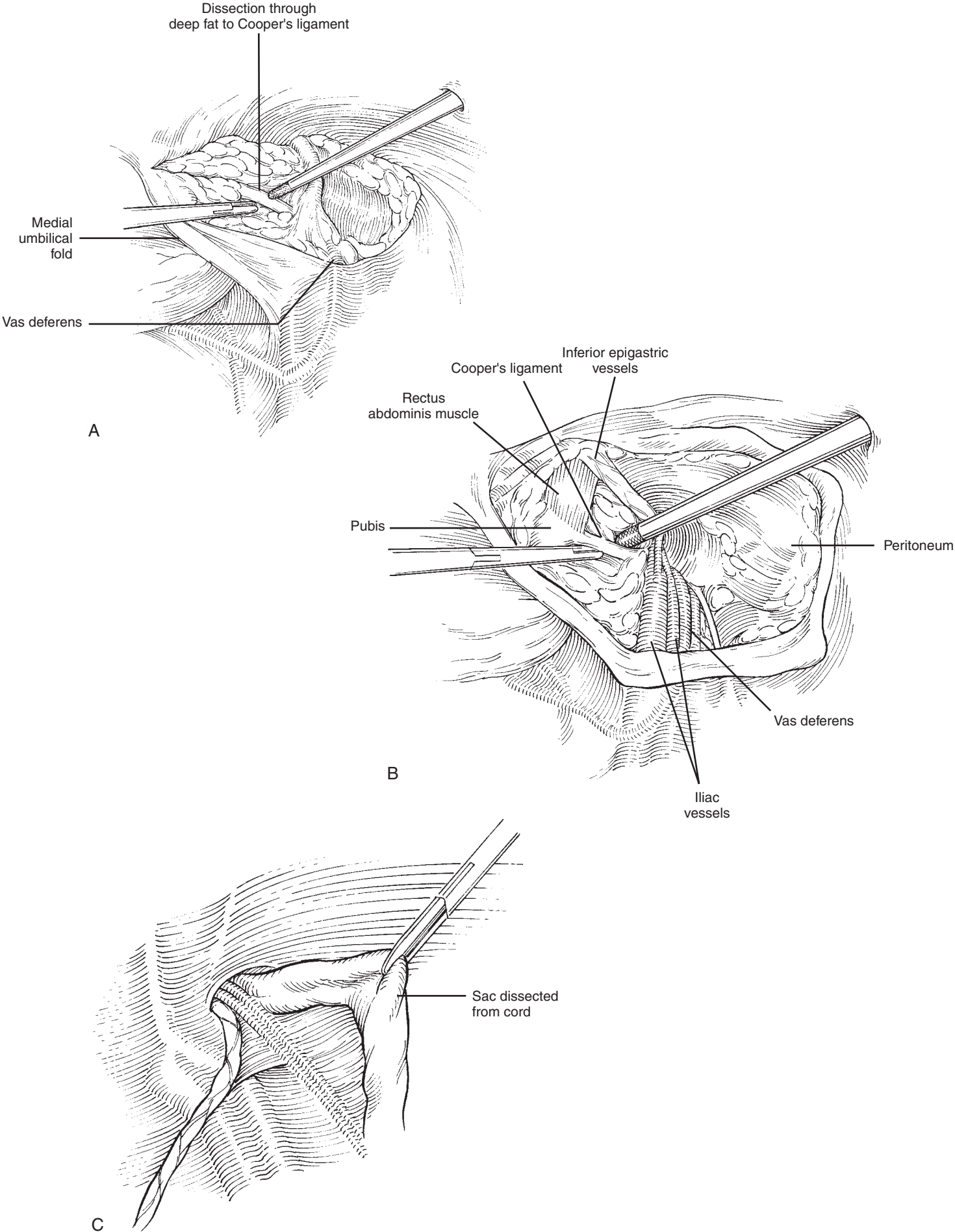


FIGURE 78-4

- ◆ A sheet of polypropylene or polyester mesh is trimmed to shape, tightly rolled, inserted through the scope port, and spread out on the deperitonealized surface.
- ◆ The mesh should be anchored at the pubic bone, for a short distance along Cooper's ligament, and along the anterior abdominal wall, taking care to avoid the important neurovascular structures (**Figure 78-5**).

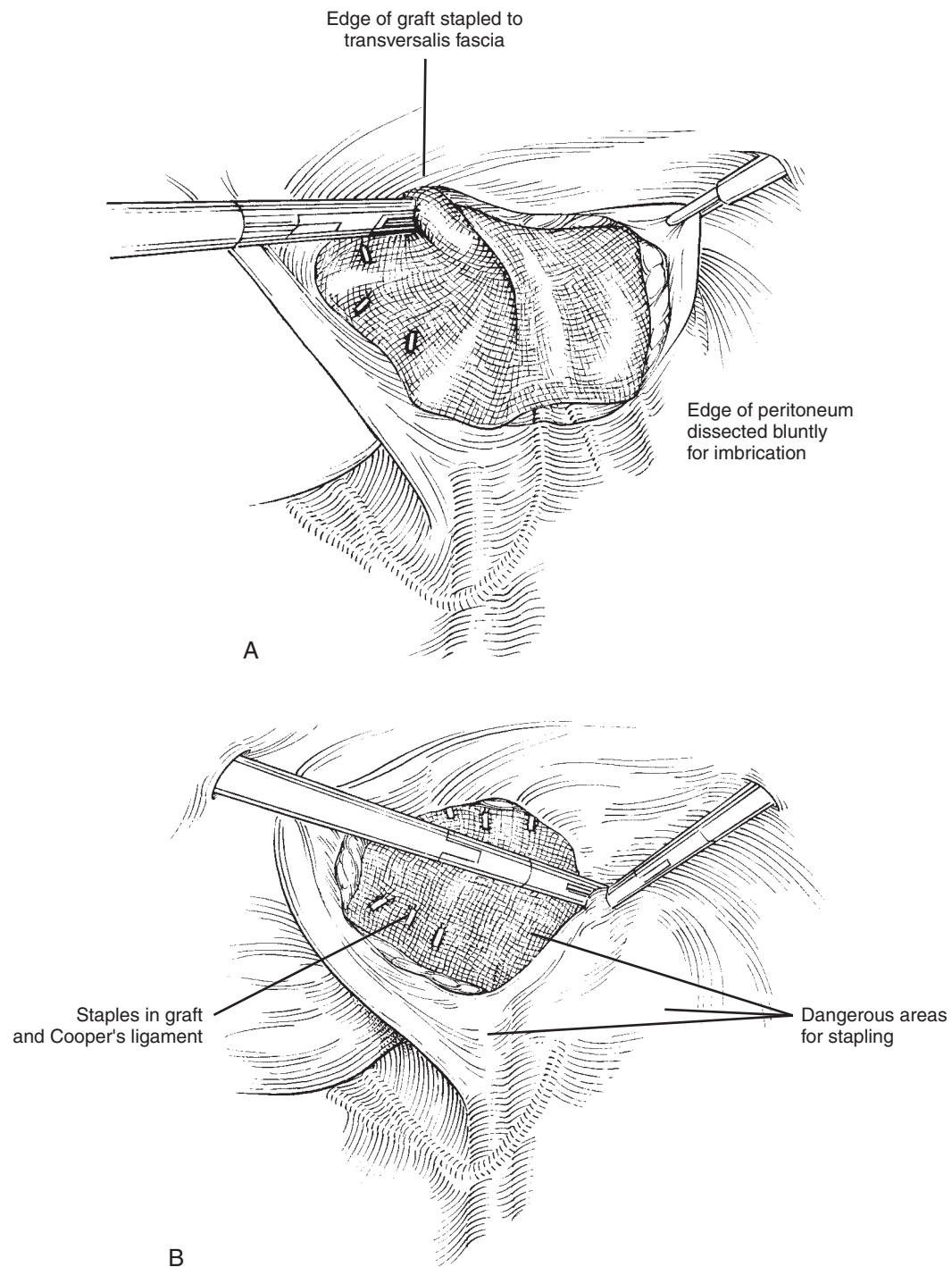


FIGURE 78-5

- ◆ The peritoneal flap is used to cover the mesh. The incision in the peritoneum is closed with clips or tacks (**Figure 78-6**).

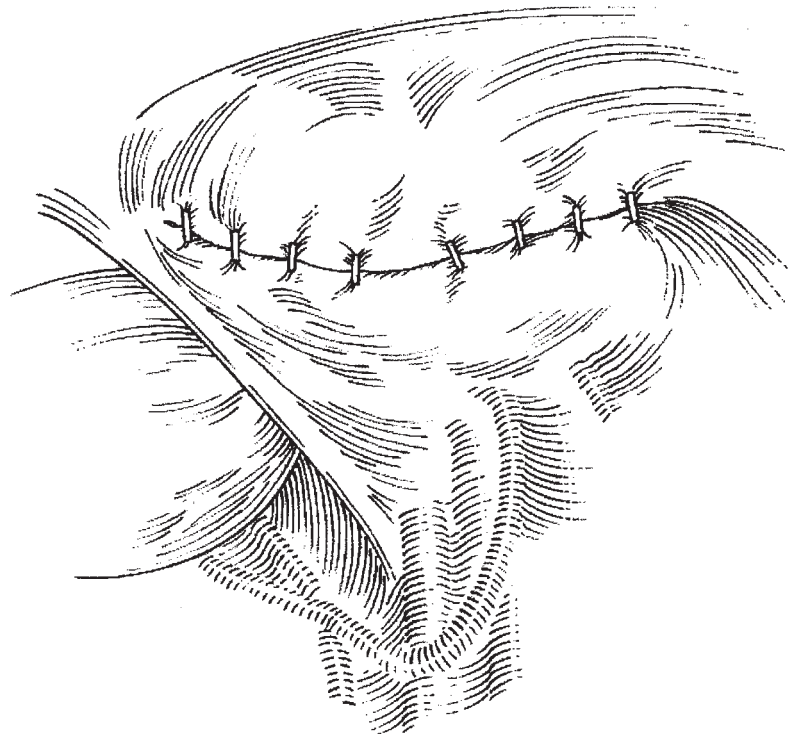


FIGURE 78-6

- ◆ Totally extraperitoneal (TEP) repair
 - ◆ The 10-mm port is first placed using an open approach. A small incision is made along the inferior edge of the umbilicus. The anterior rectus sheath is exposed and incised on either side of the midline. The rectus muscle is retracted laterally from the midline to expose the posterior rectus sheath.
 - ◆ The preperitoneal plane (between the rectus muscle and posterior rectus sheath) is bluntly dissected manually or, preferably, with a balloon dissector. This plane is developed down to the pubic bone (**Figure 78-7**).

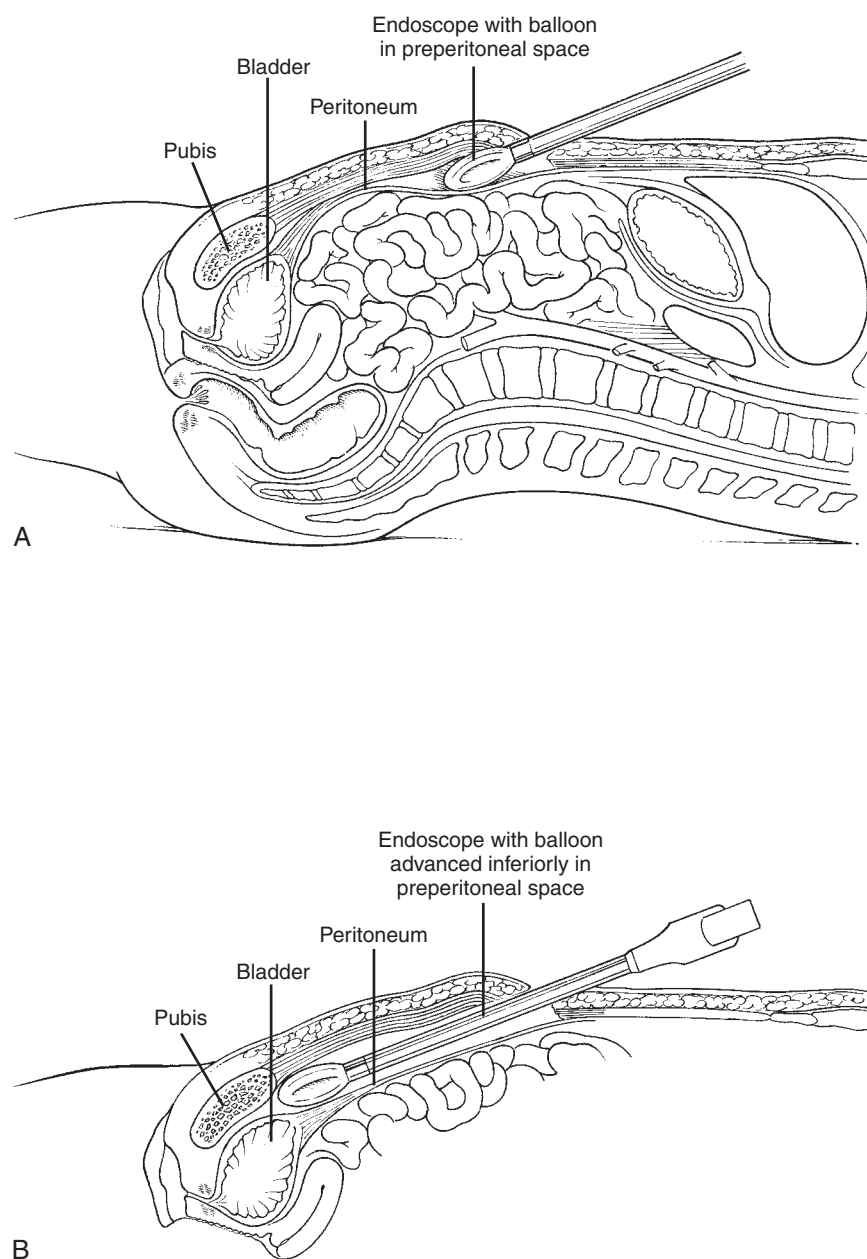


FIGURE 78-7

- ◆ The balloon dissector is inflated fully while the laparoscope is positioned to view internally. An interior view of the structures of the spermatic cord is usually observed at this point. A small indirect hernia sac may be reduced by inflation of the balloon (**Figure 78-8**).
- ◆ The balloon is deflated and the balloon dissecting port is replaced with a working 10-mm port. CO₂ insufflation at 13 to 15 mm Hg is used to maintain the expansion of the preperitoneal space. The seal of the port to the anterior fascia can be provided by cinching the fascia with suture, placing a Hassan adapter, or using a balloon trocar (illustrated) (**Figure 78-9**).
- ◆ The scope is inserted for placement of the two 5-mm ports and completion of the preperitoneal dissection. This dissection can usually be done bluntly without the need for the electrosurgical unit. However, care must be taken to avoid tearing the peritoneum, because this will lead to inflation of the peritoneal cavity and loss of adequate preperitoneal visualization. The structures entering the internal ring should be dissected until they are clearly seen (see Figure 78-1, B).

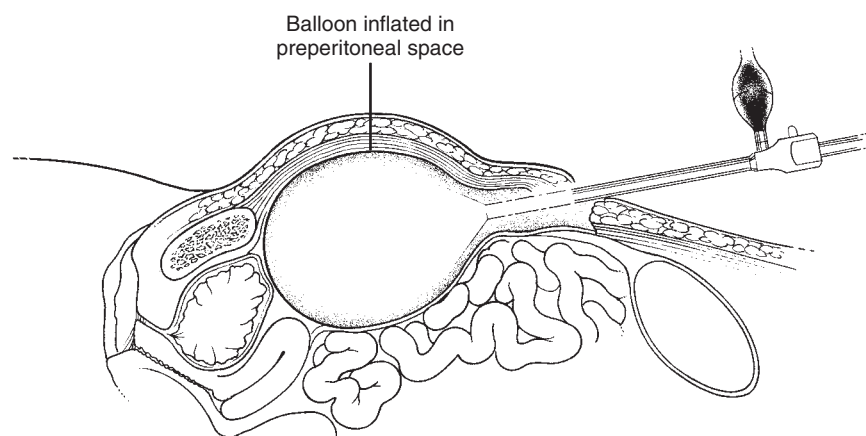


FIGURE 78-8

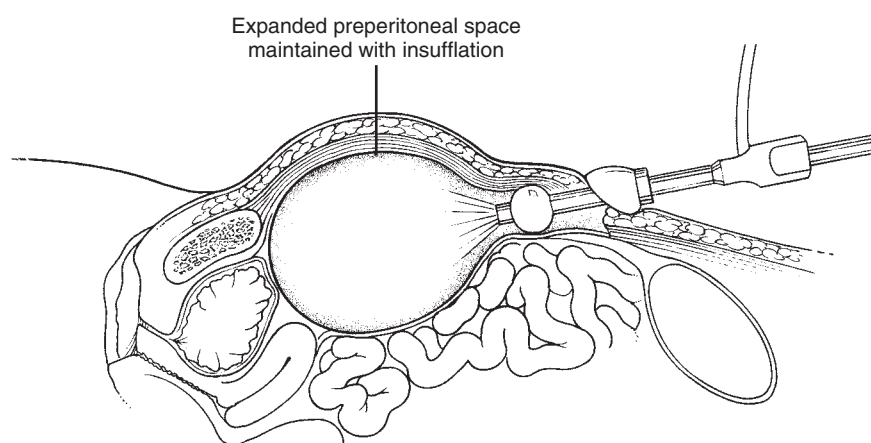


FIGURE 78-9

- ◆ The fatty contents of a direct hernia defect should be reduced and usually excised (**Figure 78-10**).
- ◆ An indirect sac, if not reduced by the balloon inflation, is gently grasped and pulled out of the internal ring and carefully dissected it from the cord structures (**Figure 78-11**).
- ◆ The margins of dissection are the midline medially, the anterior superior iliac spine laterally, the transverse arch superiorly, and Cooper's ligament inferiorly. The polypropylene or polyester mesh is trimmed to shape, rolled tightly, inserted through the 10-mm port, and laid in place to cover the femoral and both inguinal potential orifices. A notch cut for the iliac vessels allows the mesh to lay with less buckling (**Figure 78-12**).
- ◆ The mesh can be anchored, if desired, to the pubis, Cooper's ligament, and the abdominal wall above the transverse arch avoiding the sites of neurovascular structures (see **Figure 78-5**).
- ◆ The insufflation is released (while the mesh is held in place if no anchoring is used), and the ports are removed.

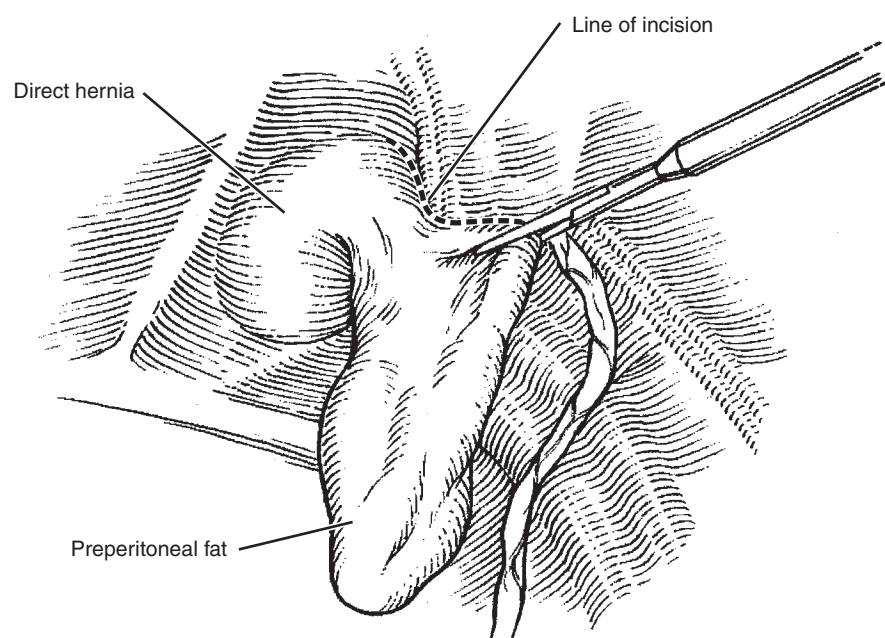


FIGURE 78-10

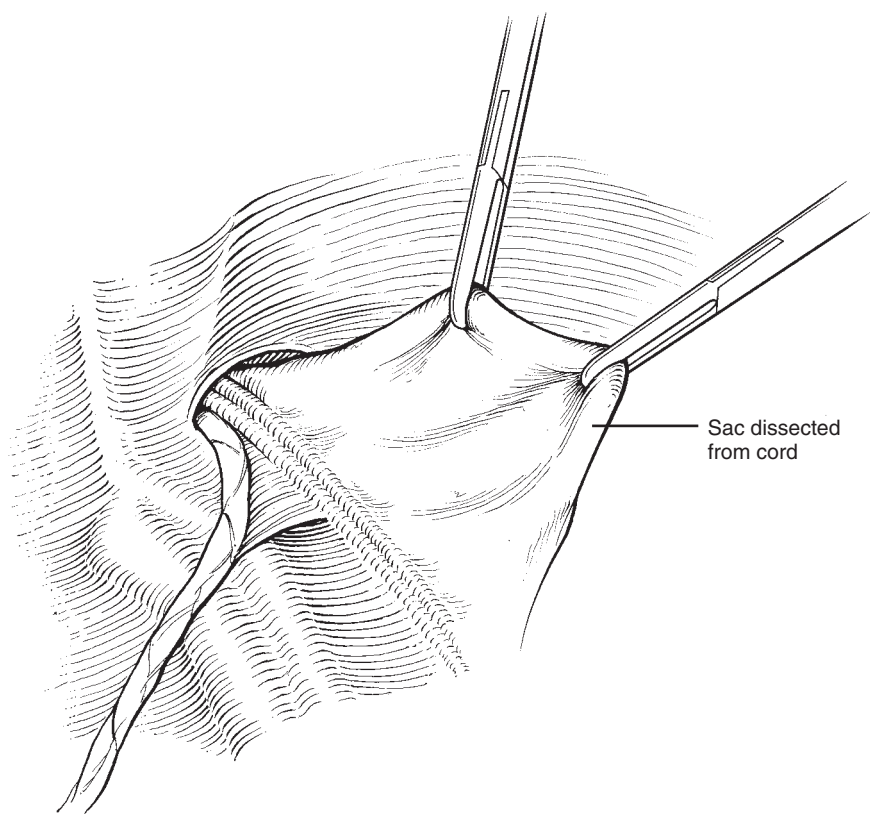


FIGURE 78-11

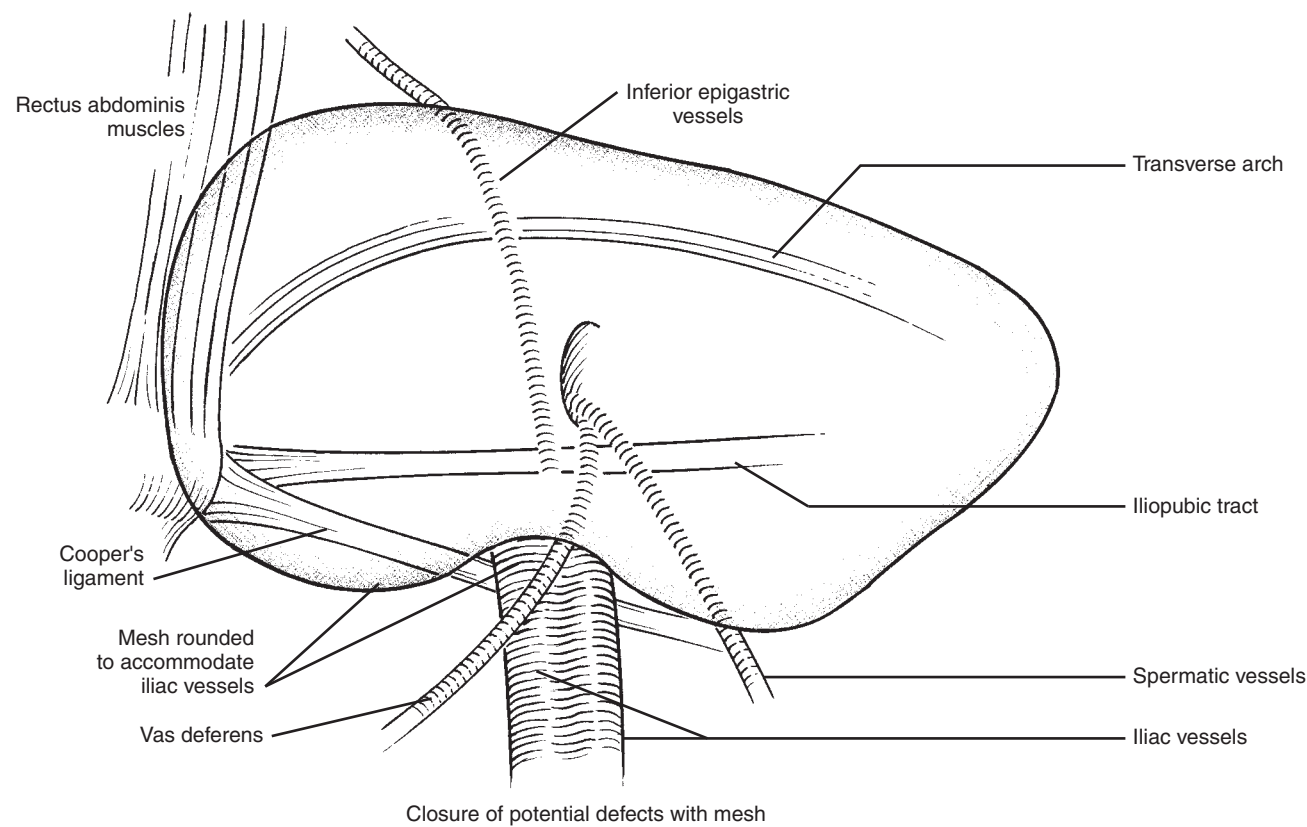


FIGURE 78-12

3. CLOSURE

- ◆ The anterior fascia is closed with absorbable suture. The skin incisions are closed with absorbable suture with a subcuticular technique.
- ◆ The incisions are dressed with tissue adhesive or tapes.

STEP 4: POSTOPERATIVE CARE

- ◆ These operations are usually performed in the outpatient setting.
- ◆ An oral narcotic such as hydrocodone is appropriate for pain management.
- ◆ Patients may return to regular activity as the surgical discomfort resolves.

STEP 5: PEARLS AND PITFALLS

- ◆ One must avoid fixation clips and tacks in the lower outer quadrant of the mesh. This is where the nerves and large vessels travel.
- ◆ If the peritoneal membrane is entered during a TEP, inflation of the preperitoneal space can be maintained by placing a Veress needle into the peritoneal cavity in the upper abdomen. Alternatively, the TEP procedure can be converted to a TAPP procedure.
- ◆ The inferior edge of the mesh should be tucked under the peritoneum as the pneumoperitoneum is released to avoid migration of the mesh.

SELECTED REFERENCES

1. McKernan JB, Laws HL: Laparoscopic repair of inguinal hernias using a totally extraperitoneal prosthetic approach. *Surg Endosc* 1993;7:26-28.
2. Stoppa RE, Warlaumont CR: The preperitoneal approach and prosthetic repair of groin hernia. In Nyhus LM, Condon RE (eds): *Hernia*, 3rd ed. Philadelphia, Lippincott, 1989, pp 199-225.
3. Liem MS, van Vroonhoven TJ: Laparoscopic inguinal hernia repair. *Br J Surg* 1996;83:1197-1204.

FEMORAL HERNIA

Thomas D. Kimbrough

STEP 1: SURGICAL ANATOMY

- ◆ **Figure 79-1** shows, in a stylized fashion, the structures that pass underneath the inguinal ligament. Although large femoral hernias may extend laterally in the subcutaneous tissues of the thigh, most should be located medial to the palpated pulse of the femoral artery.

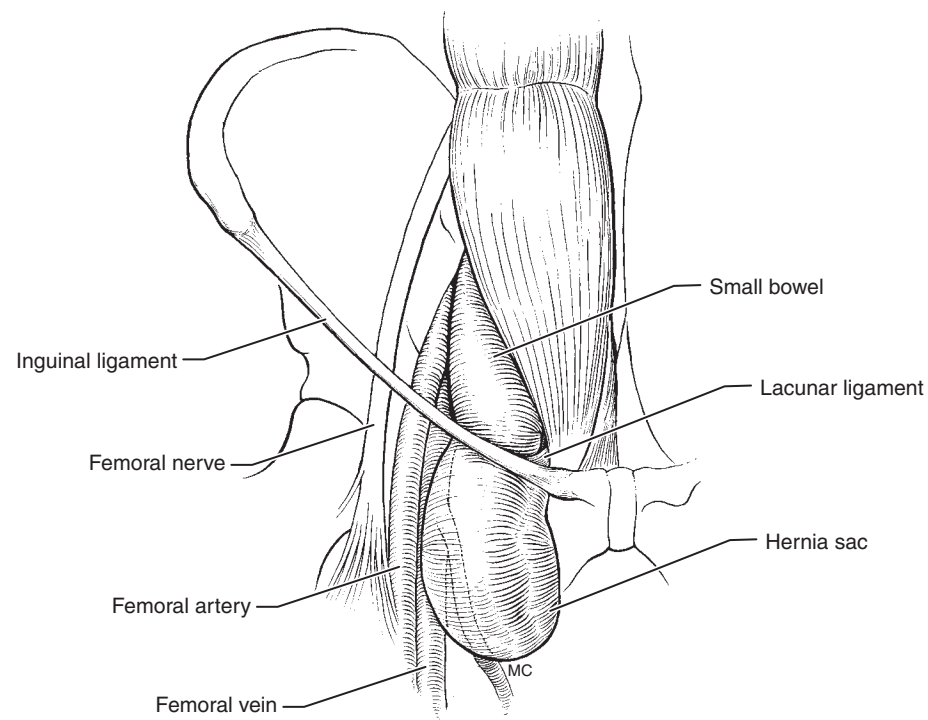


FIGURE 79-1

- ◆ The hernia is protruding through the femoral canal, which remains only a potential space in most people. The space is bounded anteriorly by the inguinal ligament, posteriorly by the pubic ramus and pectineal ligament, laterally by the femoral vein and sheath, and medially by the lacunar portion of the inguinal ligament.

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ The course of the inguinal ligament parallels a line drawn between the pubic tubercle and the anterior superior iliac spine. Any mass that lies beneath this line and medial to the femoral artery pulsation is a possible femoral hernia.
- ◆ As outlined later, some preoperative consideration regarding placement of the incision should take place.

STEP 3: OPERATIVE STEPS

1. INCISION

- ◆ The elective femoral hernia can be approached through a transverse incision, parallel to the inguinal ligament over the palpable mass in the medial thigh, just below the inguinal ligament.
- ◆ A standard incision should be considered for larger hernias, especially those that might require access to the preperitoneal space.

2. DISSECTION

- ◆ The exposed sac with surrounding structures is shown in **Figure 79-2**. In most cases, the sac can be reduced through the femoral canal back into the preperitoneal space and a repair effected.

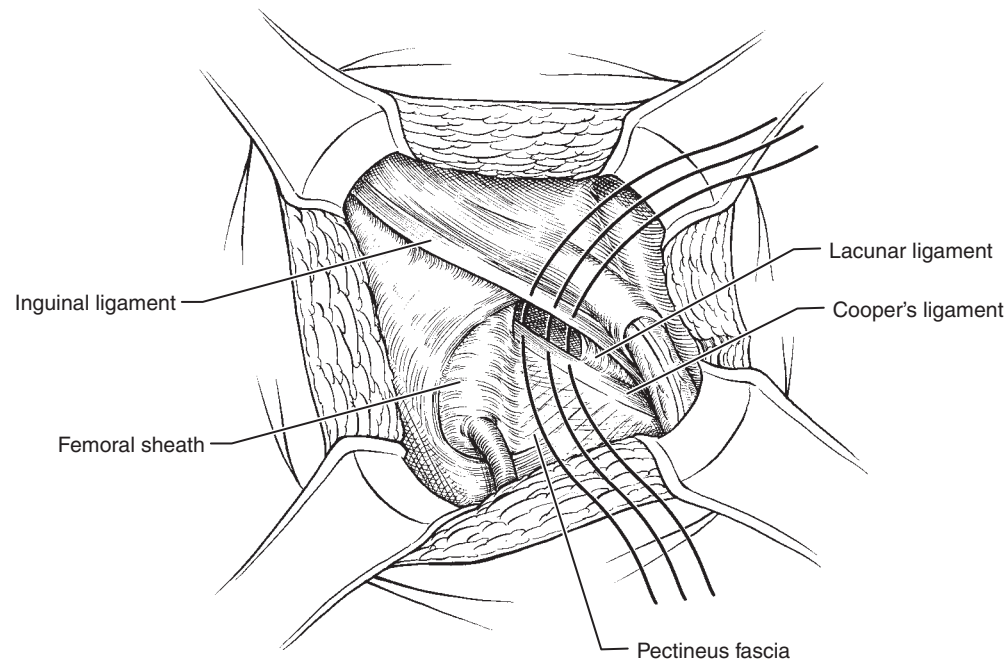


FIGURE 79-2

- ◆ The classic repair approximating the inguinal ligament to Cooper's ligament and the pectineus fascia with 2-0 monofilament polypropylene suture is shown in **Figure 79-3**.
- ◆ The relatively higher rate of recurrences associated with this and other tissue repairs has led most to use a tension-free mesh plug repair, as illustrated in **Figure 79-4**. A piece of polypropylene mesh approximately 2 inches long is rolled into a generous plug and inserted into the femoral canal. Suitable anchoring sutures are placed superiorly, medially, and inferiorly, as shown in Figure 79-4.
- ◆ In the case of emergency operations for strangulation or small bowel obstruction, careful consideration should be given to placement of the skin incision. In such cases, it may not be possible to reduce the hernia from below, and access to the preperitoneal space also through the floor of the inguinal canal may be necessary.
- ◆ **Figure 79-5** illustrates such an approach. After successful reduction, repair should include not only the mesh plug described but also a mesh repair of the inguinal floor, such as the Prolene Hernia System repair described in Chapter 72.

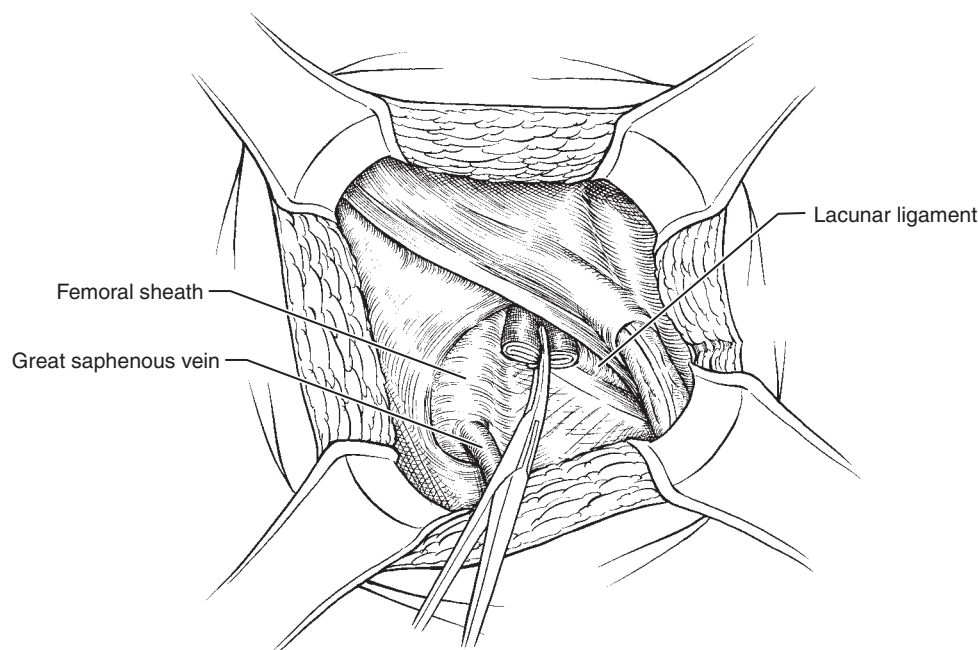


FIGURE 79-3

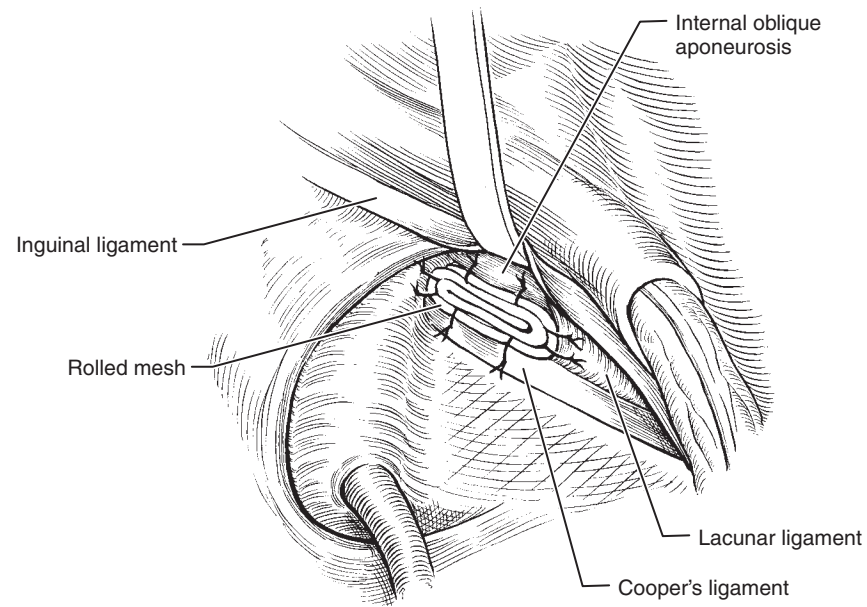


FIGURE 79-4

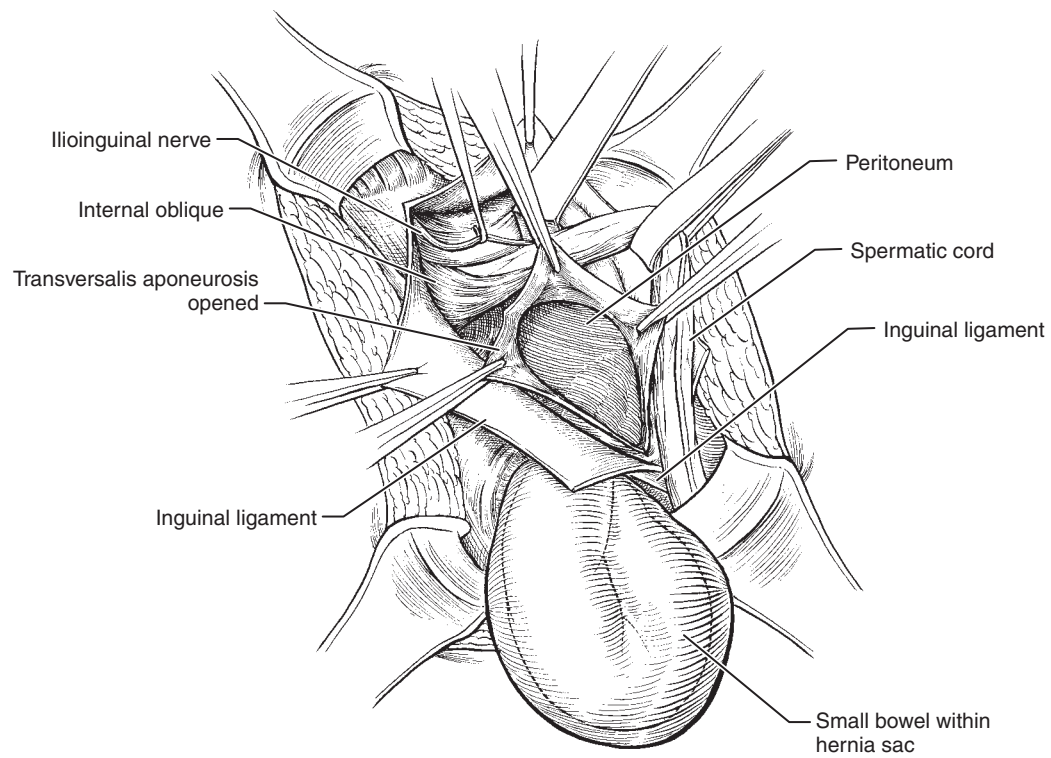


FIGURE 79-5

3. CLOSURE

- ◆ Closure is as described in Chapter 72.

STEP 4: POSTOPERATIVE CARE

- ◆ No instructions additional to those described earlier for mesh repairs of direct and indirect hernias are necessary.

STEP 5: PEARLS AND PITFALLS

- ◆ If in the approach to reduction of an incarcerated femoral hernia repair the sac cannot be reduced, the following is an option. Partial division of the lacunar ligament medially will enlarge the canal and usually allow reduction. If this is chosen, one should remember that if there is an aberrant course of the obturator artery, it can be lacerated during this maneuver. A far less desirable option is division of the overlying inguinal ligament.
- ◆ In any elderly patient with a bowel obstruction, especially women, the possibility of an incarcerated femoral hernia should be considered and looked for on physical examination. It is not only embarrassing for the surgeon to find such a hernia after a large midline laparotomy, but it is potentially quite harmful to the patient to experience the invasion of the abdominal cavity when an inguinal exploration would most likely have sufficiently solved the problem.

SELECTED REFERENCE

1. Nyhus LM, Condon RE (eds): *Hernia*, 4th ed. Philadelphia, JB Lippincott, 1995.



UMBILICAL HERNIA (CHILD AND ADULT)

Michael D. Trahan

STEP 1: SURGICAL ANATOMY

- ◆ Most umbilical hernias are congenital. Conditions that increase intra-abdominal pressure can lead to an acquired hernia later in life.
- ◆ The hernia contents protrude through a defect in the linea alba through which the fetal umbilical vessels passed.
- ◆ The linea alba is the result of the midline fusion of the external oblique, internal oblique, and transversus abdominis muscles.

STEP 2: PREOPERATIVE CONSIDERATIONS FOR A CHILD

- ◆ Umbilical hernias are common in children and usually (up to 80%) close sometime during the first 4 years of life. Unless the hernia is complicated, repair should be delayed until 4 years of age.
- ◆ General anesthesia is preferred.

STEP 2: PREOPERATIVE CONSIDERATIONS FOR AN ADULT

- ◆ Umbilical hernias should be repaired in adults unless there is a contraindication. Contraindications may include patients unfit for anesthesia and the presence of massive ascites.

Anesthesia

- ◆ General anesthesia is preferred.
- ◆ Spinal anesthesia or deep sedation with local anesthesia are possibilities for small fascial defects in cooperative patients.

STEP 3: OPERATIVE STEPS FOR A CHILD

1. INCISION

- ◆ A curvilinear incision is made at the inferior rim of the umbilicus (**Figure 80-1**).

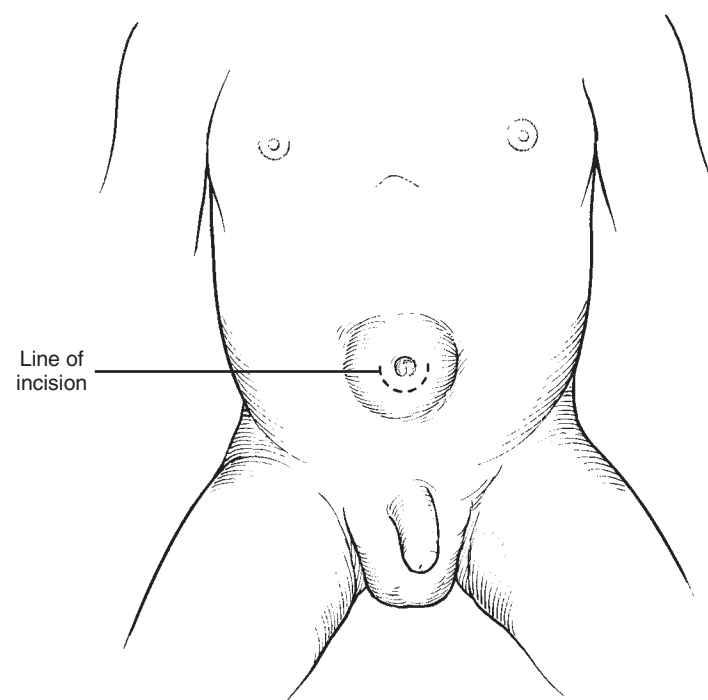


FIGURE 80-1

2. DISSECTION

- ◆ A combination of blunt and sharp dissection is used to expose the fascia of the abdominal wall and the hernia sac. The umbilical stalk is bluntly encircled with a right-angled hemostat (Figure 80-2).
- ◆ The hernia contents, if present, are reduced, and the stalk is divided (Figure 80-3).
- ◆ The excess sac is excised, and the fascia is approximated in the midline using 3-0 interrupted polypropylene suture.

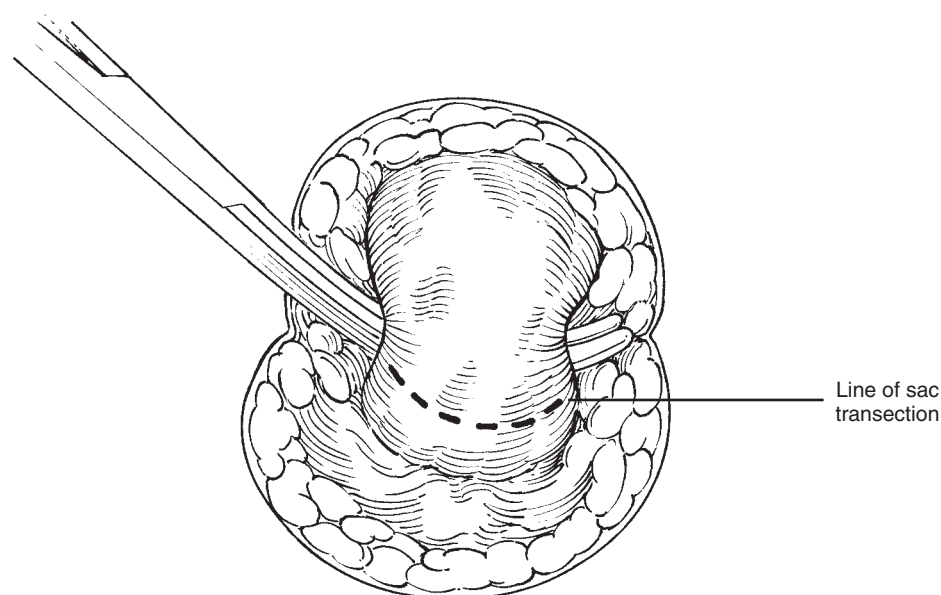


FIGURE 80-2

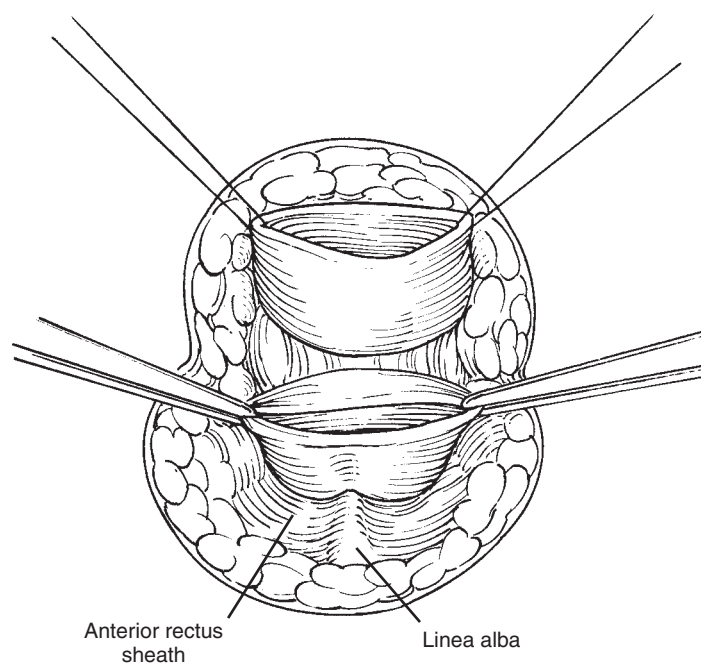


FIGURE 80-3

3. CLOSING

- ◆ The dermis of the umbilicus is tacked down to the fascia using an absorbable 3-0 suture. The incision is closed with a running subcuticular absorbable 4-0 suture and dressed with tissue adhesive.
- ◆ A cotton ball or fluffed gauze is placed within the umbilicus, covered with gauze, and taped into place.

STEP 3: OPERATIVE STEPS FOR AN ADULT

1. INCISION

- ◆ For a small defect, an incision at the rim of the umbilicus may be used, but for larger hernias, a midline approach results in better exposure for placement of the mesh and a more acceptable postoperative cosmetic result (**Figures 80-4 and 80-5**).

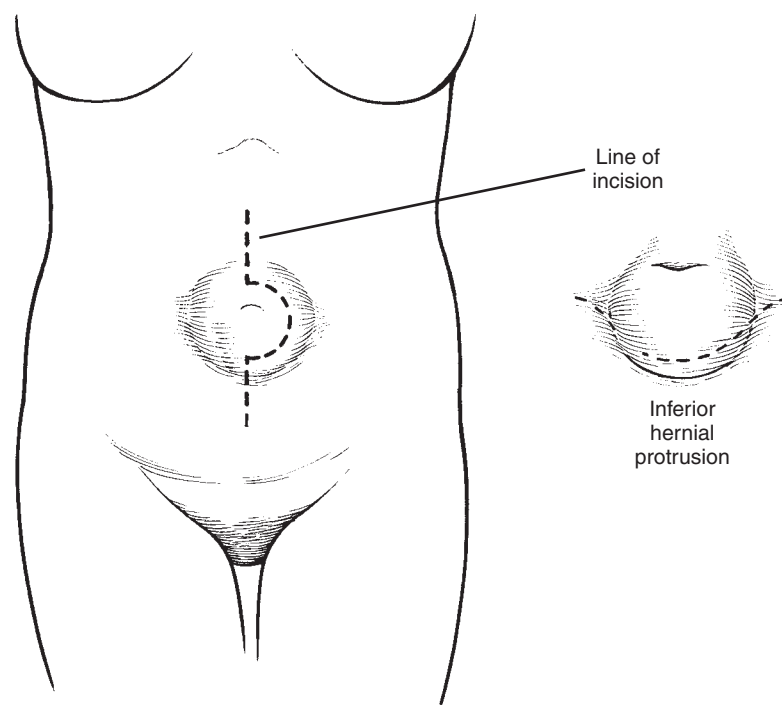


FIGURE 80-4

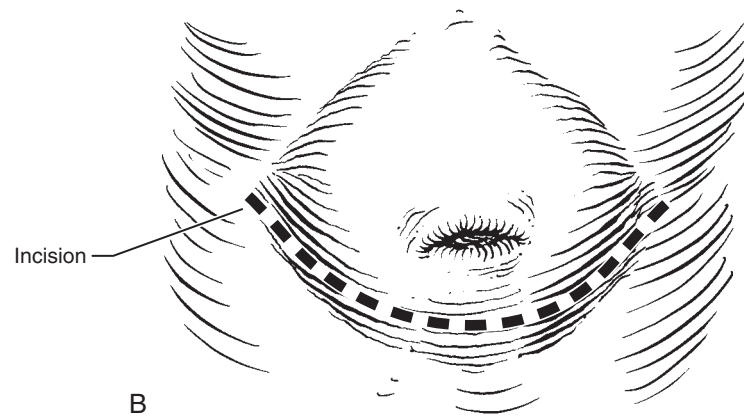
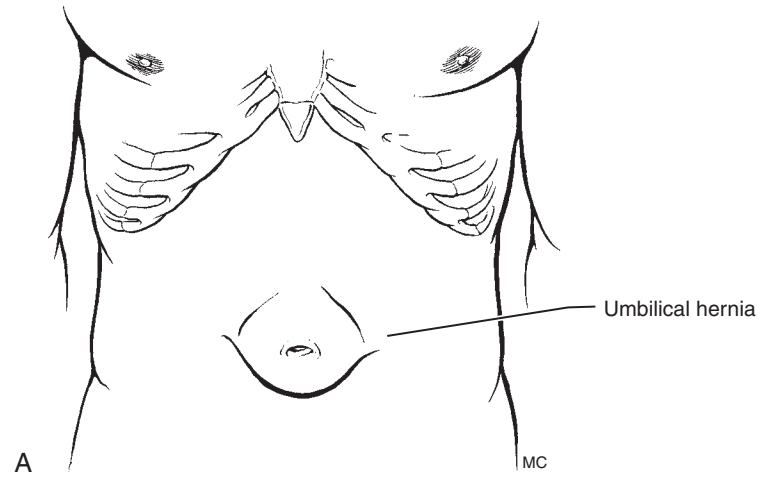


FIGURE 80-5

2. DISSECTION

- ◆ A combination of blunt and sharp dissection is used to expose the midline fascia and hernia sac. If possible, the hernia contents are reduced. If the contents are reducible or chronically incarcerated, an attempt is made to sharply dissect the sac from the overlying skin without opening the sac. For chronically incarcerated hernias, the midline fascia may need to be incised to completely reduce the contents (**Figure 80-6**).
- ◆ For an acutely incarcerated hernia, the sac is opened so that its contents can be inspected for strangulation and ischemia (**Figure 80-7**).
- ◆ The fascia may need to be carefully incised in the midline to release the incarcerated contents (**Figure 80-8**).
- ◆ If determined to be nonviable, the hernia contents are excised. The healthy remaining intestine is repaired primarily (**Figure 80-9**).

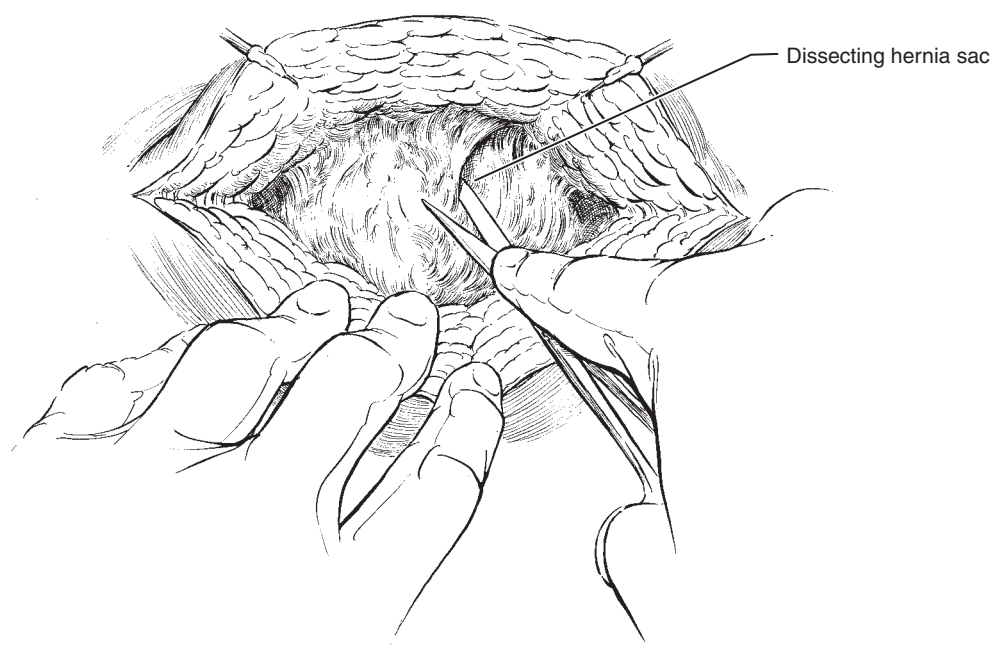


FIGURE 80-6

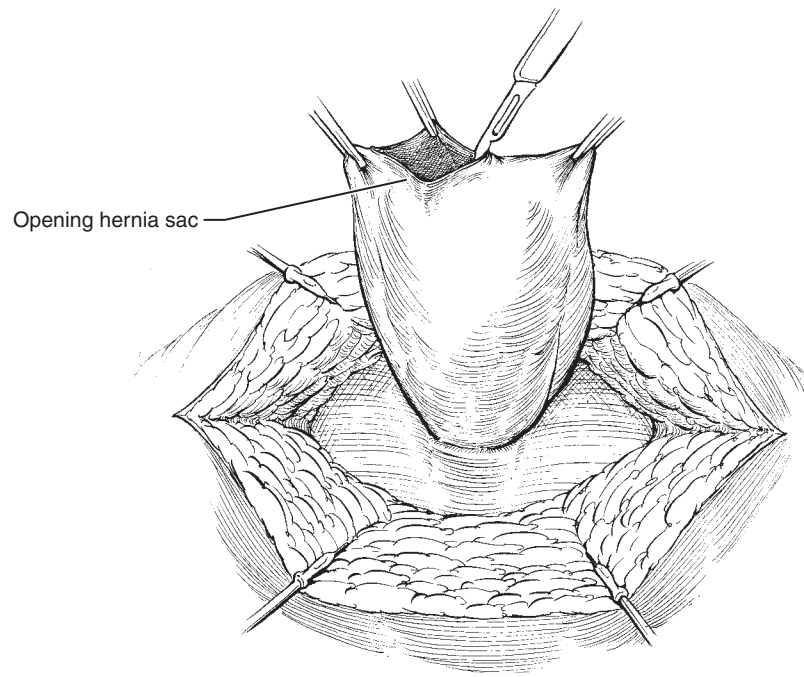


FIGURE 80-7

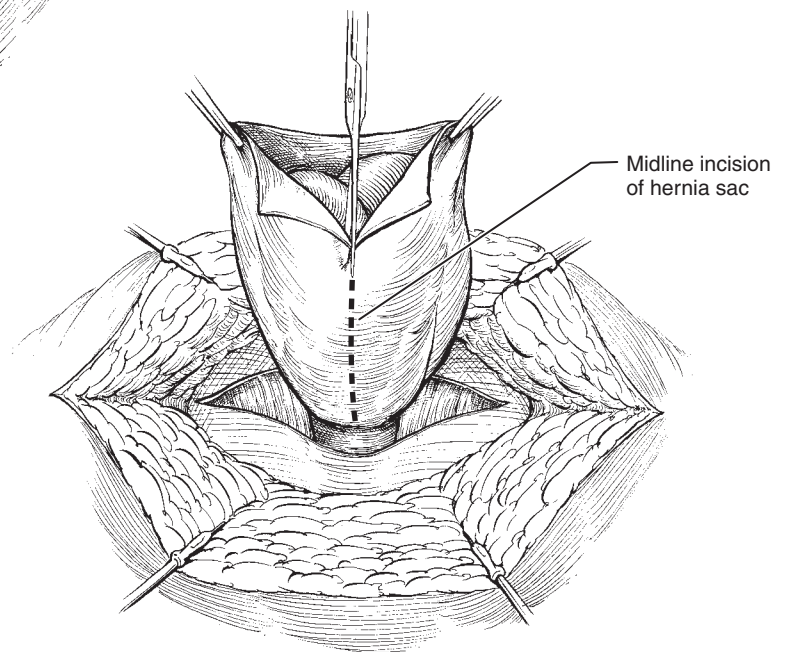


FIGURE 80-8

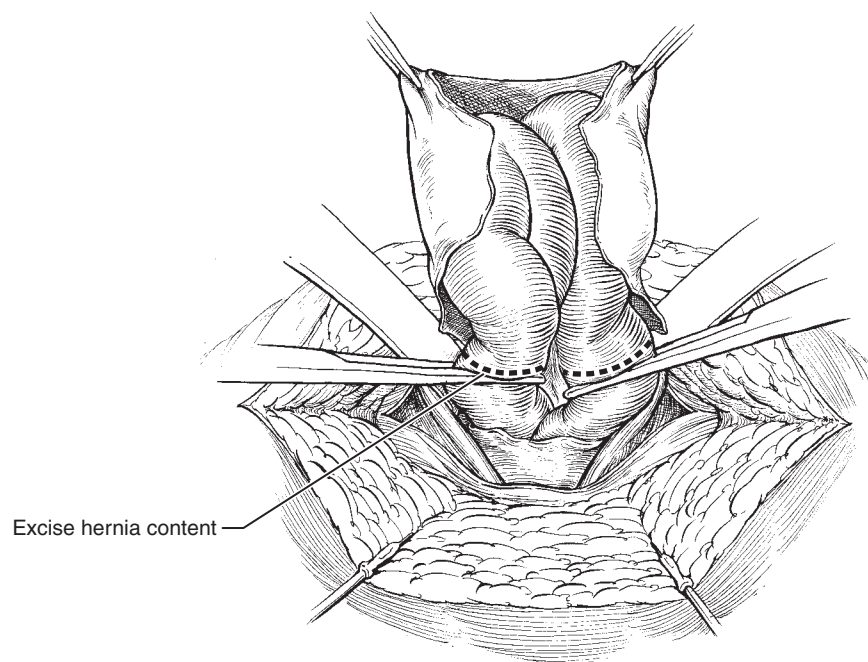


FIGURE 80-9

- ◆ After complete reduction of the hernia contents, the sac is closed and the plane between the peritoneum and the posterior abdominal fascia is sharply dissected for a distance of 3 cm circumferentially if mesh is to be used. If this plane is not accessible, the mesh may be placed on the posterior rectus sheath, posterior to the rectus muscles (**Figure 80-10**).
- ◆ Consideration may be given to primary midline approximation of smaller fascial defects using 2-0 interrupted nonabsorbable suture, especially if intestinal ischemia was encountered, but lower recurrence rates have been shown with mesh repairs.
- ◆ Polypropylene mesh is placed in the preperitoneal space with 3 cm of overlap with the fascia. The mesh is sutured circumferentially to the fascia without tension, and the edges of the fascia are sutured to the center of the mesh (**Figure 80-11**).

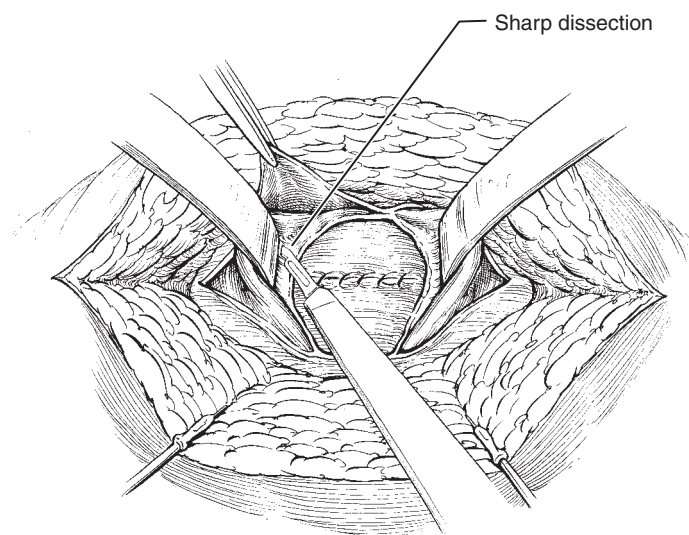


FIGURE 80-10

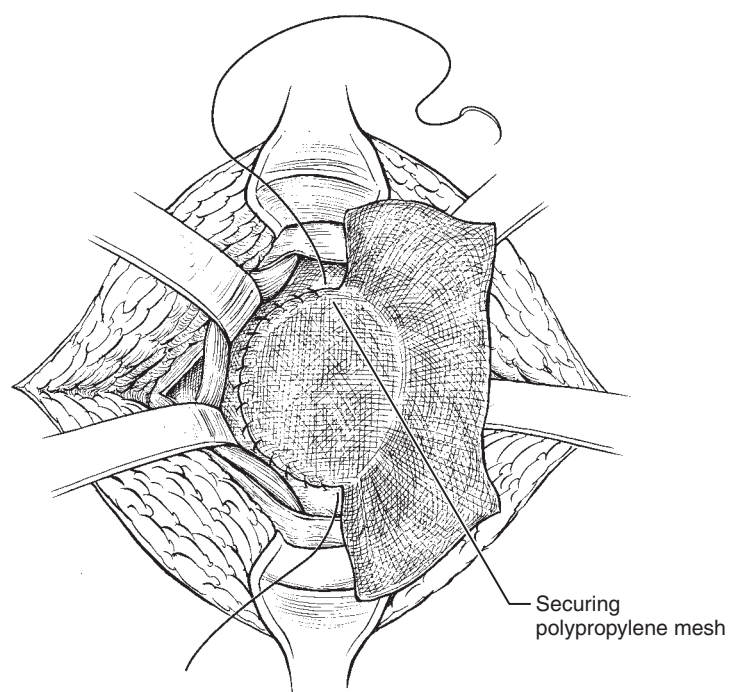


FIGURE 80-11

3. CLOSING

- ◆ A closed suction drain may be required to prevent postoperative seroma if the soft tissue space is excessive. The drain should exit the skin remote from the surgical wound. A two-layer closure using absorbable suture or skin staples completes the operation (Figure 80-12).
- ◆ To help prevent a wound seroma, a ball of cotton or gauze should be placed in the umbilicus and held in place with an abdominal binder.

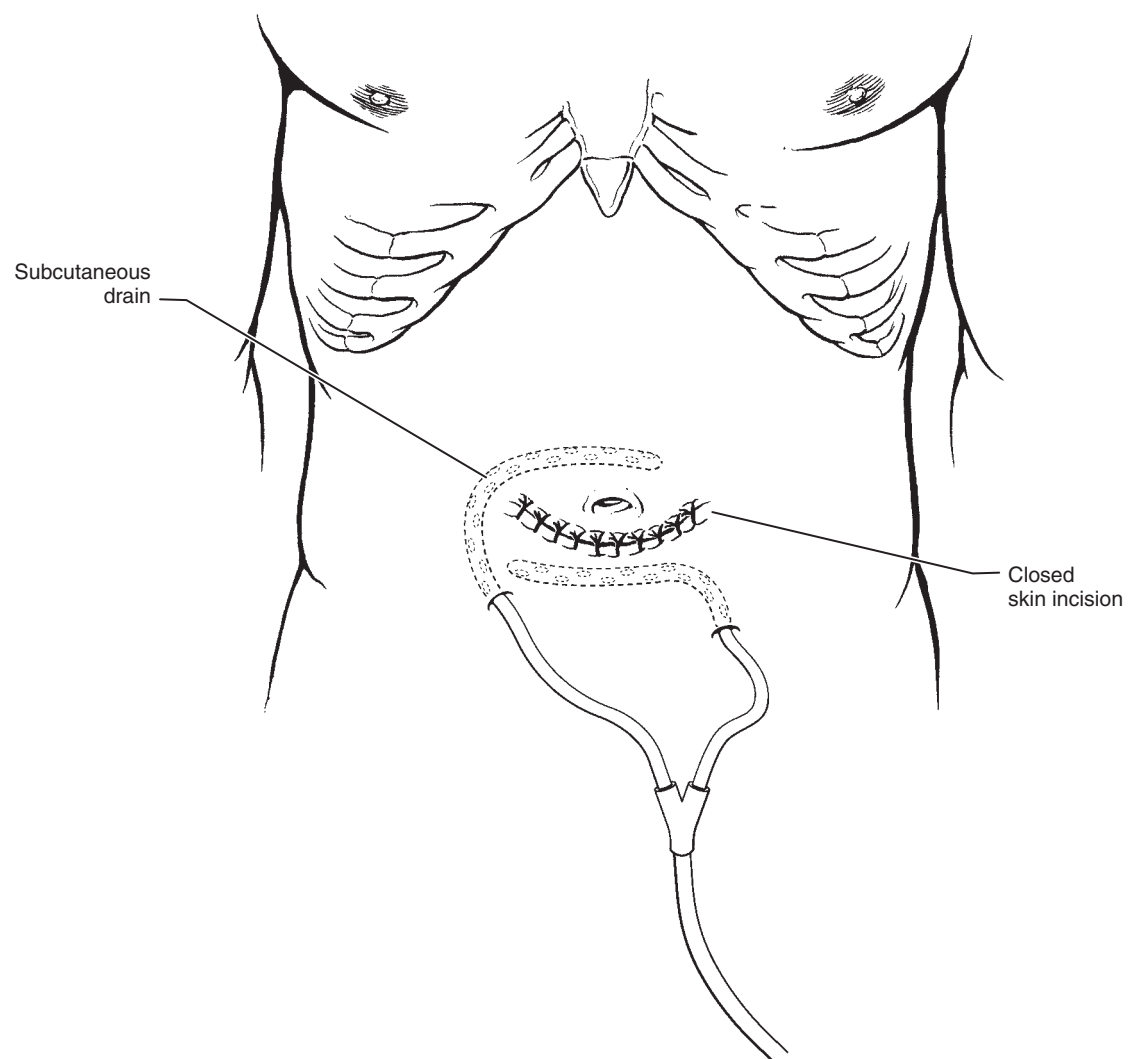


FIGURE 80-12

STEP 4: POSTOPERATIVE CONSIDERATIONS

- ◆ Drains usually may be safely removed once the output declines to less than 30 mL per day.

STEP 5: PEARLS AND PITFALLS

- ◆ In a contaminated field (e.g., resection of gangrenous intestine), an absorbable or biologic fascia substitute may be used with the expectation that a recurrence of the hernia is likely but may be repaired later after the eradication of the contamination.
- ◆ Polypropylene mesh should not be placed within the peritoneal cavity. If there is insufficient peritoneum to close in the midline, an expanded polytetrafluoroethylene mesh product may be placed, because formation of adhesions to this surface or erosion into adjacent viscera is unlikely.

SELECTED REFERENCES

1. Radhakrishnan J: Umbilical hernia. In Nyhus LM, Condon RE (eds): *Hernia*, 4th ed. Philadelphia, JB Lippincott, 1995.
2. Skinner MA, Grosfeld JL: Inguinal and umbilical hernia repair in infants and children. *Surg Clin North Am* 1993;73:439-449.

INCISIONAL/VENTRAL HERNIA— MESH AND TISSUE FLAP

Thomas D. Kimbrough

STEP 1: SURGICAL ANATOMY

- ◆ A midline epigastric incisional hernia is represented in **Figure 81-1**.
- ◆ **Figure 81-2** is a cross-section of the hernia illustrating the relevant layers.

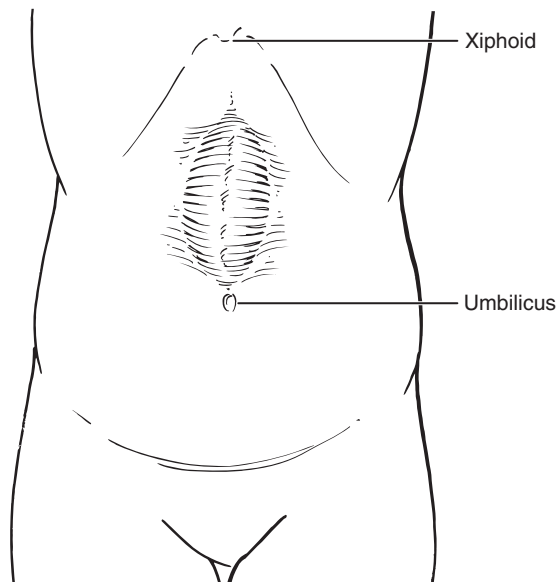


FIGURE 81-1

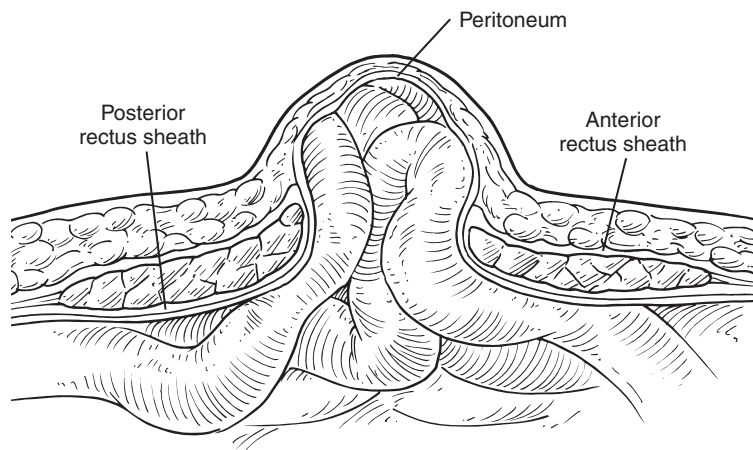


FIGURE 81-2

STEP 2: PREOPERATIVE CONSIDERATIONS

- ◆ The most important preoperative consideration is whether the hernia should be repaired. Because the risk to the patient from the hernia decreases as its diameter increases, and the chance of recurrence and other surgical complications increases, the risk-to-benefit ratio should be carefully assessed.
- ◆ In the event repair is deemed desirable, many of these patients have significant comorbidities that must be addressed preoperatively and managed perioperatively. Neglect of these can lead to failure in spite of a technically superb surgical repair.
- ◆ There are many techniques for repair of incisional hernias, illustrating among other things that no one method has been judged superior. The technique illustrated here is but one of many acceptable available.

STEP 3: OPERATIVE STEPS**1. INCISION**

- ◆ In the case of incisional hernias, the new incision is made by excising the old scar.
- ◆ In the case of a ventral hernia not related to a previous surgical procedure, the incision is best placed along the longer axis of the fascial defect.
- ◆ If the fascial defect is circular with no significant difference in the length of axes, transverse incisions leave better scars.

2. DISSECTION

- ◆ After the hernia sac is identified, its external peritoneal lining is dissected free from surrounding structures, including the innermost fascial layer of the abdominal wall.
- ◆ Although it is often necessary to open the peritoneum and even resect portions of it, preservation of enough of the peritoneum to close allows the imposition of a tissue layer between the mesh to be used and the contents of the intra-abdominal cavity. The end result is illustrated in **Figure 81-3**.
- ◆ **Figure 81-4** shows the next step, which is separation of the posterior rectus sheath from the overlying rectus abdominis muscle.
- ◆ Primarily the cut edges of the posterior sheath are then closed, even if under tension. This closure can be facilitated by application of the techniques of component separation.

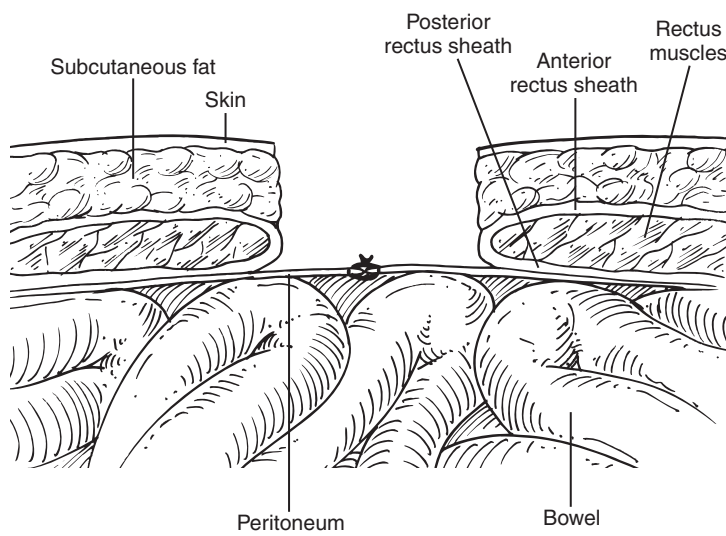


FIGURE 81-3

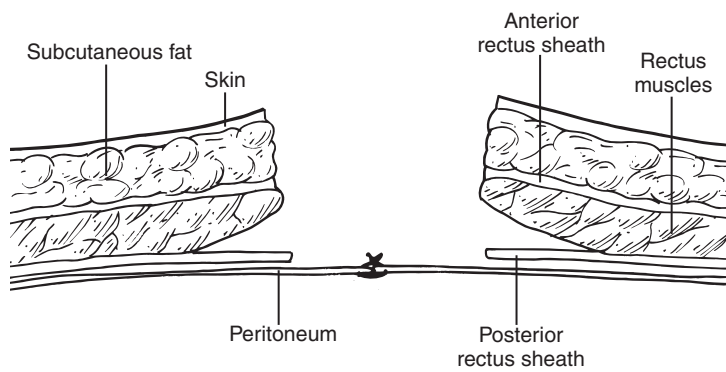


FIGURE 81-4

- ◆ A sheet of polypropylene mesh is then positioned on top of the posterior sheath and under the rectus muscle. There should be an overlap of 3 to 5 cm on all sides. Some choose to tack the mesh in place with mattress sutures through the rectus muscle and anterior sheath. I have found it sufficient to tack the mesh to the underlying posterior sheath with absorbable sutures. Either way the mesh should be tacked down as tautly as possible. The completion of these steps is illustrated in **Figure 81-5**.
- ◆ **Figure 81-6** shows the completed repair after primary closure of the anterior sheath. Again, fascial release techniques such as component separation can facilitate this process.
- ◆ It is, of course, not always possible to completely close the posterior and anterior fascial layers, even with releases. In that event, as much as possible is closed, even if under tension. Every effort is made to close some type of tissue layer between the abdominal contents and the mesh, and the mesh and the skin. As mentioned earlier, the peritoneum can be used in the first instance, and the subcutaneous tissues in the latter.

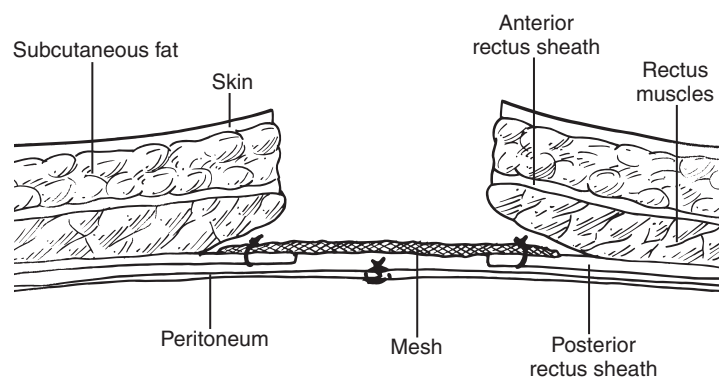


FIGURE 81-5

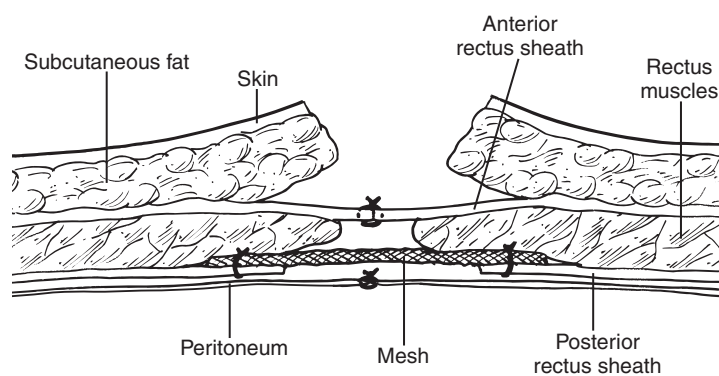


FIGURE 81-6

3. CLOSING

- ◆ Scrupulous attention should be paid to hemostasis, because postoperative hematomas are not uncommon and can create significant problems.
- ◆ Whether to use drains in any or all of the spaces created by the dissection is the choice of the individual surgeon. The drains are no substitute for good technique and offer a route for the introduction of bacteria.

STEP 4: POSTOPERATIVE CARE

- ◆ Most patients require a day or two in the hospital for adequate pain control.
- ◆ Patients are instructed to refrain from lifting or doing strenuous work for 4 to 6 weeks.

SELECTED REFERENCE

1. Zinner MJ, Schwartz SI, Ellis H: Hernias. In Maingot R, Zinner M (eds): *Maingot's Abdominal Operations*, vol 1, 10th ed. Stamford, Conn, Appleton & Lang, 1997, pp 479-580.